

Chapter 3

Shafts

Shafts are vital access ways for many underground mines. They provide passageways for personnel, materials, and ventilation air. Vertical shafts are the norm, but inclines are also used. Shafts must be engineered for a long, stable design life; they must be safe at all points along the shaft route. Where the rock mass is strong and the *in situ* stress is low, an unlined shaft may be acceptable, for example, a ventilation shaft, but usually some provision is needed for preventing loose rock from cascading down the shaft. Even small rock fragments can do great harm after falling from considerable height. Rock bolts, steel strapping, wire mesh, screen of some type, lacing, and lagging are commonly used to prevent minor falls of rock in shafts and are usually installed during shaft sinking. A relatively thin concrete liner serves the same purpose and also provides a smooth surface that reduces air friction loss in ventilation shafts.

If the safety factor of the unlined, unsupported shaft wall is acceptable, then no additional support is needed beyond that provided by shaft framing which may be wood or steel, although steel is the most commonly used material today. Shaft framing partitions the shaft space into compartments for hoisting personnel and material, and for utilities (electrical power, compressed air, drill water, and pump water) as well as for ladders that allow passage of personnel. Spacing of shaft framing is variable but generally of the order of feet or meters. Ends of shaft framing are usually placed in notches excavated in the shaft walls (bearing sets), but may also rest on bearing plates (ledges) bolted to the walls.

3.1 Single unlined naturally supported shafts

Calculation of a safety factor (FS) for an unlined shaft wall is straightforward in principal, but requires attention to geological details. Thus,

$$\begin{aligned} \text{FS} &= \frac{\text{Strength}}{\text{Stress}} \\ \text{FS} &= \frac{C_o}{\sigma_c} \\ \text{FS} &= \frac{T_o}{\sigma_t} \end{aligned} \tag{3.1}$$

where C_o and T_o are the unconfined compressive and tensile strengths of the shaft wall rock, and σ_c and σ_t are the peak compressive and tensile stresses at the shaft wall. If tension or compression is absent, then the associated safety factor is not an issue. These safety factors

use stress and strength at a point and therefore are *local*. However, when the most highly stressed point at a shaft wall is within the *elastic limit*, then all other points at the shaft wall have higher safety factors and thus overall or *global* safety is assured. Safety factors of 2 to 4 in compression and 4 to 8 in tension have been suggested (Obert *et al.*, 1960, chapter 3).

Although a safety factor calculation as such is simple in the extreme, determining the requisite strength and stress requires thoughtful attention to details of rock mass strength and stress and how they may be influenced by shaft route geology. The safety factor approach implies that the rock mass response to load is initially elastic, but that the elastic range of deformation is limited by rock mass strength. Within the elastic range of stress, shaft wall displacements remain small. These conditions may not always be met. For example, an important exception occurs when a shaft route penetrates bedded salt that behaves much like a highly viscous fluid. Shales may also exhibit a significant time-dependent component of displacement and lead to “squeezing” ground. Densely jointed rock masses may tend to flow because of joint alteration mineralogy and inelasticity. The presence of acid mine waters, saline solutions, and liquid and gaseous hydrocarbons may further complicate a rock mass response to stresses induced by shaft excavation. Nevertheless, elastic behavior is the usual expectation and is the model followed in the absence of field evidence to the contrary.

Shaft wall stress concentration

In elastic ground, the peak stresses act tangentially at the shaft wall and may be obtained from stress concentration factors K_c and K_t for compression and tension, respectively. Thus,

$$\begin{aligned}\sigma_c &= K_c S_1 \\ \sigma_t &= K_t S_1\end{aligned}\tag{3.2}$$

where S_1 is the major principal stress before excavation. If a stress concentration factor is negative, then the associated stress is opposite in sense to S_1 . Stress concentration factors are obtained from solutions to problems in the mathematical theory of elasticity and are simply the ratios of peak to reference stresses. The usual reference stress in rock mechanics is the preexcavation or applied major principal stress. Usually compression is considered positive.

Peak stresses occur at hole boundaries and depend on:

- 1 hole shape
- 2 aspect ratio
- 3 principal stress ratio M
- 4 orientation.

Common shapes are rectangular and circular. New shafts are quite likely to be circular and for this reason the circular shaft is important. Older shafts are likely to be rectangular, and so extension of an existing shaft may involve a rectangular shape. Elliptical sections are possible but rare, although a circle may be considered a special ellipse. Aspect ratio refers to the ratio of section dimensions, for example, “width” W_o to “height” H_o . The principal stress ratio M is the ratio of minor to major premining stress S_3/S_1 . Orientation is determined by the angle the long axis of the opening makes with the major principal premining stress.

Unlined circular shafts

A model of an unlined shaft with circular cross section is a circular hole excavated in a slab of unit thickness that is loaded by uniformly distributed stresses over the sides of the slab. Relative displacement between slab faces is zero (plane strain). In linearly elastic, homogeneous, isotropic rock, the circumferential (tangential) normal stress at the shaft wall is

$$\sigma_t = (S_1 + S_3) - 2(S_1 - S_3)\cos(2\theta) \quad (3.3)$$

where S_1 and S_3 are the major and minor applied (preshaft) principal stresses and θ is the angle from the direction of S_1 to the point of action of σ_t (see, for example, Obert and Duvall, 1967, chapter 3) Circumferential stress concentration K at the shaft wall is then

$$K = (1 + M) - 2(1 - M)\cos(2\theta) \quad (3.4)$$

where M is the ratio of preshaft minor to major principal stress and θ is a counterclockwise angle from S_1 to the point of stress concentration at the shaft wall. Figure 3.1 illustrates a compressive stress concentration at the wall of a circular shaft according to Equations (3.3) and (3.4).

Equation (3.4) is similar in each quadrant, so only the first quadrant needs to be examined. Maximum and minimum values of K occur at 0° and 90° in this quadrant and have values $K_{\max} = (3 - M)$ and $K_{\min} = (-1 + 3M)$. By definition, $M < 1$. Although this restriction allows for very large but negative values of M , a range between -1 and $+1$ is reasonable, while a range between 0 and $+1$ includes most practical cases. Table 3.1 describes the state of stress for selected values of M . The first case in Table 3.1 (pure shear or “diagonal tension”) is unlikely, while the last case is highly improbable in nature, but is included for the sake of completeness.

Figure 3.2(a) is a plot of (3.4) for a range of M values and shows the vanishing of tension at the shaft wall for $M \geq 1/3$. This result suggests that tensile failure of an unlined shaft is often not a consideration. All curves have a common point of $K = 2$ at $\theta = 60^\circ$. Figure 3.2(b) is a

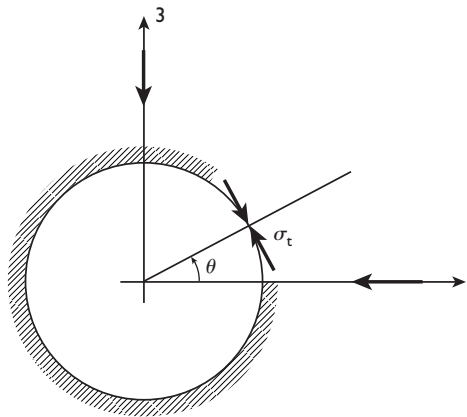


Figure 3.1 Compressive stress concentration at the wall of a circular shaft.

Table 3.1 Principal stresses for selected M values^a

Case M	Principal stresses	Physical description
-1	$S_3 = -S_1$	Pure shear
0	$S_3 = 0$	Uniaxial compression
+1	$S_3 = +S_1$	Hydrostatic compression
$-\infty$	$S_1 = 0$	Uniaxial tension

Note

a Compression is considered positive.

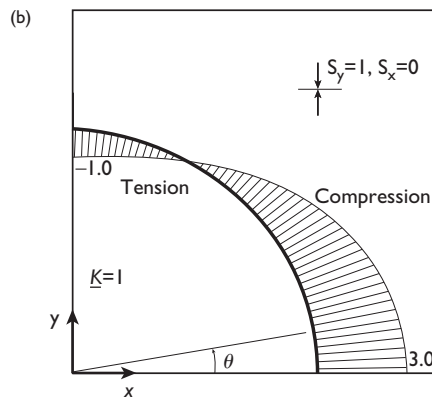
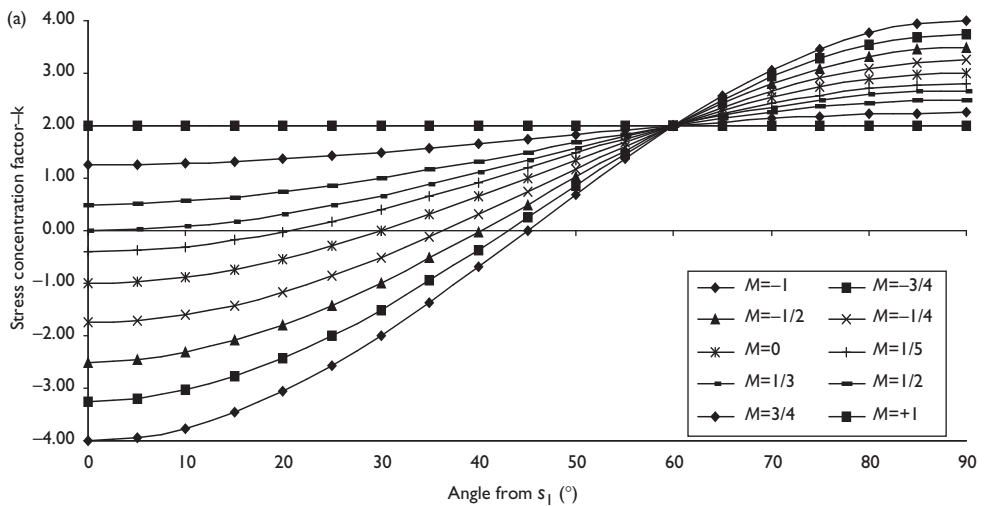


Figure 3.2 (a) Stress concentration at the wall of a circular shaft at 5° intervals for a range of principal preshaft stress ratios (M values) and (b) graphical representation of stress concentration about a circular hole.

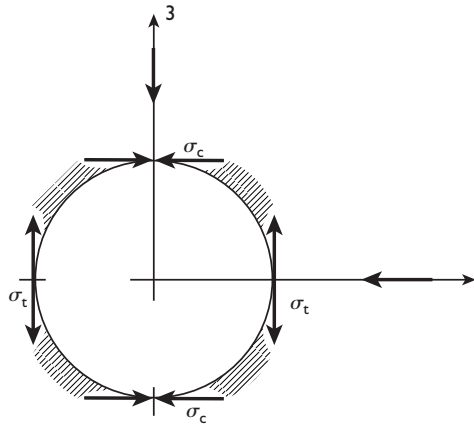


Figure 3.3 Peak tensile and compressive stress concentration about a circular hole.

graphical representation of (3.4) for $M = 0$ (uniaxial compression). Stress concentration magnitude is proportional to radial line length; tension (negative) is plotted inside the excavation boundary, while compression (positive) is plotted outside. The scale bar under $K = 1$ shows a stress concentration of one.

The peak compressive stress concentration K_c always occurs parallel to the direction of S_1 at $\pm 90^\circ$ and ranges from 2.0 to 3.0, as a practical matter. When $M > 1/3$, the minimum stress concentration factor is compression and only safety with respect to compressive stress failure is of concern. However, when tension is present ($M < 1/3$), the peak tensile stress concentration factor K_t occurs perpendicular to S_1 at 0° (and 180°) and ranges from -1 to nil, as a practical matter. Figure 3.3 shows the location and orientation of these peak compressive and tensile stress concentrations.

Consider a hypothetical case of a vertical, circular shaft sunk in a preshaft stress field attributable to gravity alone. The vertical preshaft stress is then unit weight of rock times depth ($S_v = \gamma H$); the horizontal stress in any direction is some fraction of the vertical stress. Assume compass coordinates, so $x = \text{east}$, $y = \text{north}$ and $z = \text{up}$, then in the x - and y -directions, the horizontal stresses S_H and S_h are some fraction of the vertical stress, that is, $S_H = S_h = K_o S_v$. In plan view, $S_H = S_h = S_1 = S_3$, $M = 1$ (hydrostatic case) and the shaft wall is in a uniform state of compression. The compressive stress concentration factor $K_c = 2$, hence $FS = FS_h = C_o/2K_o S_v$ in plan view (horizontal cross section). In vertical section, a vertical compressive stress acts at the shaft wall that is equal to the preshaft vertical stress. Thus, in vertical section, $S_v = S_1$ and $S_3 = 0$, but there is no stress concentration of the vertical stress. In this view, the unlined shaft wall safety factor is simply $FS = FS_v = C_o/S_v$. In a gravity-only initial stress field; FS_v is often less than FS_h and therefore governs the design, but both safety factors should always be computed.

Example 3.1 A vertical, unlined circular shaft is sunk in a preexcavation stress field caused by gravity alone. Estimate the *peak stresses* in horizontal and vertical sections as functions of depth. Note: The ratio of horizontal to vertical stress is $1/3$.

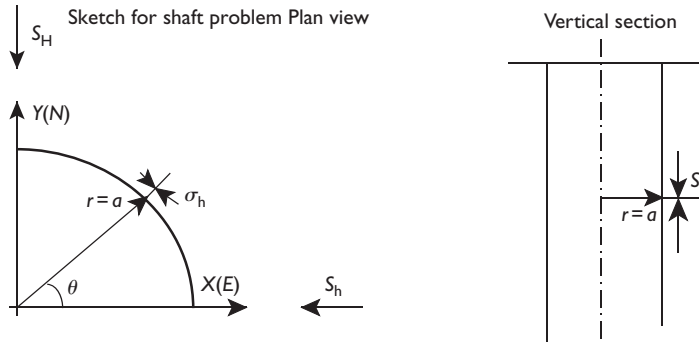
Solution: Peak stress may be obtained from stress concentrations and reference stresses. There are no tensile stresses. In plan view, the stress concentration factor is 2 and in section is 1.

However, the reference stress in plan view is the horizontal preexcavation stress, while in section, it is the vertical preexcavation stress. Thus, the peak compression in plan view is

$$\begin{aligned}\sigma_c &= 2S_h \\ &= 2K_o S_v \\ &= (2)(1/3)\gamma h \\ \sigma_c &= (2/3)(\gamma/144)h\end{aligned}$$

where stress σ_c is in psi, specific weight γ is in pcf, and depth h is in ft. The peak stress in psi in this case is about 2/3 times the depth in ft.

In vertical section, the preexcavation and post-excavation stresses are equal, so the post-stress compression at the shaft wall $\sigma_v = \gamma h$ or about 1 psi per foot of depth.

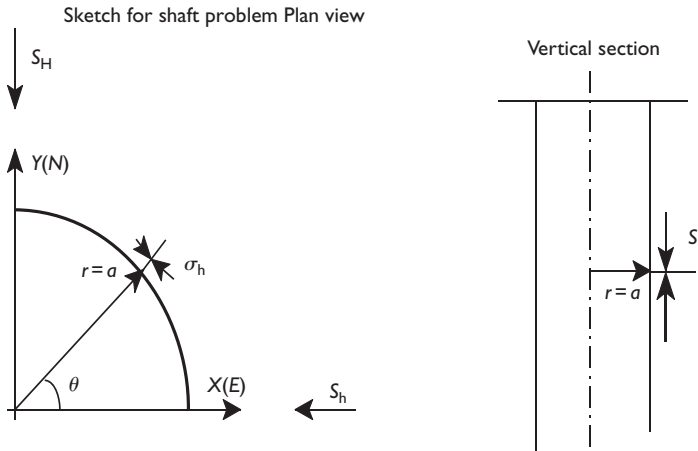


Example 3.2 A vertical, unlined circular shaft is sunk in a preexcavation stress field caused by gravity alone. Estimate the *peak stress concentrations* in horizontal and vertical sections as functions of depth.

Solution: In a gravity stress field, the vertical preexcavation normal stress is unit weight of rock times depth, so $S_v = \gamma h$ where γ is unit weight of rock and h is depth from the surface. Compression is considered positive. The unit weight is not given, so an estimate is necessary for a numerical calculation. Horizontal normal stresses, S_h and S_H , say, in the east and north directions (compass coordinates, z is positive up) are equal in a gravity stress field and are some fraction of the vertical stress. Thus, $S_H = S_h = K_o S_v$ where K_o is some decimal fraction. An estimate of K_o can be obtained by considering the gravity field to be applied to an elastic rock mass under complete lateral restraint. This assumption implies zero lateral strains. In consideration of Hooke's law

$$\begin{aligned}E\varepsilon_h &= 0 \\ &= S_h - \nu S_H - \nu S_v \\ 0 &= (1 - \nu)S_h - \nu S_v \\ \therefore S_h &= S_H = \left(\frac{\nu}{1 - \nu}\right)S_v\end{aligned}$$

A reasonable value for Poisson's ratio is 0.25, so that $K_o = 1/3$.



A formula for stress concentration about a circular hole can be obtained from the expression for the tangential stress. Thus, at the shaft wall of radius a

$$\begin{aligned}\sigma_\theta &= \left(\frac{S_h + S_H}{2} \right) (2) - \left(\frac{S_h - S_H}{2} \right) (4) \cos(2\theta) \\ &= 2S_h \\ &= 2K_o S_v \\ \sigma_\theta &= (2)(1/3)\gamma h\end{aligned}$$

as shown in the sketch.

The compressive stress concentration K_c in a horizontal plane (plan view) is by definition the ratio of peak stress to reference stress which is the preexcavation major principal stress *in the considered view*, that is, S_h . Hence, $K_c = 2$ in plan view and is independent of depth.

In vertical section, the postshaft vertical stress is equal to the preshaft vertical stress. Hence the stress concentration factor (in compression) $K_{cv} = S_1/S_v = 1$ and is also independent of depth.

There are no tensile stresses in either view, so tensile stress concentration need not be considered.

Example 3.3 A vertical circular shaft is proposed in a region of high horizontal stress. Measurements along the shaft route indicate the preshaft principal stresses in psi (compression positive) are given by

$$S_v = 1.1(\gamma/144)h, S_h = 100 + 1.5(\gamma/144)h, S_H = 500 + 2.2(\gamma/144)h$$

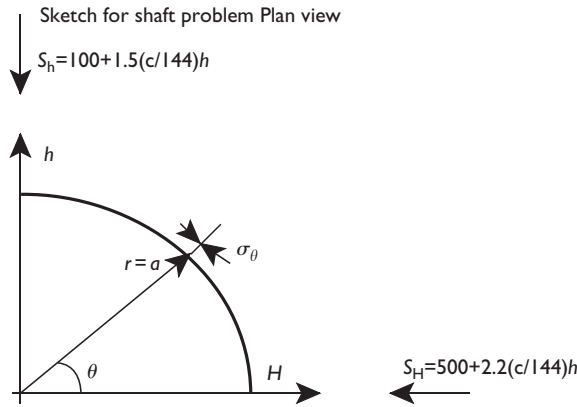
(h is depth in feet).

Determine the strengths (C_o , T_o) needed for safety factors with respect to compression and tension of 1.5 and 2.5, respectively, at 3,750 ft, with $\gamma = 160$ pcf.

Solution: A sketch illustrates the situation in plan view. In this view, the tangential stress is given by

$$\sigma_\theta = (1/2)(S_H + S_h)(2) - (1/2)(S_H - S_h)(4)\cos(2\theta)$$

The extreme values of this function with respect to θ are at zero at 0° and 90° . The extreme values are $-S_H + 3S_h$ ($\theta = 0^\circ$) and $3S_H - S_h$ ($\theta = 90^\circ$). Because $S_H > S_h$, the first is the minor stress which is tensile when $3S_h < S_H$; the second is the major compression.



Thus,

$$\sigma(0) = -[500 + 2.2(160/144)3,750] + (3)[100 + 1.5(160/144)3,750] = 9,383 \text{ psi}$$

$$\sigma(90) = (3)[500 + 2.2(160/144)3,750] - [100 + 1.5(160/144)3,750] = 22,650$$

By definition $FS_c = C_o/\sigma_c$ and $FS_t = T_o/\sigma_t$. No tension is present, so tensile strength is not a factor. The compressive strength needed is $(1.5)(22,650) = 33,975$ psi, a relatively high value even for intact core tested in the laboratory. Reinforcement and support would be needed before reaching the considered depth. In vertical section $\sigma = S_v = 1.1(160/144)(3,750) = 4,583$ psi which is considerably less than the peak compression in plan view.

Unlined elliptical shafts

An elliptical shape introduces two additional considerations for unlined shaft safety factor calculations: aspect ratio and orientation. In case of elliptical shafts, when the axes of the ellipse coincide with the preshaft principal directions, if a is the semi-axis parallel to S_1 and b is parallel to S_3 , then the stress concentration tangential to the shaft wall is given by

$$K = \frac{(1 - k^2)(1 - M) + 2k(1 + M) - (1 - M)(1 + k)^2\cos(2\alpha)}{(1 + k^2) - (1 - k^2)\cos(2\alpha)} \quad (3.5)$$

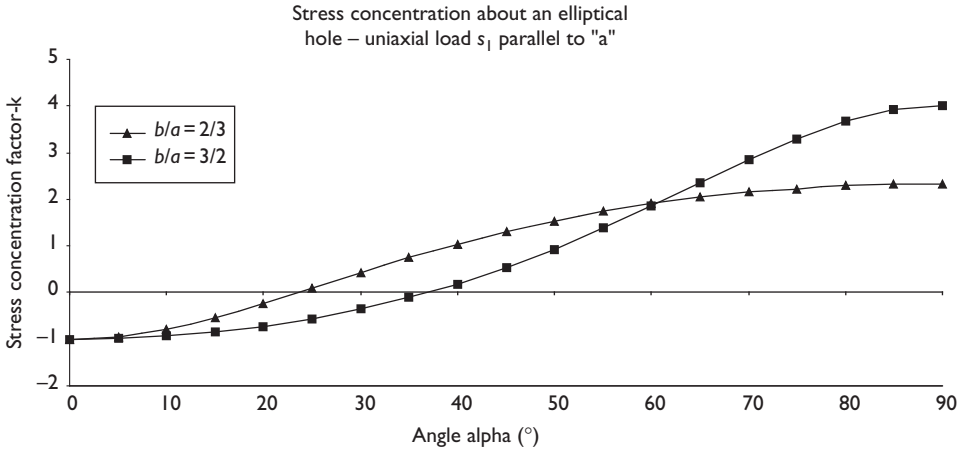


Figure 3.4 Stress concentration about an elliptical shaft with semi-axes ratios of 2/3 and 3/2 under uniaxial stress ($M = 0$) S_1 applied parallel to the semi-axis a .

where k is the aspect ratio b/a , which may be greater or less than one (see, e.g. Jaeger and Cook, 1969). The angle α is related to the polar angle by $\tan(\theta) = (b/a) \tan(\alpha)$ where angles are measured counterclockwise from the direction of S_1 . In the case of a circular shaft, $k = 1$ and (3.5) reduces to (3.4). Figure 3.4 shows how K varies with α for two selected values of k (2/3 and 3/2). Inspection of Figure 3.4 suggests that peak stress concentrations occur at the ends of the semi-axes, that is, at the sides and ends of an ellipse ($\theta = \alpha = \pm 90^\circ$, $\theta = \alpha = 0^\circ$ and 180°) as in the case of a circular shaft. Indeed, this is the case. In fact, stress concentrations at the ends of an ellipse semi-axes with S_1 parallel to semi-axes a are

$$K_a = -1 + M \left(\frac{2}{k} + 1 \right) \quad (3.6)$$

$$K_b = (1 + 2k) - M$$

which reduce to the circular case when $k = 1$. Figure 3.5(a) shows the meaning of stress concentration factors K_a and K_b when the long axis of an ellipse is parallel to S_1 ($k < 1$); Figure 3.5(b) shows the situation when the long axis is perpendicular to S_1 ($k > 1$). In case $M = 0$, the tension under S_1 at the ends of the a semi-axis is equal to S_1 (but opposite in sign) regardless of k , while the compression at the ends of the b semi-axes is much greater when the long axis of the ellipse is perpendicular to S_1 ($k > 1$). Figures 3.5(c) and (d) show graphically two uniaxial cases: (c) when the applied load is parallel to the long axis of an ellipse with semi-axes ratio $b/a = 2/3$ and (d) when the applied load is perpendicular to the long axis of an ellipse with semi-axes ratio $b/a = 3/2$. These cases are those plotted in Figure 3.4. If both stress concentration factors are positive, then tension is not a factor.

Consider an elliptical shaft with a semi-axes ratio $k = 1/3$ in a preshaft stress field characterized by $M = 1/4$. The long axes of the shaft cross section is parallel to S_1 and according to (3.6), $K_a = +0.75(+3/4)$, $K_b = +1.42(+1 + 5/12)$, and tension is absent. If the long

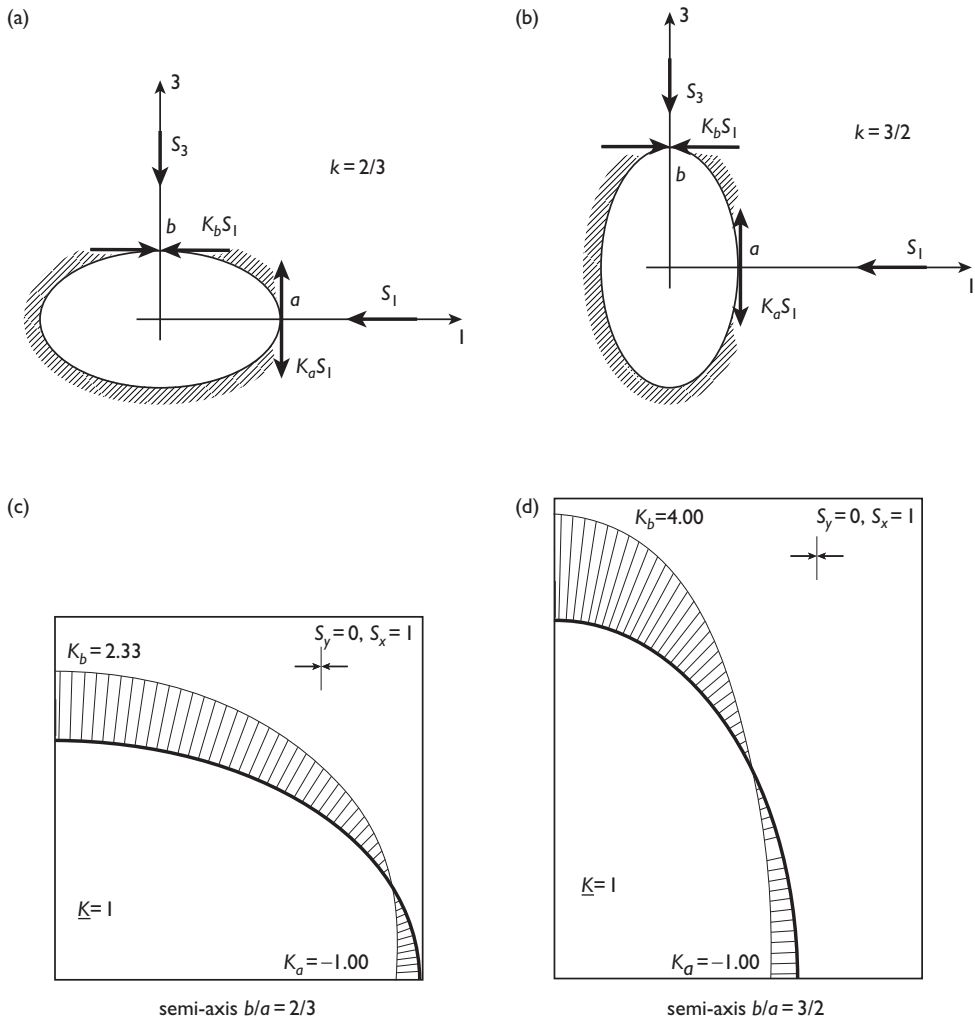


Figure 3.5 (a) and (b) Stress concentration factors K_a and K_b about an elliptical shaft. (c) and (d) Stress concentration about two uniaxially loaded ellipses.

axis is perpendicular to S_1 , so $k = 3$, then $K_a = -0.59(-7/12)$ and $K_b = +6.75(+6 + 3/4)$. These numbers illustrate a *rule of thumb* that states *the most favorable orientation of an elliptical excavation is with the long axis parallel to the major preshaft compression*.

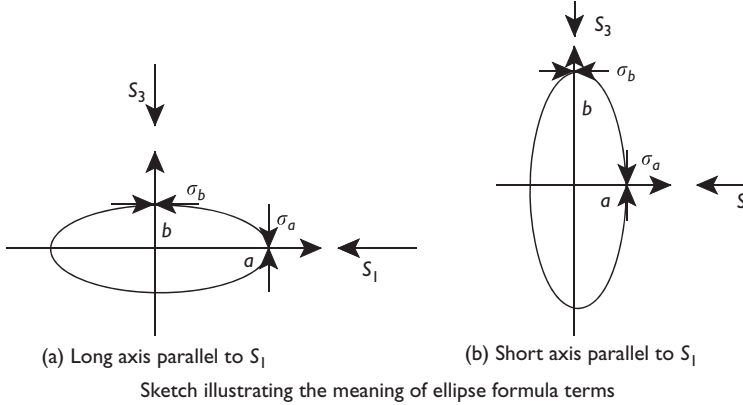
Example 3.4 An elliptical cross section may be advantageous in a stress field where the major and minor preexcavation stresses S_1 and S_3 are aligned with the geometric axes of the section. Show that the preferred orientation is with the long axis of the ellipse parallel to S_1 .

Solution: The peak stress concentrations at the wall of an elliptical section occur at the ends of the geometric axes. The peak stresses are given by the formulas

$$\sigma_a = K_a S_1 = -S_1 + (1 + 2/k)S_3 = S_1[(1 + 2/k)M - 1]$$

$$\sigma_b = K_b S_1 = -S_3 + (1 + 2k)S_1 = S_1[(1 + 2k) - M]$$

where the meaning of terms is shown in the sketch, $k = b/a$ and $M = S_3/S_1$.



In the uniaxial case, $M = 0$, so when the long axis is parallel to S_1 , $k < 1$, $K_a = -1$ and $K_b = 1 + 2k$, and when the short axis is parallel to S_1 , $k = k' > 1$, $K_a = -1$, as before, and $K_b = 1 + 2k'$, which is a larger number than the first case. These observations suggest that when the long axis is parallel to S_1 , the peak compression will be minimal. A change in axis orientation corresponds to using the reciprocal of k . If k is the aspect ratio when the long axis is parallel to the x -axis (Case 1), then use of the reciprocal $k' = 1/k$ corresponds to the case where the long axis is normal to the x -axis (Case 2). Examination of the stress concentration factors for points a and b shows that $K'_b > K_b$ and $K_a > K'_a$ always and that $K_b > K_a$ when $k > M$. With this proviso, the stress concentration factors are ordered: $K'_b > K_b > K_a > K'_a$. Thus, stress concentrations in Case 1 are bounded by those in Case 2, provided $k > M$.

For a given value of preexcavation stress ratio M , one may choose an ellipse aspect ratio k that meets this condition. The best orientation is then with the long axis parallel to the major compression before excavation. High aspect ratio ellipses are unlikely to be considered in practice, so the condition $k > M$ is not too restrictive. For example, given $M = 1/3$, then for $k = 1/2$, $K'_b = 1 + 2(2) - 1/3 = 4.67$, $K_b = 1 + 2(1/2) = 2$, $K_a = [1 + 2/(1/2)](1/3) - 1] = 0.67$, $K'_a = -0.33$. Thus, when the long axis of the ellipse is oriented parallel to S_1 , the stress concentrations at b and a are 2 and 0.67, respectively. If the long axis were normal to S_1 , the stress concentrations would be 4.67 and -0.33 , respectively, where the negative sign indicates a tension (at a).

Consider a problem of selecting the best elliptical shape for a given *in situ* stress field. The problem is to determine the optimum k for a given M . Optimum from the ground control view means lowest stress concentrations. Suppose the given stress state is hydrostatic, so $M = 1$, then $K_a = 2/k$ and $K_b = 2k$ according to (3.6). Both stress concentration factors are positive. Thus tension is not a factor. Optimum or least compressive stress

concentration can now be obtained by reducing the larger of K_a or K_b . Ideally, one has neither $K_a > K_b$ nor $K_b > K_a$. Hence, the optimum occurs when $K_a = K_b$, that is, when $k = 1$. Thus, a circular shape, a special case of an ellipse, is optimum when the preshaft stress state is hydrostatic. This situation occurs for vertical shafts in ground where the stress state is caused by gravity alone.

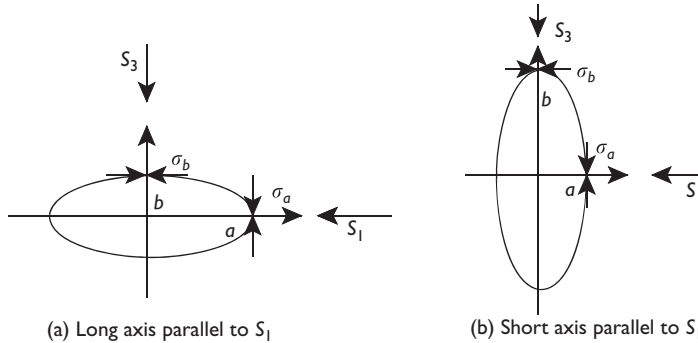
Example 3.5 A vertical shaft of elliptical section is being considered in a region of high horizontal stress. Measurements along the shaft route indicate the preshaft principal stresses in psi (compression positive) are given by $S_v = 1.1h$, $S_h = 100 + 1.5h$, $S_H = 500 + 2.2h$ (h is depth in feet). Determine the optimum orientation and aspect ratio of the section.

Solution: The peak stress concentrations at the wall of an elliptical section occur at the ends of the geometric axes. The peak stresses are given by the formulas

$$\sigma_a = K_a S_1 = -S_1 + (1 + 2/k)S_3 = S_1[(1 + 2/k)M - 1]$$

$$\sigma_b = K_b S_1 = -S_3 + (1 + 2k)S_1 = S_1[(1 + 2k) - M]$$

where the meaning of terms is shown in the sketch, $k = b/a$ and $M = S_3/S_1$.



Sketch illustrating the meaning of ellipse formula terms

An optimum orientation would be one that reduces stress concentration to a minimum, so the stress concentration at b is no greater or less than at a and tension is absent. Thus,

$$K_a = K_b \geq 0$$

$\therefore$

$$(1 + 2/k)M - 1 = (1 + 2k) - M$$

$\therefore$

$$k = M = \frac{100 + 1.5h}{500 + 2.2h}$$

so the aspect ratio varies with depth, but is equal to the preexcavation stress ratio at any particular depth. At the surface $k = 0.200$ and at great depth $k = 1.5/2.2 = 0.682$.

The best orientation is with the long axis a parallel to S_H . In this orientation, with $k = M$, $K_a = K_b = 1 + M$ and indeed the wall of the ellipse is in a uniform compression. At the surface, the stress concentration is 1.2; at great depth the stress concentration is 1.682.

With the short axis parallel to S_H , $K'_a = 2M^2 + M$ and $K'_b = 1 - M + 2/M$, which is greater than $1 + M$ and therefore less favorable at any depth.

If a circular section were considered, the stress concentrations would be $-1 + 3M$ and $3 - M$ or -0.4 and 2.8 at the surface; 1.046 and 2.318 at great depth. The elliptical section shows a considerable advantage over the circular section in this stress field. However, other considerations such as difficulty in excavation to the proper shape would need to be considered before final selection of the shaft shape.

In vertical section, the shaft wall stress is equal to the preshaft stress and stress concentration is one. This value is less than stress concentration in plan view of the best orientation and aspect ratio $(1+M)$ for $M = k > 0$. Hence, the plan view section governs design.

Example 3.6 Suppose a vertical shaft of elliptical section $14 \text{ ft} \times 21 \text{ ft}$ is being considered for sinking to a depth of $4,350 \text{ ft}$. The horizontal stress in the north–south direction is estimated to be twice the vertical stress caused by rock mass weight, while the east–west stress is estimated to be equal to the “overburden” stress. Unconfined compressive strength of rock is $22,000 \text{ psi}$, as measured in the laboratory; tensile strength is $2,200 \text{ psi}$. Determine shaft wall safety factors as functions of depth with the section oriented in a way that minimizes compressive stress concentration.

Solution: Shaft wall safety factors with respect to compression and tension, by definition, are

$$\text{FS}_c = \frac{C_o}{\sigma_c}, \quad \text{FS}_t = \frac{T_o}{\sigma_t}$$

In case of an ellipse, the peak stresses at the ends of the semi-axes are:

$$\sigma_a = K_a S_1 = -S_1 + (1 + 2/k)S_3 = S_1[(1 + 2/k)M - 1]$$

$$\sigma_b = K_b S_1 = -S_3 + (1 + 2k)S_1 = S_1[(1 + 2k) - M]$$

$$k = 14/21 = 2/3 \leq 1, \quad M = S_{EW}/S_{NS} = 1/2,$$

$$\sigma_a = K_a S_1 = -S_1 + (1 + 2/k)S_3 = S_1[(1 + 2/(2/3))(1/2) - 1] = S_1(1.00)$$

$$\sigma_b = K_b S_1 = -S_3 + (1 + 2k)S_1 = S_1[(1 + 2(2/3)) - 1/2] = S_1(1.83)$$

The maximum and minimum stress concentration factors are both positive, so no tension is present. With respect to compressive stress, the shaft wall safety factor as a function of depth is

$$\begin{aligned} \text{FS}_c &= \frac{C_o}{1.83 S_1} \\ &= \frac{22,000}{1.83(2)(158/144)h} \\ \text{FS}_c &= \frac{5,478}{h} \end{aligned}$$

where h is in feet and the specific weight of the rock mass is estimated to be 158 pcf. The long axis of the section is in a north–south alignment.

The safety factor in vertical section is

$$FS_{cv} = \frac{22,000}{(158/144)h} = \frac{20,050}{h}$$

When the ellipse axes do not coincide with the preshaft principal axes, an orientation effect is introduced into the stress concentration formula (3.6). Thus,

$$K = \frac{(1 - k^2)(1 - M)\cos(2\beta) + 2k(1 + M) - (1 - M)(1 + k)^2\cos(2\beta - 2\alpha)}{(1 + k^2) - (1 - k^2)\cos(2\alpha)} \quad (3.7)$$

where β is an inclination angle measured counterclockwise from S_1 to a as shown in Figure 3.6(a). Alternatively, the applied stresses include a shear stress relative to $x - y$ axes coincident with the $a - b$ semi-axes, as shown in Figure 3.6(b). Principal stresses and directions seen in the shaft cross section are given by the usual equations. Thus,

$$\left. \begin{matrix} S_1 \\ S_3 \end{matrix} \right\} = \frac{S_{xx} + S_{yy}}{2} \pm \left[\left(\frac{S_{xx} - S_{yy}}{2} \right)^2 + (T_{xy})^2 \right]^{1/2}$$

$$\tan(2\beta) = \left[\frac{T_{xy}}{(S_{xx} - S_{yy})/2} \right] \quad (3.8)$$

where S_{xx} , S_{yy} , T_{xy} are the preshaft stresses in the shaft cross section and β is measured counterclockwise from the x -axis. There are two solutions for β ; care must be exercised in selecting the correct one. One method is to consider that the major principal stress will be “nearest” to the largest given normal stress and then to measure the angle obtained from a calculator using the arctan function from the direction of the largest given normal stress using

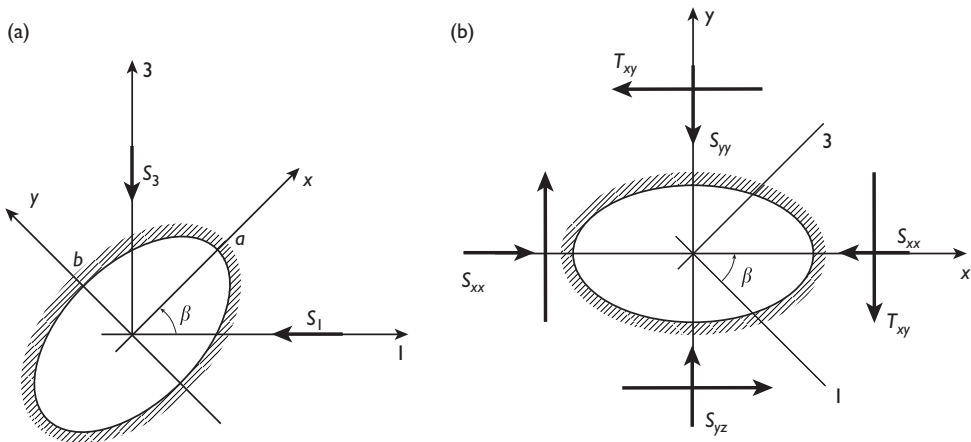


Figure 3.6 Orientation of an elliptical shaft when shear stress is present.

the sign convention that counterclockwise is positive. For example, suppose stress measurements show $S_{xx} = 1,500$, $S_{yy} = 2,500$, $T_{xy} = 866$ in unspecified units. According to Equation (3.8), $S_1 = 3,000$, $S_3 = 1,000$, $\beta = -30^\circ$. Hence, the direction of S_1 is at a 30° clockwise angle from the y -axis (60° counterclockwise from the x -axis).

An interesting exception to the rule of peak stresses occurring at the semi-axes ends occurs when tension is present at the walls of an inclined ellipse. Figure 3.7(a) shows stress concentration at the walls of an elliptical shaft with semi-axes ratio $1/4$ under uniaxial compression ($M = 0$) as a function of polar angle θ (measured counterclockwise from the a semi-axis) for inclinations up to 90° . In the worst case shown in Figure 3.7(a), $\beta = 30^\circ$ and the peak tensile stress concentration is about -1.5 which occurs about 8° away from the a semi-axis. This tension is 50% greater than the frequent peak tension of -1 . The peak tension and angle away from the semi-axis increase with decreasing aspect ratio. A similar compressive stress concentration effect is also evident in Figure 3.7(a). As the ellipse is rotated from being parallel to S_1 ($\beta = 0^\circ$) and having the most favorable orientation perpendicular to S_1 ($\beta = 90^\circ$), which is the least favorable orientation, the compressive stress concentration factor increases from 1.5 to 9.0 . The peak tensile stress concentration ranges from -1.0 to about -1.5 at $\beta = 30^\circ$ and back to -1.0 . Figures 3.7(b) and (c) show graphically the distribution of stress concentration in this case ($\beta = 30^\circ$) and in the case when $\beta = 60^\circ$. However, these effects are not of much practical importance for shafts because the elliptical shape itself is not very practical and uniaxial stress *in situ* is rare, while tension is greatly reduced under biaxial stress.

Example 3.7 Consider a vertical shaft with an elliptical cross section that has a semi-axis ratio $k = 1/3$ and is excavated in a stress field characterized by a principal stress ratio $M = 1/4$ in plan view. The section of interest is at a depth of $1,200$ ft where the vertical stress is $1,350$ psi. Intact rock strengths are: $C_o = 13,500$ psi and $T_o = 1,250$ psi. Suppose the a -axis is inclined 30° to S_1 and that the xy compass directions coincide with the principal 1-3 directions. Determine shaft wall safety factors with respect to compression and tension.

Solution: According to Equation (3.7), (with $M = 1/4$, $k = 1/3$, $\beta = 30^\circ$, $\alpha = 0^\circ, 30^\circ$), stress concentration factors at the ends of the ellipse are: $K_a = 9/4$ and $K_b = 11/12$. Thus, tension is absent and peak compression is at a . If $S_1 = 1,200$, estimated at 1 psi/ft of depth, then $\sigma_t = 2,700$ psi at the ends of the a semi-axes and $1,100$ psi at the ends of the b semi-axes. The shaft wall safety factor with respect to compression is

$$\text{FS}_c = \frac{13,500}{(9/4)(1,200)} = 5.00$$

$$\text{FS}_v = \frac{13,500}{1,350} = 10.00$$

where the first calculation is in plan view and the second is in vertical section. In this example, safety is determined in horizontal section.

Unlined rectangular shafts

In the case of rectangular shapes, rounding of corners is an important consideration. Amathematical sharp corner leads to infinite stress at a corner point. A rounded corner, characterized by a specified radius of curvature, results in finite stress. The greater the

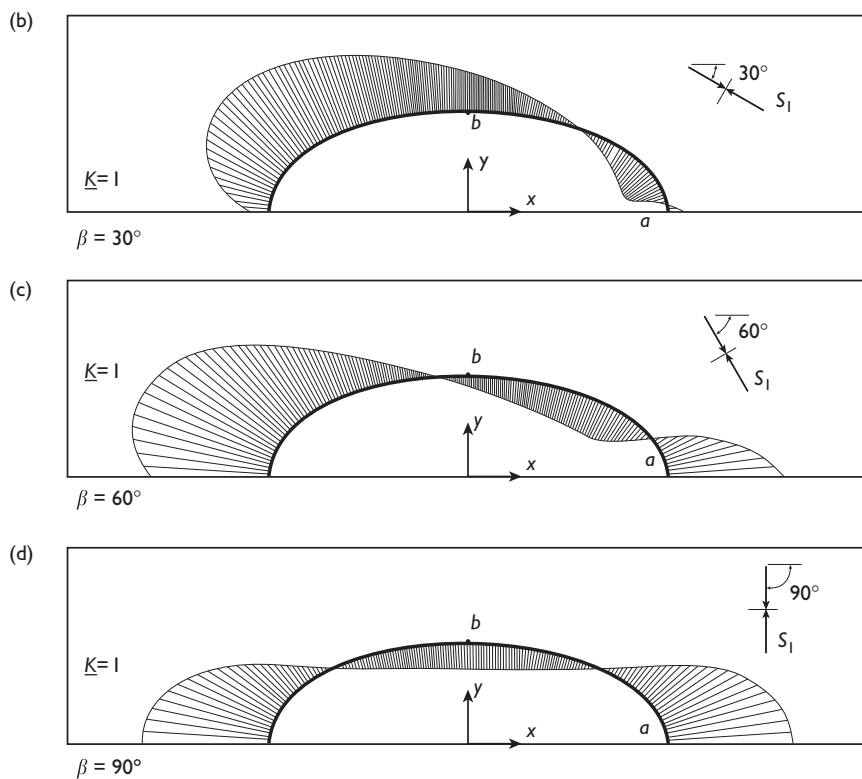
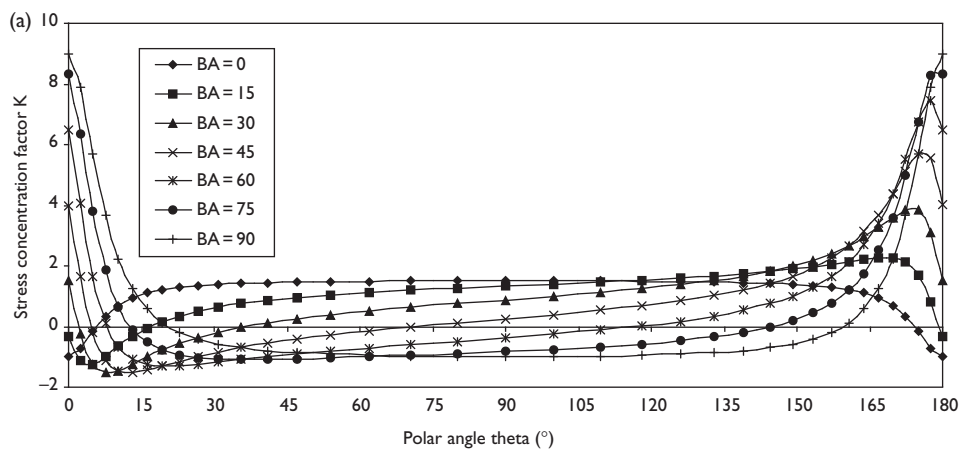


Figure 3.7 (a) Stress concentration at the wall of an inclined elliptical shaft under uniaxial stress ($M = 0$). BA = inclination angle of the a semi-axis to S_1 , $b/a = 0.25$. Semi-axes ratio $k = 1/4$. (b), (c), and (d) stress concentration about an ellipse of semi-axes ratio $b/a = 1/2$. β = ccw angle from S_1 to semi-axis a . Tension inside, compression outside section.

rounding, the smaller the stress concentration. The maximum corner radius possible is one-half the least dimension of the section and results in an ovaloid. As with ellipses, one may characterize rectangular shafts by an aspect ratio k where $k = b/a$, a is the semi-axis parallel to S_1 and b is parallel to S_3 . A range of k between $1/8$ and 8 more than covers the range of practical rectangular shaft dimensions.

A *new* phenomenon occurs in stress concentration about rectangular shafts. Unlike elliptical shaft shapes, rectangular and ovaloidal sections experience peak compressive stress concentration near corners, not at the end of the semi-axes. However, when tension is present, peak tension still occurs at the ends of a semi-axis, that is, at mid-sides; which sides (short or long) depends on orientation with respect to the preshaft principal axes. These observations are made with the understanding that the preexcavation principal stress directions are parallel to the semi-axes of the section.

Simple mathematical expressions for stress concentration about rectangular shafts are not available as they were for circular and elliptical shaft sections. Although stress concentration about rectangular sections in homogeneous, isotropic, linearly elastic ground also depends on the ratio of preshaft principal stresses, aspect ratio, and orientation, useful stress concentration formulas are lengthy and complex in derivation. Numerical evaluation is required that introduces a degree of approximation to symbolic but exact results from the mathematical theory of elasticity.

Figure 3.8 shows two cases of stress concentration about a rectangular shaft with an aspect ratio of $1/2$ and 2 . Data in Figures 3.8(a) and (b) were obtained numerically from an approximate analyses (Heller *et al.*, 1958). Inspection of Figure 3.8 shows that indeed the peak compressive stress concentration always occurs at the corners, regardless of preshaft principal stress ratio M , and that peak tension, when present, occurs at the ends of a semi-axis. Figure 3.8 also suggests that for likely preshaft principal stress ratios ($M > 0$), tension is relatively constant until the considered point nears the corner where tension rapidly diminishes and stress concentration rises sharply to peak compression. Comparisons of peak compressive stress concentration (compression positive) suggests that the most favorable orientation of a rectangular section is with the long axis parallel to the major preshaft principal stress S_1 , as was the case with an elliptical shaft section, for the practical range of $M > 0$. In this regard, the long axis is parallel to S_1 for $k < 1$ and is perpendicular to S_1 when $k > 1$. Examination of many such plots shows that this rule of thumb does indeed apply to rectangular shafts when the choice is whether to orient the long axis of a section parallel or perpendicular to the major preshaft compression.

Maximum and minimum stress concentration about rectangular sections with corners rounded to $1/12$ th the short dimension are shown in Figure 3.9 for a range of preexcavation principal stress ratios. Figure 3.9(a) shows an *almost linear increase in peak compressive stress* (maximum stress) concentration when the semi-axis ratio $k > 1$ (long axis perpendicular to S_1), that is, when unfavorably orientated. The case $M = 1/3$ falls in the center of the cluster of almost parallel lines to the right of $b/a = 1$ and has the equation $K_c = 4.06 + 1.19[b/a - 1]$. Extrapolation to $b/a = 20$ gives $K_c = 26.7$. At this aspect ratio the difference between $M = 0$ and $M = 1$ from $M = 1/3$ is about 4% less and 8% more, respectively.

Figure 3.9(b) shows an *almost constant tensile stress* (minimum) concentration when the semi-axis ratio $k > 1$; the minimum stress concentration is tension for values of

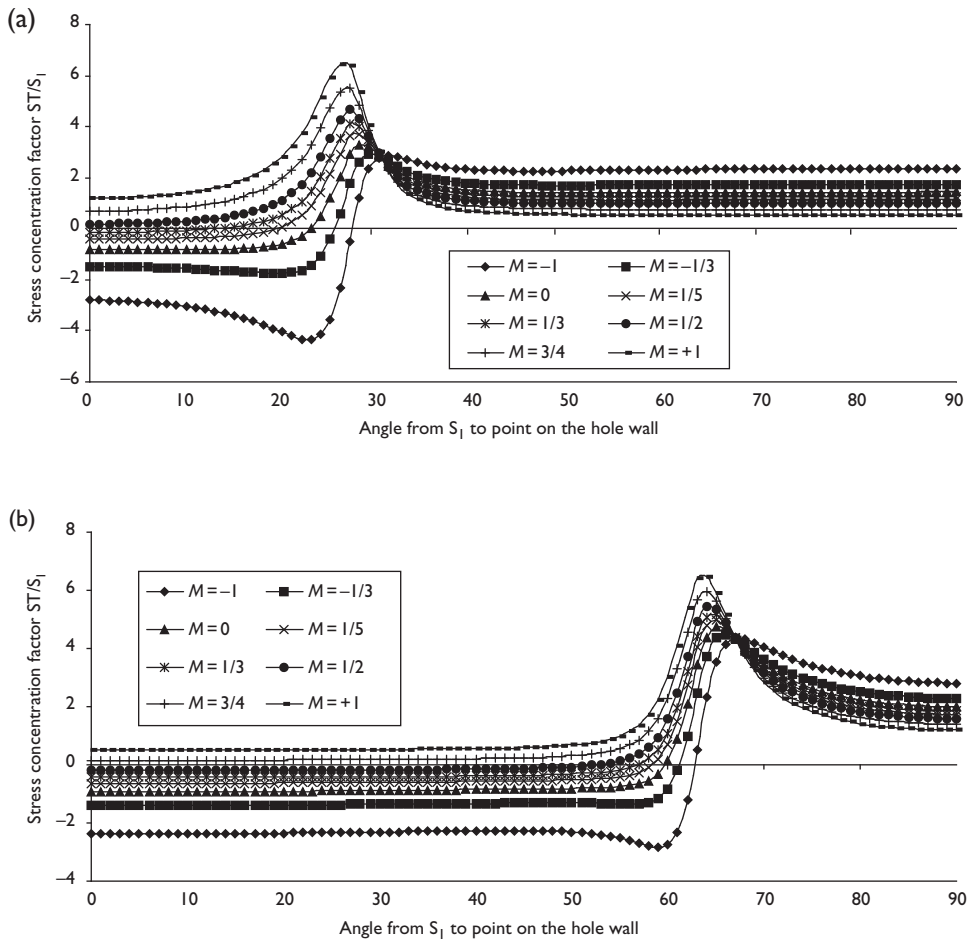
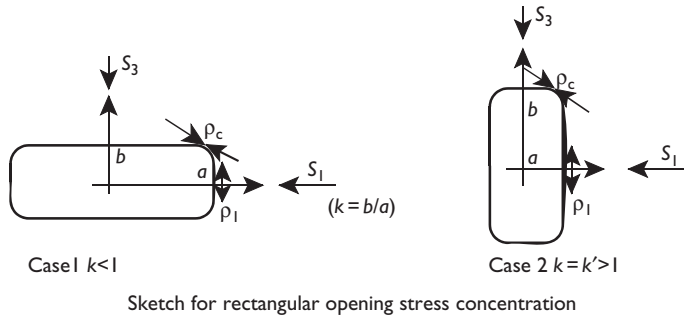


Figure 3.8 Examples of stress concentration about a rectangular shaft (a) S_1 is parallel to the long axis a and $k = 1/2$ (b) S_1 is perpendicular to the long axis b and $k = 2$, for various preshaft principal stress ratios M .

$M < 3/4$. For $M = 0$, an extreme value, as a practical matter, the peak tensile stress concentration is about -1 . For a more likely value of $M = 1/3$, the peak tension is about -0.5 . When $k < 1$ (favorable orientation), peak tension is even less in the practical range of $M > 0$.

The data in Figure 3.9 are summarized in an abbreviated tabular form in Table 3.2 for M in the practical range of zero to one. Generally the maximum stress concentration occurs at the corner of the section, while the minimum stress concentration occurs at a midside. Two cases are presented: Case 1 where the major compression acts parallel to the long axis of the section ($k < 1$) and Case 2 where the major compression acts perpendicular to the long axis

($k = k' > 1$), as illustrated in the sketch. In both cases, tension when present, occurs “under the load” at the midside normal to S_1 .



Comparisons of peak compressive and tensile stress concentrations at the walls of elliptical and rectangular sections are presented in Table 3.2(b), respectively, for a practical value of preexcavation principal stress ratio ($M = 1/3$) and two values of semi-axis ratios ($k = 1/2$ and 2). Table 3.2(b) data show the superiority of the ellipse when in the favorable orientation with the long axis of the section parallel to the major principal stress before excavation. The tension is small for the rectangle and nil for the ellipse, so tension is not an important factor. However, the difference in compressive stress concentration is considerable, more than double in the case of the rectangle. When the long axis of the considered section is perpendicular to the preexcavation principal stress, the differences are much less and the ellipse shows only a slight advantage in compression and tension. In this case, the rectangle would be favored as a practical matter because of greater ease of construction.

Another comparison can be readily made in the case of hydrostatic preexcavation stress ($M = 1$). In this case there are no orientation effects on peak stresses. Tension is absent from both sections and is not a factor. Peak compressive stress concentration for the ellipse is 4.00 and for the rectangle is 6.46 (circle, 2.0).

Figure 3.9(c) shows the variation of maximum stress concentration about a rectangle as a function of aspect ratio for M in the range $[0, 3/4]$. The regression model is a multilinear function of aspect ratio k , reciprocal $1/k$ and preexcavation principal stress ratio M . The constants in the regression function were determined using all the data in Table 3.2(a), and for this reason, the fit at $M = 0$ is best. For this fit: $a_1 = 2.00$, $a_2 = 1.22$, $a_3 = 0.51$, and $a_4 = 2.28$. Changing the range of the regression analysis would enable one to obtain better fits at other M values. For example, Figure 3.9(d) shows a fit using M in the practical range $[0, 1]$.

Example 3.8 Compare the peak compressive and tensile stress concentrations for elliptical and rectangular shaft sections when the sections are 10×20 ft and are oriented in the most favorable way and the preexcavation ratio of minor to major principal stress is 1/3. If the preexcavation major principal stress is 2,500 psi in plan view and compression is positive, determine the peak shaft wall compressive stresses for the two sections in both orientations.

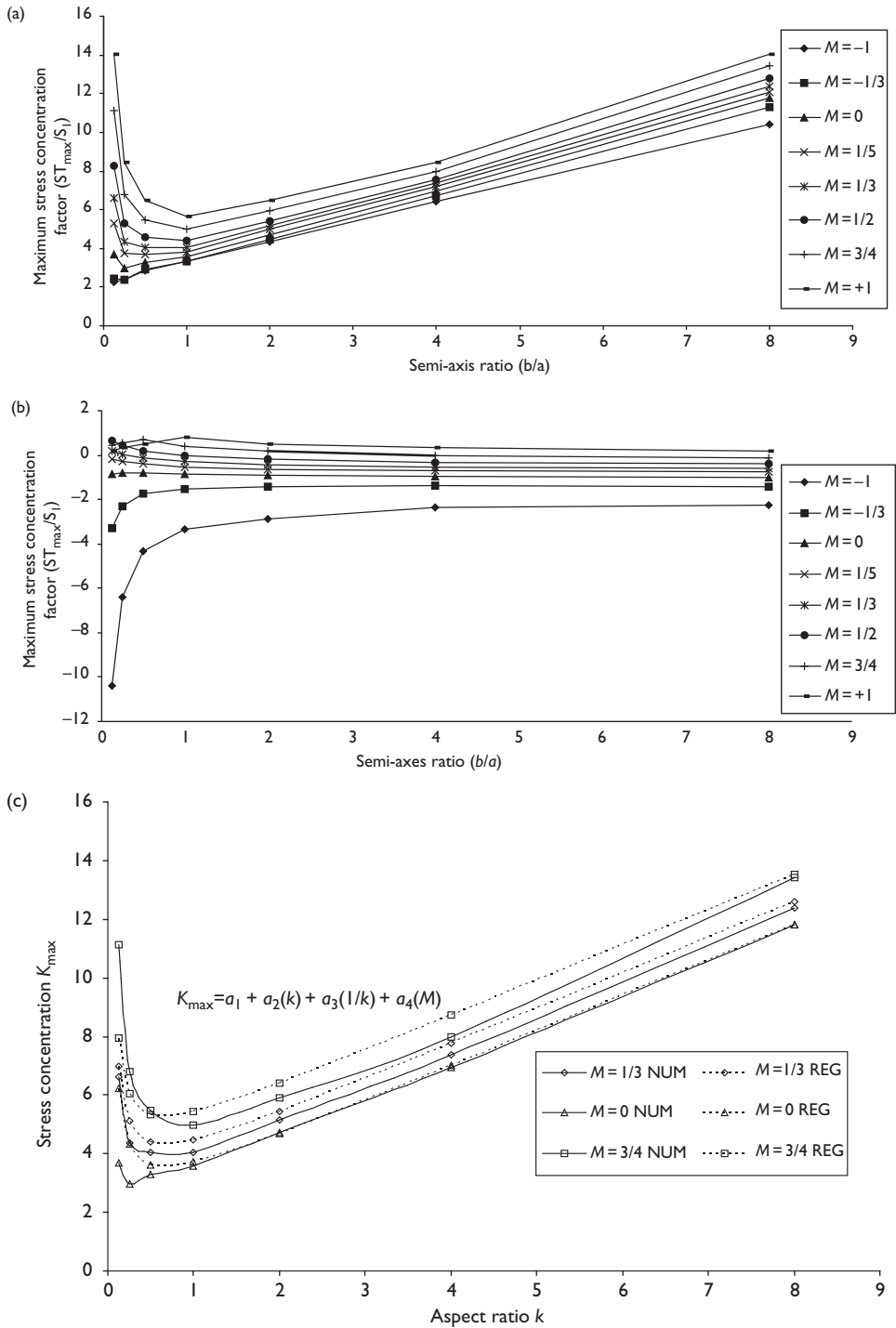


Figure 3.9 (a) Maximum and (b) minimum stress concentration about a rectangular section as a function of aspect ratio k for various preexcavation principal stress ratios (M), (c) maximum stress concentration about a rectangle as a function of aspect ratio. Solid line from numerical analysis. Dotted line from regression analysis using the equation model shown in the plot, (d) using M in the range $[0, 1]$. For this fit: $M = 1/3$ and $a_1 = 0.69$, $a_2 = 1.27$, $a_3 = 0.68$, $a_4 = 3.83$.

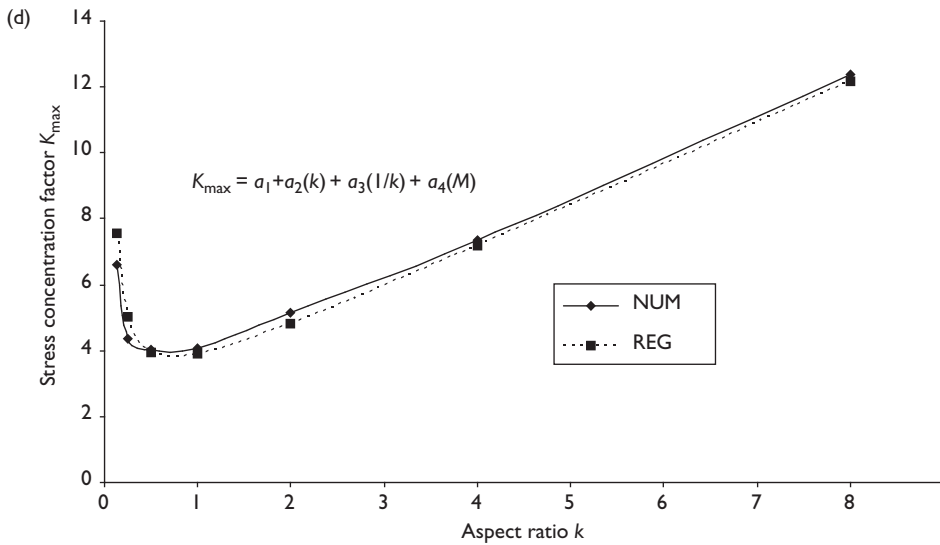
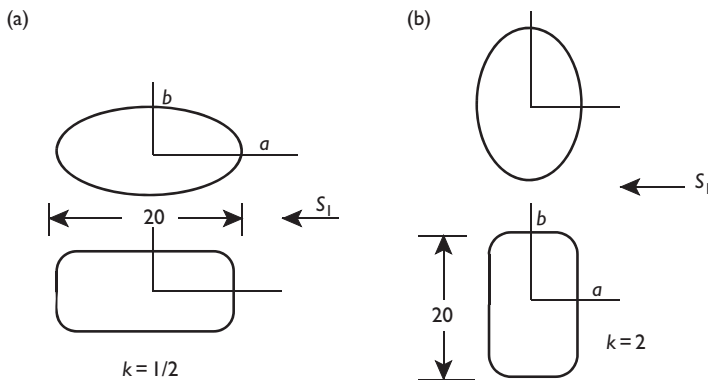


Figure 3.9 Continued.

Solution: The semi-axis ratio for both sections is either 1/2 or 2 depending on orientation, as shown in the sketch.



The most favorable orientation, according to a rule of thumb is to position the long axis of the section parallel to the major compression before excavation. This orientation is (a) in the sketch. In this orientation, the semi-axis ratio b/a is 1/2 for both sections. With $M = 1/3$, one may use data from Table 3.2(b) that gives a peak compressive stress concentration factor of 1.67 for the ellipse and 4.05 for the rectangle. Tension is absent about the ellipse, while the tensile stress concentration factor for the rectangle is 0.14.

In this orientation, peak compression for the ellipse is $\sigma_c = K_c S_1 = (1.67)(2,500) = 4,175$ psi and for the rectangle $\sigma_c = (4.05)(2,500) = 10,125$ psi.

In the adverse orientation, peak compression for the ellipse is $\sigma_c = K_c S_1 = (4.67)(2,500) = 11,675$ psi and for the rectangle $\sigma_c = (5.15)(2,500) = 12,875$ psi.

Table 3.2a Stress concentration factors about a rectangular section

$\frac{k}{k'}$	M										
	0		1/5		1/3		1/2		3/4		1
	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$K_{\max}$	$K_{\min}$	$\frac{K_{\max}}{K_{\min}}$
1/8	3.70	-0.84	5.30	-0.18	6.62	0.20	8.28	0.64	11.1	0.42	14.04
8	11.8	-1.01	12.08	-0.77	12.4	-0.60	12.8	-0.40	13.4	-0.10	0.18
1/4	2.96	-0.81	3.72	-0.29	4.36	0.03	5.28	0.42	6.80	0.56	8.43
4	6.96	-0.96	7.21	-0.71	7.37	-0.53	7.58	-0.32	7.99	0.00	0.32
1/2	3.28	-0.82	3.70	-0.41	4.05	-0.14	4.57	0.19	5.48	0.69	6.46
2	4.72	-0.92	4.97	-0.64	5.15	-0.44	5.44	-0.20	5.92	0.16	0.52
1	3.58	-0.85	3.83	-0.52	4.06	-0.29	4.39	-0.02	4.97	0.40	5.65
											0.80

Notes

M = principal stress ratio (S_3/S_1).

K = aspect ratio (b/a), S_1 is applied parallel to the a -axis (long axis of the section).

$k' = 1/k$ corresponding to application of S_1 parallel to the short axis of the section.

$K(\max)$ = maximum stress concentration factor.

$K(\min)$ = minimum stress concentration factor.

Table 3.2b Peak stress concentration factor comparison of elliptical and rectangular sections at $M = 1/3$

Section	Case			
	$M = 1/3, k = 1/2$		$M = 1/3, k = 2$	
	K_c	K_t	K_c	K_t
Ellipse	1.67	—	4.67	-0.33
Rectangle	4.05	-0.14	5.15	-0.44

An interesting orientation effect occurs in the case of rectangular sections when the preexcavation principal stress directions do not coincide with the section semi-axes. Let the a and b semi-axes coincide with x and y axes, respectively, and the *in situ* stresses be S_{xx} , S_{yy} , and T_{xy} , then the preexcavation principal stresses and directions are given by the usual Equations (3.8) where, as before, β is the angle between the x -axis and the direction of S_1 .

Orientation effects on a rectangular section with semi-axes ratio of 1/8 are shown in Figure 3.10 for a range of principal stress ratios. Figure 3.10(a) shows that the maximum stress concentration factor $K_{\max}$ is compressive at all inclinations (compression positive) and that there is a *peak* $K_{\max}$ at about 35° for non-negative M . In the hydrostatic case there is no orientation effect, of course, and this result is also shown in Figure 3.10(a). Figure 3.10(b) shows that the minimum stress concentration factor $K_{\min}$ is generally tensile except for small inclinations at high principal stress ratios. There is a *peak* $K_{\min}$ at about 35° that increases with M . Again, there is no orientation effect in the hydrostatic case.

Figures 3.10(c), (d), and (e) show graphically stress concentration about a rectangular opening with semi-axes ratio of 0.5 ($b/a = 1/2$) when the a or x semi-axis is inclined at $\beta = 30^\circ$, 60° , and 90° to S_1 . The corners in these figures are rounded off to one-twelfth the short dimension b and $M = 0$ in all cases. Interestingly, peak tension and compression occur at the corners when $\beta = 30^\circ$. When $\beta = 60^\circ$, peak compression remains at a corner while tension peaks occur at a corner and at the end of the long semi-axes a . When $\beta = 90^\circ$ and the preexcavation compression is perpendicular to the long dimension of the section, peak tension occurs at the centerline of the section, that is, at the end of the b semi-axis. Peak compression still occurs at the corner of the section.

These results call into question the rule of thumb that the most favorable orientation is with the long axis parallel to the preexcavation major principal stress. The orientation effect is to reduce the peak compressive stress while increasing the peak tension for practical M values

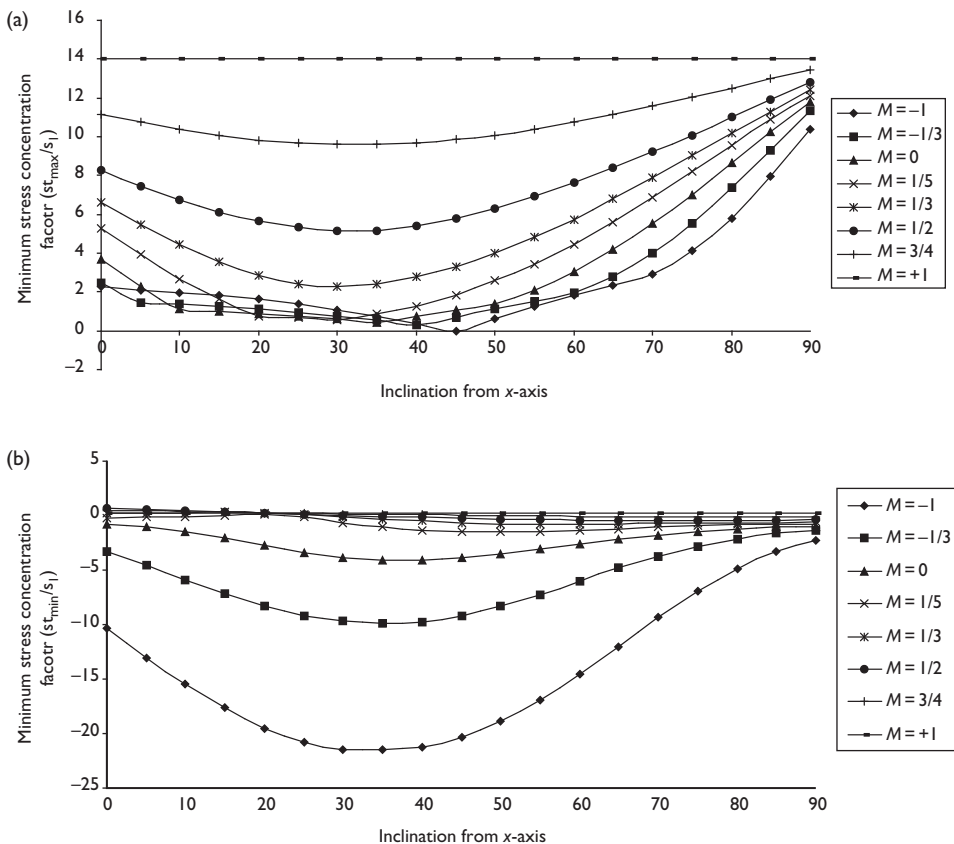


Figure 3.10 (a) and (b) Orientation effect on a rectangular section with semi-axis ratio $k = 1/8$. (c),(d), and (e) Stress concentration about a rectangle of semi-axis ratio $b/a = 1/2$. $\beta =$ ccw angle from S_1 to semi-axis a . Tension inside, compression outside section. Corners rounded to $1/6$ th semi-axis b .

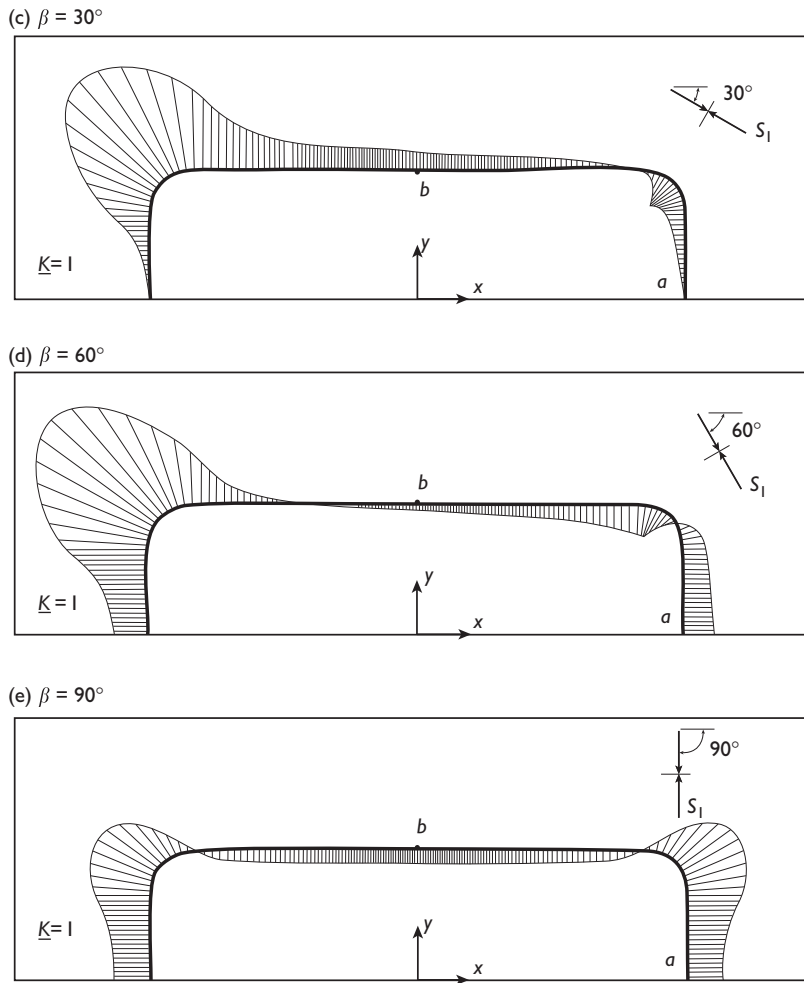


Figure 3.10 (Continued)

(positive M). For example, at $M = 1/3$, the peak compressive stress concentration is about 6.6 in favorable orientation (inclination is zero), while at an inclination of about 30° , the peak is only about 2.3. There is no tension in the favorable orientation (inclination is zero), while at 30° , a slight tension appears with a concentration of -0.08 . Whether the compressive stress effect remains large at lesser and more practical semi-axes ratios is an open question at this juncture as is the question of the most favorable orientation of a rectangular shaft with respect to the *in situ* stresses.

Returning to the important practical case of a vertical shaft in a preexcavation stress field caused by gravity alone, one recalls that in horizontal section, the principal stresses are equal and therefore $M = 1$ (hydrostatic case). Hence there are no orientation effects and the question of the most favorable orientation is moot. The peak stresses can then be readily determined

Table 3.3 Peak compressive (K_c) stress concentration factors for rectangular sections^a in a hydrostatic stress field ($M = 1$). Tension is absent

$k =$	1/8	1/4	1/2	1	2	4	8
$K_c =$	14.04	8.43	6.46	5.65	6.46	8.43	14.04

Note

a Corners rounded to 1/12 radius of least dimension.

from stress concentration factors in Figure 3.9. Data for this case are given in Table 3.3. These data are symmetric about $k = 1$ (square section) because there is no unique orientation with respect to principal directions. A simple straight line fit of Table 3.3 data for $k > 1$ leads to a $K_c = 5.65 + 1.2(b/a - 1)$ which allows for estimation of peak compressive stress concentration at higher aspect ratios (for $M = 1$).

A square is the best shaped rectangle in a hydrostatic stress field according to the data in Table 3.3. By comparison, the optimum elliptical shaft excavated in a hydrostatic stress field was a circle. Increasing corner sharpness increases peak stress; rounding decreases peak stress. In fact, rounding the corners of a square to maximum ($1/2$ least dimension) creates a special ovaloid. This ovaloid is actually a circle which has a peak stress concentration of 2.0. Reduction of peak stress by rounding corners is significant, by almost a factor of 3. One may speculate if corners are not deliberately rounded by design, then stress reduction will occur naturally in small, localized regions that yield under high stress at corners. There is also a practical trade-off between more usable cross-sectional area in a square and less stress concentration about a circular shaft and between a square and other rectangles in a gravity-only stress field.

In plan view (horizontal cross section), the safety factor FS associated with a square with $1/12$ corner rounding is $FS_h = C_o/5.65K_oS_v$. In vertical section, a vertical compressive stress acts at the shaft wall that is equal to the preshaft vertical stress. Thus, in vertical section, $S_v = S_1$ and $S_3 = 0$. In this view, the unlined shaft wall safety factor is simply $FS = FS_v = C_o/S_v$. In a gravity-only initial stress field; FS_v is less than FS_h when $5.65K_o < 1$ or K_o about less than $1/5$ which is seldom the case. Thus, the safety factor computed in plan view usually governs rectangular shaft section design, although both safety factors should always be computed.

Example 3.9 Stress measurements with respect to compass coordinates $x = \text{east}$, $y = \text{north}$, $z = \text{up}$ show $\sigma_{xx} = 1,200$; $\sigma_{yy} = 2,350$; $\sigma_{zz} = 3,420$; $\tau_{xy} = -760$; $\tau_{yz} = 0$; $\tau_{zx} = 0$ where compression is positive. A 12×24 ft rectangular section is scheduled to pass through this region. What orientation of the shaft is best and what are the shaft wall safety factors in compression and tension, given unconfined compressive and tensile strengths of 18,500 and 1,250 psi, respectively?

Solution: The most favorable orientation is one that induces the least stress concentration at the shaft wall, and according to a rule of thumb, is one with the long axis of the section parallel to major compression before excavation.

By inspection, the z -direction is a principal direction because of the vanishing z -direction shear stresses. Orientation of the major principal stress in the xy -plane is given by

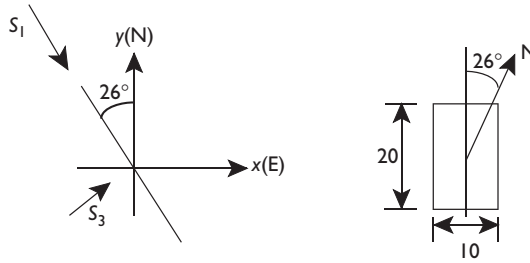
$$\begin{aligned}\tan(2\theta) &= \frac{\tau_{xy}}{(1/2)(\sigma_{xx} - \sigma_{yy})} \\ &= \frac{-760}{(1/2)(1,200 - 2,350)}\end{aligned}$$

$$\tan(2\theta) = 1.322$$

$\therefore$

$$\theta = 26, 116^\circ$$

The larger angle is the correct solution. One may intuitively see that this is correct by supposing the major principal stress is nearest to the largest given normal stress which is σ_{yy} in this case. Measuring 26° , positive counterclockwise from the y -axis gives the desired orientation, as shown in the sketch.



The major and minor principal stresses before excavation are (in psi):

$$\begin{aligned}S_1 &= (1/2)(S_{xx} + S_{yy}) \pm [(1/4)(S_{xx} - S_{yy})^2 + (T_{xy})^2]^{1/2} \\ &= (1/2)(1,200 + 2,350) \pm [(1/4)(1,200 - 2,350)^2 + (-760)^2]^{1/2} \\ &= 1,775 \pm 938\end{aligned}$$

$\therefore$

$$S_1 = 2,713$$

$$S_3 = 837$$

The principal stress ratio $M = 837/2,713 = 0.31$, a value close to $1/3$ which is associated with known stress concentration factors in Table 3.2(b). Thus, to a reasonable approximation, $K_c = 4.07$ and $K_t = 0.14$. Hence, $FS_c = 18,500/(4.07)(2,713) = 1.68$ and $FS_t = 1,250/0.14(2,713) = 3.29$.

Inspection of Table 3.2(a) indicates that the compressive stress concentration would be somewhat less, while the tensile stress concentration would be somewhat more than the values used. The actual safety factors would be somewhat greater with respect to compression and somewhat less with respect to tension.

Example 3.10 A rectangular shaft is being considered for extension to 5,250 ft from 4,200 ft. The *in situ* stresses are given by

$$S_v = 1.1h$$

$$S_{NS} = 100 + 0.7h$$

$$S_{EW} = 1,000 + 0.5h$$

where h is depth in feet and stress is in psi. The shaft is 12×24 ft in section and is oriented in the most favorable direction. Unconfined compressive and tensile strengths are 18,000 and 1,200 psi, respectively. Determine the shaft wall safety factors as functions of depth to 5,250 ft from the surface.

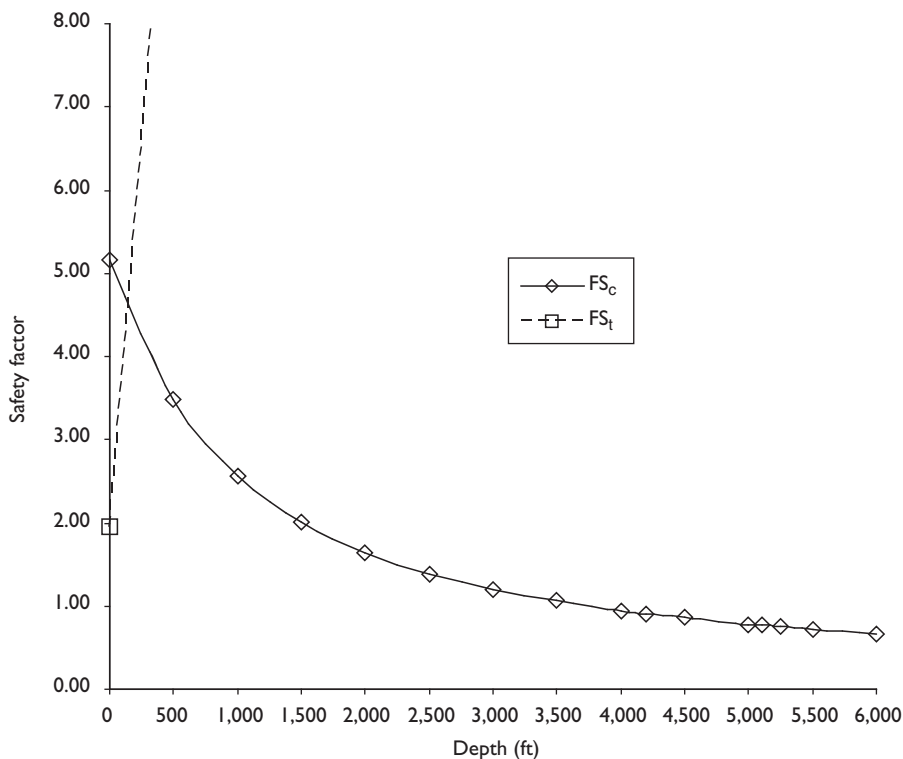
Solution: In plan view, the EW stress is larger than the NS stress above 4,500 ft, so the long axis of the shaft is in the EW direction (most favorable) and the EW stress is the major compression in plan view. The plan view preexcavation principal stress ratio M varies with depth. Consequently, the peak stress concentration factors also vary with depth as do the shaft wall safety factors. A tabulation of data is shown in the table and a plot is given in the sketch. Table data were interpolated from Table 3.2(a) in the text and extended to 6,000 ft.

Depth ft	S_v psi	S_{NS} psi	S_{EW} psi	M	k	K_c	K_t	$FS_c =$	$FS_t =$
0	0	100	1,000	0.10	0.5	3.49	-0.62	5.16	1.95
500	550	450	1,250	0.36	0.5	4.13	-0.09	3.48	11.08
1,000	1,100	800	1,500	0.53	0.5	4.69	0.26	2.56	No tens
1,500	1,650	1,150	1,750	0.66	0.5	5.14	0.50	2.00	
2,000	2,200	1,500	2,000	0.75	0.5	5.48	0.69	1.64	
2,500	2,750	1,850	2,250	0.82	0.5	5.76	0.64	1.39	
3,000	3,300	2,200	2,500	0.88	0.5	5.99	0.60	1.20	
3,500	3,850	2,550	2,750	0.93	0.5	6.17	0.57	1.06	
4,000	4,400	2,900	3,000	0.97	0.5	6.33	0.54	0.95	
4,200	4,620	3,040	3,100	0.98	0.5	6.38	0.53	0.91	
4,500	4,950	3,250	3,250	1.00	0.5	6.46	0.52	0.86	
5,000	5,500	3,600	3,500	0.97	2	6.40	0.54	0.78	
5,100	5,610	3,670	3,550	0.97	2	6.39	0.47	0.77	
5,250	5,775	3,775	3,625	0.96	2	6.37	0.46	0.75	
5,500	6,050	3,950	3,750	0.95	2	6.35	0.45	0.72	
6,000	6,600	4,300	4,000	0.93	2	6.31	0.42	0.66	

Inspection of the table and figure data show that tension is not a concern, especially at depth. However, below 3,500 ft failure in compression becomes possible as the shaft wall safety factor decreases below 1.0. One may presume that the original shaft required support and further support will be needed with the extension to depth.

Shaft wall strengths

Strengths that enter shaft wall safety factor calculations limit the range of elastic deformation. Indeed, strength may be defined as stress at the elastic limit. A naturally supported shaft wall is unconfined; applicable strengths are therefore unconfined compressive and tensile



Shaft wall safety factors as functions of depth

strengths, C_o and T_o , respectively. These strengths should be consistent with the concept of a homogeneous material used in elastic analyses for stress and stress concentration. Homogeneity is easy to define in the context of mechanics and implies identical properties throughout a considered region, for example, same Young's modulus, same compressive strength at every point.

In this regard, a remarkable feature of two-dimensional stress analyses in isotropic elastic material obscures the question of homogeneity. This feature is the absence of elastic properties, such as Young's modulus and Poisson's ratio, in the distribution of stress. Consequently, associated stress concentrations about shaft walls are independent of elastic moduli. However, displacements induced by excavation are not independent of elastic moduli even in two-dimensional analyses, but are tacitly assumed to be small because of the expected elastic response.

Although easy to define, the concept of homogeneity raises important practical concerns and is central to much of theoretical and applied rock mechanics. Natural rock masses generally contain numerous cracks, fractures, joints, and faults ("joints" for brevity) that significantly alter the behavior of a rock mass from the mechanical response of intact rock. The issue of homogeneity can be simply posed by asking what should be done about the

“joints.” There are two distinct approaches to addressing this question: (1) address joint failure and failure of intact rock between separately, or (2) homogenize the rock mass using equivalent moduli and strengths and calculate safety factors based on these properties.

Calculation of stress in a jointed rock mass using the first approach is necessarily simplified by first considering the joints to be absent and then using the resulting stresses for analysis of intact rock and joint safety. This approach is homogenization in the extreme that is convenient, obviously inconsistent and therefore quite approximate, but still useful for the guidance provided. Safety factors for intact rock between joints can be calculated immediately following specification of C_o and T_o , preferably from laboratory measurements, but also possibly from estimates based on tabulated values of similar rock types. The latter specification is certainly cheaper but obviously much less certain than measured values. Safety factors for joints require consideration of joint orientation in addition to joint strength and the stress field at the shaft wall after excavation.

The well-known Mohr–Coulomb criterion is the most frequently used failure criterion for “joints” and is adopted here in the traditional form

$$\tau_j = \sigma_j \tan(\phi_j) + c_j \quad (3.9)$$

where τ_j , σ_j , ϕ_j , and c_j are joint *shear strength*, normal stress, friction angle, and cohesion, respectively, and the subscript j indicates joint. A joint safety factor FS_j is then

$$FS_j = \frac{\tau_j(\text{strength})}{\tau_j(\text{stress})} = \frac{\sigma_j \tan(\phi_j) + c_j}{\tau_j} \quad (3.10)$$

which is simply the ratio of shear strength to shear stress. Joint properties, ϕ_j and c_j are usually measured in direct shear test apparatus.

Joint normal and shear stresses must be calculated from the post-excavation stress state in consideration of joint orientation. The calculations are straightforward but require several steps:

- 1 Rotate the post-excavation stress state at a point on the shaft wall from $x' y' z'$ reference axes to compass coordinates xyz (x = east, y = north, z = up) about the vertical z axis with $z' = z$. Stresses relative to the prime system are: $\sigma_{x'x'} = 0$, $\sigma_{y'y'} = \sigma_t = KS_1$, $\sigma_{z'z'} = S_v$ where the stress concentration factor K is obtained from (3.4) in case of circular shafts and shear stresses are nil.
- 2 Compute direction cosines of the joint normal from the dip direction α and dip δ of the considered joint relative to compass coordinates xyz . These direction cosines are $n_x = \sin(\delta) \cos(\alpha)$, $n_y = \sin(\delta) \sin(\alpha)$, $n_z = \cos(\delta)$.
- 3 Compute joint tractions $T_x = \sigma_{xx}n_x + \tau_{xy}n_y + \tau_{xz}n_z$, $T_y = \tau_{xy}n_x + \sigma_{yy}n_y + \tau_{yz}n_z$, $T_z = \tau_{xz}n_x + \tau_{yz}n_y + \sigma_{zz}n_z$.
- 4 Compute magnitude $T = [(T_x)^2 + (T_y)^2 + (T_z)^2]^{1/2}$.
- 5 Compute $\sigma_j = N = T_xn_x + T_yn_y + T_zn_z$.
- 6 Compute $\tau_j = [T^2 - N^2]^{1/2}$.

The first step may be executed from formulas for rotation of reference axes about the given z -axis. In symbolic form this rotation is

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} [\sigma'] [R]^t \quad (3.11)$$

where $[\sigma']$ is a 3×3 array of stresses in the prime system, θ is the angle of rotation (positive counterclockwise) about the z' -axis from x' to x , and $[R]^t$ is the transpose of the rotation matrix that precedes $[\sigma']$. The array $[\sigma']$ is diagonal because of the absence of shear stresses at the shaft wall. Results of (3.11) are the stresses in compass coordinates:

$$\begin{aligned} \sigma_{xx} &= \sigma_t \left(\frac{1 - \cos(2\theta)}{2} \right), & \tau_{xy} &= \sigma_t \left(\frac{\sin(2\theta)}{2} \right) \\ \sigma_{yy} &= \sigma_t \left(\frac{1 + \cos(2\theta)}{2} \right), & \tau_{yz} &= 0 \\ \sigma_{zz} &= S_v & \tau_{xz} &= 0 \end{aligned} \quad (3.12)$$

where σ_t is the post-excavation normal stress acting tangentially to the shaft wall, as before.

The actual stress field that results from excavation in the presence of joints differs from the simplified view to an unknown degree and there is no certainty that joint safety factors computed from the assumed stress field are less than they would be in actuality. Thus, there is no assurance that a shaft wall joint safety factor greater than one guarantees joint safety in this simplified approach. However, low safety factors are unambiguously indicative of the need to consider installation of rock reinforcement and support as excavation continues.

Where cracks, fractures, and joints are too numerous to be treated individually, homogenization is indicated. Homogenization is a process that results in a heterogeneous sample volume responding *on average* as an *equivalent* homogeneous sample. Most equivalent rock properties models assume the sample is *representative* of the rock mass. A representative rock mass volume must contain so many joints that addition or elimination of a joint does not significantly affect the results of homogenization. Consider a cubical sample of edge length L containing a joint set with spacing S , then to be representative, $L \gg S$, say, L should be an order of magnitude greater than S . At the same time, the homogenized sample volume must be much smaller than the size of the considered excavation for valid stress analyses. For example, the diameter of an unlined shaft D_o should satisfy the inequality $D_o \gg L$. Cracks and fractures spaced a few inches or centimeters indicate samples of a foot or somewhat less than a meter on edge. Shaft diameters of ten feet or meters are common. Homogenization using representative volumes is useful in such cases of relatively closely spaced joints. However, joint sets with joint spacings of a meter or more lead to much larger representative rock mass volumes. A meter joint spacing indicates a representative volume element L of ten meters and therefore unrealistically large shaft sizes D_o for applicability of the representative sample volume approach to equivalent rock mass properties. In such cases, a nonrepresentative volume element approach to equivalent properties is necessary, one that takes into account the unique geological structure in relative small volumes about the shaft wall.

Generally, joints reduce moduli and strengths from intact rock testing values. Scale factors that multiply intact rock elastic moduli and strengths to give rock mass moduli and strengths

are often used for homogenization. For example, scale factors for Young's modulus R_E and unconfined compressive strength R_C are simply the ratios E_f/E_l and $(C_o)_f/(C_o)_l$, respectively, where subscripts f and l stand for field and laboratory scales. Clearly, such scale factors must be less than one and greater than zero, that is, they must be in the interval $[0,1]$. If no data are available to guide scale factor estimates, then one might simply choose the mid-point and reduce laboratory moduli (Young's modulus, E , shear modulus, G) and strengths (C_o and T_o) by $1/2$. Other estimates may be based on pseudodimensional analyses. For example, one may suppose the strain to failure under uniaxial compressive stress is the same at the laboratory scale and at the scale of the rock mass, that is, $\varepsilon_f = C_o/E$. According to this concept, scale factors for moduli and strength are the same. Alternatively, one may suppose that under uniaxial compressive stress, the strain energy to failure is the same, so $(\sigma\varepsilon)_f = (C_o)^2/E$. Hence, the scale factor for moduli is the square of the scale factor for strength. These simple guides relate compressive strength and moduli scale factors but do not indicate how either are to be estimated. A similar analysis could be applied to tensile strength and related modulus.

Example 3.11 Given a reasonable 0.3% strain to failure and corresponding laboratory test values of 30,000 psi and $10(10^6)$ psi for unconfined compressive strength and Young's modulus, respectively, determine rock mass compressive strength when displacement measurements in the field show Young's modulus to be $2.5(10^6)$ psi. Use (a) strain to failure scaling and (b) energy to failure scaling.

Solution:

(a) Strain to failure is

$$\varepsilon_f = (0.003) = [C_o/E]_l = [C_o/E]_m$$

$\therefore$

$$C_o(\text{rock mass}) = (0.003)(2.5)(10^6) = 7,500 \text{ psi}$$

(b) Energy to failure is

$$U = (1/2)[(C_o)^2/E]_l = (1/2)[(C_o)^2/E]_m$$

$$= (1/2)[3(10^4)]^2/10(10^6) = (9/2)10$$

$$U = 45$$

$\therefore$

$$(C_o)^2 = (45)(2)(2.5)(10^6)$$

$\therefore$

$$C_o(\text{rock mass}) = 15,000 \text{ psi}$$

In case (a) strength and modulus scale in direct proportion. In case (b), strength scales as the square root of the modulus scaling. In this example, Young's modulus for the rock mass was 1/4th the laboratory value, so the rock mass strength is 1/4 laboratory strength in case (a) and 1/2 laboratory strength in case (b), as the computations show.

The Terzaghi jointed rock mass strength model is one approach to estimating rock mass strength directly (Terzaghi, 1962). A strength scale factor can then be computed and subsequently a modulus scale factor using strain to failure or energy at failure guidelines. The Terzaghi model recognizes an important feature of rock masses and that is jointing is generally *discontinuous*. Shear failure along a plane of discontinuous jointing requires joint shear and shear of intact rock bridges between joint segments. If areas of intact rock and joints are A_r and A_j , then the total sample area $A = A_r + A_j$. The ratio of joint area to total area is *joint persistence* p . Thus, $p = A_j/A$ which ranges over the interval $[0,1]$. Shear strength along the joint is

$$\begin{aligned}\tau &= \tau_r(A_r/A) + \tau_j(A_j/A) \\ &= \tau_r(1 - p) + \tau_j(p) \\ &= (1 - p)[\sigma \tan(\phi) + c]_r + (p)[\sigma \tan(\phi) + c]_j \\ \tau &= \sigma \tan(\phi) + c\end{aligned}\tag{3.13}$$

where Mohr–Coulomb strength criteria are assumed for intact rock and joint which are indicated with subscripts r and j , respectively. Comparison of the last two equations in (3.13) shows that the rock mass angle of internal friction and cohesion are given by

$$\begin{aligned}\tan(\phi) &= (1 - p)\tan(\phi_r) + (p)\tan(\phi_j) \\ c &= (1 - p)c_r + (p)c_j\end{aligned}\tag{3.14}$$

which are persistence weighted averages. Equations (3.14) allow direct estimation of rock mass strengths using the Mohr–Coulomb strength relationships:

$$\left. \begin{matrix} C_o \\ T_o \end{matrix} \right\} = \frac{2(c)\cos(\phi)}{1 \mp \sin(\phi)}, \quad \sin(\phi) = \left(\frac{C_o/T_o - 1}{C_o/T_o + 1} \right), \quad c = \frac{\sqrt{C_o T_o}}{2}\tag{3.15}$$

that also apply to intact rock and joints. When the same strength scale factor is applied to unconfined compressive and tensile strengths, then the angle of internal friction remains unchanged between laboratory test values and rock mass values, while cohesion is scaled by the same strength scale factor. In fact, Terzaghi suggested equality of friction angles for joints and intact rock and that joints be considered cohesionless. Cohesionless joints are a limiting case that allows unconfined compressive and tensile strengths to approach zero while the ratio C_o/T_o remains a constant equal to $[1 + \sin(\phi)]/[1 - \sin(\phi)]$. Rock mass cohesion in this special case is given by the second equation of (3.14).

Example 3.12 Laboratory test data show that unconfined compressive and tensile strengths are 15,000 and 1,500 psi, respectively. Field measurements indicate a joint persistence of 0.75. Further laboratory testing shows that joint cohesion and friction angle are 8 psi and 25° , respectively. Estimate cohesion and angle of internal friction for intact rock tested in the laboratory, then determine rock mass values of cohesion, friction angle, unconfined compressive strength, and tensile strength, (c , ϕ , C_o , T_o).

Solution: Assuming Mohr–Coulomb criteria,

$$\begin{aligned}
\sin(\phi) &= \frac{C_o - T_o}{C_o + T_o} = \frac{1,5000 - 1,500}{1,5000 + 1,500} = \frac{9}{11} \\
\therefore \\
\phi &= 54.9^\circ \\
c &= \frac{\sqrt{C_o T_o}}{2} = \sqrt{(1,5000)(1,500)}/2 \\
\therefore \\
c &= 2,372 \text{ psi}
\end{aligned}$$

for the intact rock from laboratory strength values.

For the rock mass

$$\begin{aligned}
\tan(\phi) &= (1 - 0.75)\tan(54.9) + 0.75\tan(25) = 0.7054 \\
\therefore \\
\phi(\text{rock mass}) &= 35.2^\circ \\
\therefore \\
c &= (1 - 0.75)(2372) + 0.25(8) = 599 \\
\therefore \\
c(\text{rock mass}) &= 599 \text{ psi}
\end{aligned}$$

Also,

$$\begin{aligned}
\left. \begin{matrix} C_o \\ T_o \end{matrix} \right\} &= \frac{2c \cos(\phi)}{1 \mp \sin(\phi)} = \frac{2(2,372)\cos(35.2)}{1 \mp \sin(35.2)} \\
\therefore \\
\left. \begin{matrix} C_o \\ T_o \end{matrix} \right\}(\text{rock mass}) &= \left. \begin{matrix} 9,152 \\ 2,459 \end{matrix} \right\} \text{psi}
\end{aligned}$$

A check on the ratio of unconfined compressive strength to tensile strength is

$$\begin{aligned}
\frac{C_o}{T_o} &= \frac{1 + \sin(\phi)}{1 - \sin(\phi)} \\
\therefore \\
\frac{9,152}{2,459} &= 3.7218 = (?), \quad \frac{1 + \sin(35.2)}{1 - \sin(35.2)} = \frac{1.5764}{0.4236} = 3.7218(\text{checks})
\end{aligned}$$

The Terzaghi model implies directionally dependent shear strength for the rock mass because of the association with a joint plane of specific orientation and persistence measured over the considered joint plane. However, rock mass strengths computed from (3.15) imply isotropy, equal strengths in all directions. This inconsistency is sometimes addressed by adopting a *ubiquitous* joint model that assumes joints are seemingly everywhere, are fully persistent and of all orientations. This assumption also leads to a *micro-plane model*. Both concepts are invoked to explain and justify reducing laboratory values of elastic moduli and strengths to better represent rock mass behavior when joints are dominant and the role of intact rock between joints is negligible. In essence, these concepts

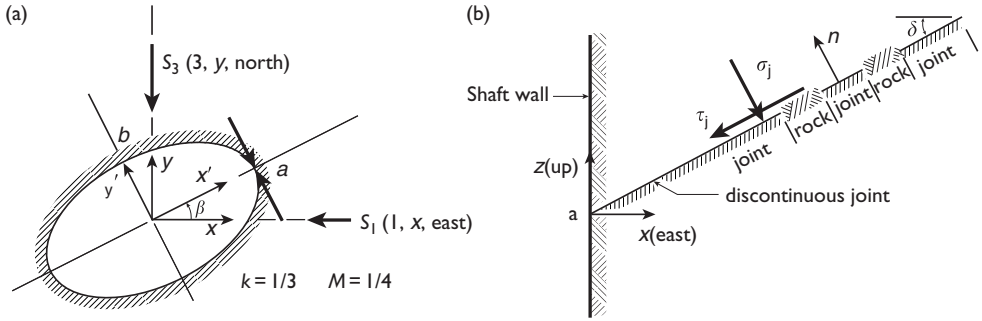


Figure 3.11 Elliptical shaft wall and joint geometry: (a) plan view and (b) vertical N-S section at a .

assign joint properties to the rock mass when site conditions justify the assumptions involved.

Example 3.13 Consider the data in Example 3.7 and a joint set with dip direction $\alpha = 0^\circ$ and dip $\delta = 60^\circ$ and suppose joint cohesion $c = 135$ psi and joint angle of friction $\phi = 35^\circ$. Given these data and joint intersections of the shaft wall at the ends of the semi-axes, a and b , compute joint safety factors according to the steps outlined previously.

Solution: The first step involves an angle of rotation from x' to x of -30° as seen in Figure 3.11. According to (3.12) stresses at a relative to compass coordinates are

$$\sigma_{xx} = 2,700 \left(\frac{1 - 0.5}{2} \right) = 675$$

$$\sigma_{yy} = 2,700 \left(\frac{1 + 0.5}{2} \right) = 2,025$$

$$\sigma_{zz} = 1,350$$

$$\tau_{xy} = 2,700 \left(\frac{-\sqrt{3}/2}{2} \right) = -1,170$$

Direction cosines from Step 2 are

$$n_x = \left(\frac{\sqrt{3}}{2} \right) (1) = 0.8667$$

$$n_y = \left(\frac{\sqrt{3}}{2} \right) (0) = 0.0$$

$$n_z = \left(\frac{1}{2} \right) = 0.5$$

Joint tractions are then

$$T_x = (0.8667)(675) + (0)(-1170) + (0.5)(0) = 585$$

$$T_y = (0.8667)(-1,170) + (0)(2,025) + (0.5)(0) = -1,013$$

$$T_z = (0.8667)(0) + (0)(0) + (0.5)(1,350) = 675$$

and magnitude $T = 1,351$. Joint normal stress from Step 5 is

$$\sigma_j = N = (585)(0.8667) + (-1,013)(0) + (675)(0.5) = 845$$

and joint shear stress from Step 6 is

$$\tau_j = [(1,351)^2 - (845)^2]^{1/2} = 1,054$$

The last two calculations enable an estimate of the joint safety factor. Thus,

$$FS_j = \frac{(845)\tan(35^\circ) + 135}{1,054} = 0.69$$

which indicates that joint slip is highly probable as shaft excavation proceeds past the considered depth.

Example 3.14 Data from Examples 3.7 and 3.13 may be used to compute rock mass strengths following the Terzaghi jointed rock mass model. Use these data to estimate shaft wall safety factors.

Solution: Tension is absent, so only safety with respect to compressive strength is of concern. For this site, persistence $p = 0.75$. From Equations (3.15), intact rock cohesion $c_r = 2,134$ and angle of internal friction $\phi_r = 54.9^\circ$. Rock mass cohesion $c = 635$ and angle of internal friction $\phi = 41.4^\circ$ from (3.14). Rock mass unconfined compressive strength $C_o = 2,813$ from (3.15). Shaft wall safety factor $FS = 2,813/2,700 = 1.04$ in plan view and indicates marginal safety. In vertical section, $FS = 2,813/1,350 = 2.08$.

Joint persistence is measured over a representative area and is thus an average quantity. Locally, at the shaft wall, a joint segment is present or absent. In the worst case then, $p = 1.0$ and therefore the rock mass is assigned joint properties. Thus, $C_o = 519$ from (3.15) and $FS = 519/2,700 = 0.19$ in plan view and 0.38 in vertical section. Both values indicate lack of safety and potential for failure if excavated without support provisions. If the joint is absent, then the FS's is associated with values for intact rock computed previously (5, 10), Example 3.7.

Similar analyses may be carried out for the opposite end of semi-axis a and for the ends of the b semi-axes. In this regard, there is a kinematic aspect of joint slip direction that should be considered. At the a point opposite to the one considered in detail, joint dip is into the rock mass rather than into the excavation and thus poses less of a threat to shaft wall failure even if the joint safety factor is less than one indicating slip.

3.2 Shaft wall support and liners

Rock bolting is a common method of preventing rock falls during shaft sinking. Rock bolts are steel rods that are secured in a drill hole on a spot basis, as needed in the judgment of the

sinking crew, or systematically according to an engineered plan. In blocky ground, steel strapping may also be used in conjunction with bolts for surface support to prevent small rock falls between bolts from developing into larger wall failures. Other types of surface support may be used. Permanent support is often in the form of a concrete liner. Occasionally, in ground difficult to hold, smooth or corrugated steel lining and steel rings may be used either separately or in combination with concrete.

Shaft wall bolting

Rock bolts are secured in drill holes mechanically, with cement of some type, or frictionally. Figure 3.12 illustrates a variety of rock bolts. Bolting plans may be designed to reinforce

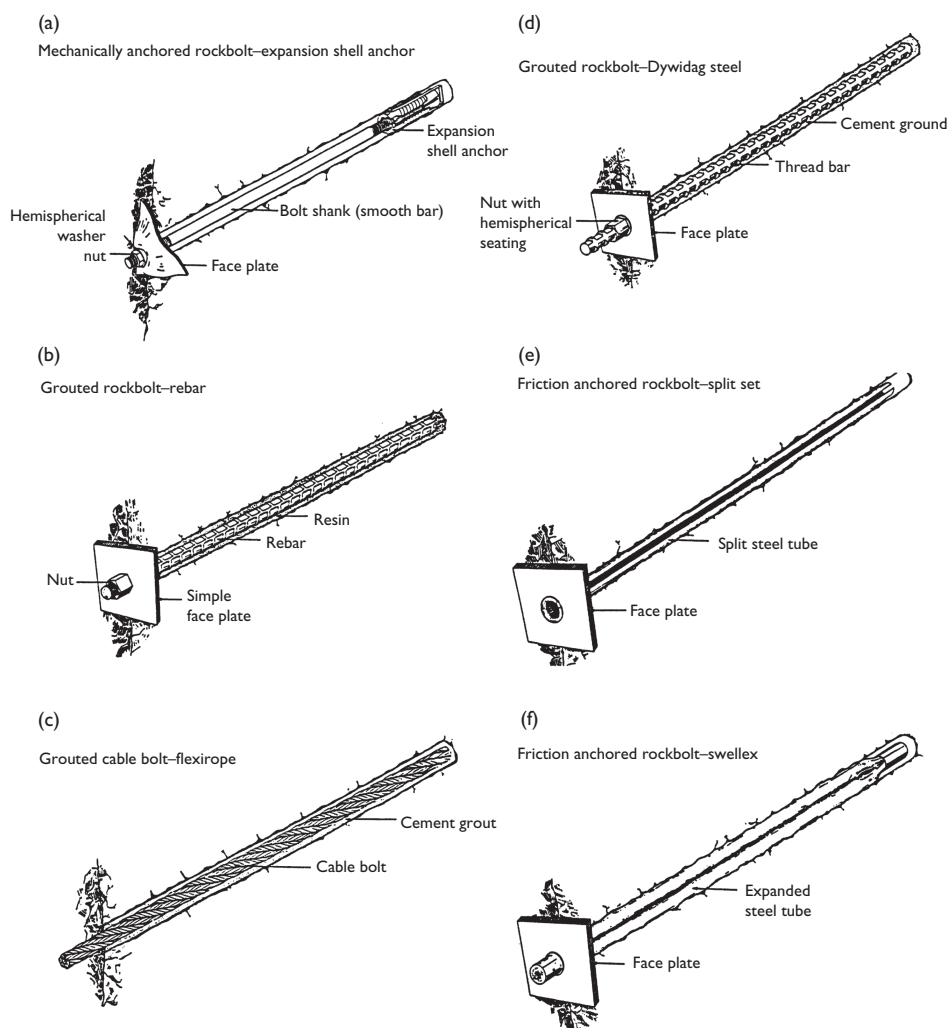


Figure 3.12 Several types of rock bolts (after Stillborg, 1986): (a) mechanical point anchored, (b), (c), (d) cemented anchorage, (e), (f) friction anchorage.

joints in a given joint set, to secure rock blocks formed by intersecting joints and to systematically reinforce the rock mass. Bolt length, diameter, steel grade, installation tension, and spacing are essential elements of such plans. Some data for bolt strengths according to steel grade are presented in Table 3.4. These data are part of the American Society for Testing and Materials (ASTM) Standard F432. Table 3.5 from the same standard gives additional bolt steel information.

Point anchored mechanical bolts are threaded at both ends. The end that goes into the drill hole first has a serrated wedge that is expanded against the drill hole wall as the bolt is spun into the hole. The wedge grips the rock and thus anchors the bolt in the hole. A small steel plate at the collar of the drill hole secures the other end of the bolt as a nut is tightened against the collar plate that bears on wall rock. Torque applied to the nut tensions the bolt that thus clamps and confines the rock mass along the bolt length. Bolts are generally tensioned to about 2/3 to 3/4 of the elastic limit (yield point). Associated bolt safety factors at installation time are therefore 3/2 to 4/3. Safety factors with respect to ultimate strength are higher, of course. Bolt spacing is generally in the range of two to eight feet; four feet is a common spacing; bolt spacing is also related to bolt length.

Bolt length may be more or less one-half shaft diameter. Eight foot long bolts could be used in a systematic plan for reinforcing a ten foot diameter shaft, while four foot long bolts could be used for spot bolting in a thirty foot diameter shaft. Bolt length combinations may also be used, short bolts for securing steel strapping, longer and larger bolts for reinforcement. The type of bolt plan dictates bolt length that in any case should be less than shaft diameter in consideration of the "1D" rule of thumb for extent of the zone of stress concentration about the shaft.

Consider a dipping joint set in the vicinity of a shaft wall and bolts passing through joints of the set as shown in Figure 3.13. The joint safety factor before bolting is given by Equation (3.10). Bolting reinforces the joints against slip directly through an uphill bolt force component and indirectly through a normal force that mobilizes frictional resistance to slip. The bolted joint safety factor $FS = FS_j + \Delta FS_j$. The improvement obtained by bolting is

$$\Delta FS_j = \frac{N_b \tan(\phi_j) + T_b}{\tau_j(\text{stress})A_j} \quad (3.16)$$

where N_b , T_b , and A_j are bolt normal force, bolt tangential force, and reinforced joint area, respectively. A reasonable estimate of A_j depends on bolt length L_b and spacings l_h and l_v in the horizontal and vertical directions, respectively. Bolt length must be sufficient to cross a joint and terminate in secure ground adequate for anchorage. The spacing product defines a bolt area of influence A_r half the distance to adjacent bolts, so $A_r = l_h l_v$. With reference to Figure 3.13, $A_j = A_r \cos(\beta) / \sin(\delta - \beta)$ where δ is joint dip and β is the bolting angle shown in Figure 3.13. Other estimates of A_j could be formulated, for example, $A_j = L_b \cos(\delta - \beta)$ per foot of shaft perimeter or $A_j = L_b / \cos(\delta - \beta)$ per foot of shaft perimeter. Rearrangement of (3.16) in terms of bolt tensile force F_b is helpful. Thus,

$$\Delta FS_j = \frac{F_b \sin(\delta - \beta) \tan(\phi_j) + F_b \cos(\delta - \beta)}{\tau_j(\text{stress})A_j} \quad (3.17)$$

where F_b is the bolt force. Bolting angle β is a design variable and thus may be optimized for maximum safety factor improvement. Inspection of (3.17) shows that the optimum occurs

Table 3.4 Load support requirements of steel bolts, threaded bars, threaded deformed bars, and threaded slotted bars whose slot has been produced without metal Removal^a N = Newton kN = kiloNewton 1 lb = 4.448 N or 0.004448 kN (adapted with permission, see ASTM in References)

Nominal diameter in.	Thread stress area in. ² , (cm ²)	Grade				60 ^b				75				100			
		Yield	Ultimate	Yield	Ultimate	Yield	Ultimate	Yield	Ultimate	Yield	Ultimate	Yield	Ultimate	Yield	Ultimate	Yield	Ultimate
		lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c	lbs(kN) ^c
5/8	0.226 (1.46)	12,400 (55.16)	19,200 (85.4)	13,600 (60.50)	20,300 (90.30)	17,000 (75.6)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	22,600 (100.5)	26,300 (130.8)
3/4	0.334 (2.55)	18,400 (81.8)	28,400 (126.3)	20,000 (89.0)	30,100 (133.9)	25,100 (111-6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	33,400 (148.6)	41,800 (193.5)
7/8	0.462 (2.96)	25,400 (113.0)	39,300 (174.8)	27,700 (123.2)	41,600 (185.0)	34,700 (154.3)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	46,200 (205.5)	57,800 (267.3)
1	0.606 (3.91)	33,300 (148.1)	51,500 (229.0)	36,400 (161.9)	54,500 (242.4)	45,500 (202.4)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	60,600 (269.5)	75,800 (350.5)
1 1/2	0.763 (4.92)	42,000 (186.8)	64,900 (286.7)	45,800 (203.7)	68,700 (305.6)	57,200 (254.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	76,300 (339.4)	95,400 (441.2)
1 1/4	0.969 (6-25)	53,300 (237.1)	82,400 (366.5)	58,100 (258.4)	87,200 (387.9)	72,700 (323.4)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	96,900 (431.0)	121,100 (560.4)
1 3/8	1.155 (7.45)	63,500 (282.4)	96,200 (436.8)	69,300 (308.2)	104,000 (462.6)	86,600 (385.2)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	115,500 (513.7)	144,400 (668.1)
1 1/2	1.405 (9.06)	77,300 (343.8)	119,400 (531.1)	84,300 (375.0)	126,500 (562.7)	105,400 (468.8)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	140,500 (624.9)	175,600 (814.6)

Notes

- a Tests of bolts and threaded bars shall be performed using full-diameter products.
b Information for Gr 60 only applies to deformed bars.
c Required yield and tensile loads shown are calculated by multiplying thread stress areas times the yield point and tensile strength values shown in Table 3.4.
Thread stress area is calculated from the mean root and pitch diameters of adrenal threads as follows:
 $A_s = 0.7854(D - 0.9743/n)^2$

where:

- A_s^2 = stress area, in.²
 D = nominal diameter, in., and
 n = number of threads per inch
 $D^{5/8}$ in. Gr 40 products are not covered.

Table 3.5 Steel strength and grade^a

Grade	Yield point psi (MPa)	Tensile strength psi (MPa)
40	40,000 (276)	70,000 (483)
55	55,000 (379)	85,500 (586)
60	60,000 (414)	90,000 (621)
75	75,000 (517)	100,000 (688)
100	100,000 (689)	125,000 (862)

Note

a After ASTM Standard F432.

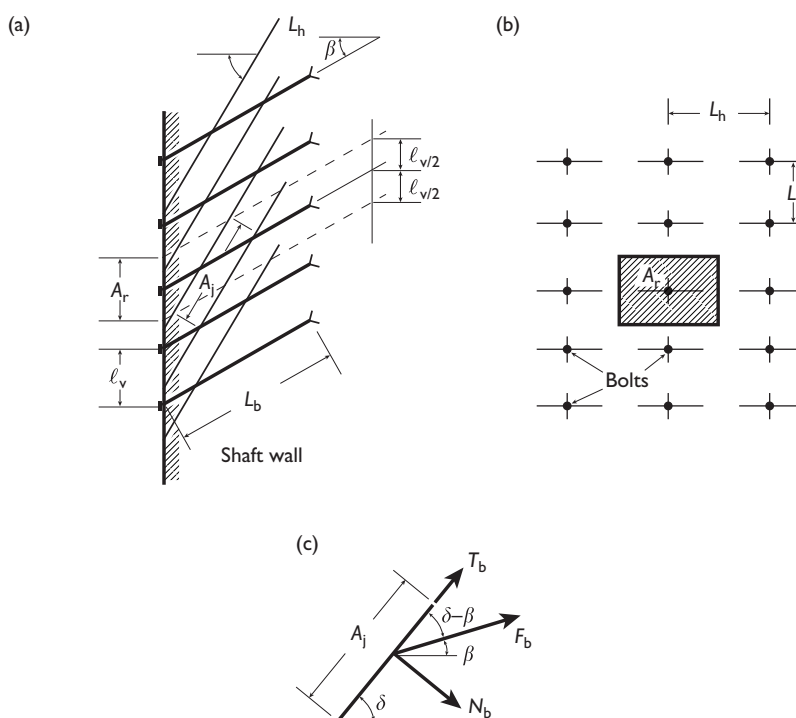


Figure 3.13 Shaft wall joint bolting geometry: (a) vertical section, (b) wall view, and (c) bolt force detail.

when the numerator is greatest, that is, when $\cos(\delta - \beta - \phi_j) = 1$, provided A_j is fixed. The optimum bolting angle $\beta = \delta - \phi_j$. Angling a bolt hole upwards has the advantage of producing a cleaner hole by allowing drill cuttings to flow freely from the hole. However, this advantage may be small compared with the disadvantage of orienting drill machines to drill upward rather than the more natural horizontal direction in a vertical shaft. In the horizontal direction, which is likely to be the case in practice, $\beta = 0$. If the joint is then vertical, $\delta = \pi/2$, and $\Delta FS_j = F_b \tan(\phi_j)/\tau_j(\text{stress})A_r$ as may be obtained directly.

The rock bolting safety factor improvement formula (3.17) masks a feature of bolting and that is bolting improvement is generally very small. Indeed, this is the case for much of rock

support and reinforcement. The reason lies in the small “pressures” that support and reinforcement are capable of supplying to the rock mass.

Example 3.15 Consider a vertical shaft excavated in a stress field where $S_v = 1,350$, S_h (north) = 300 psi and S_h (east) = 1,200 psi and a joint set that strikes due north and dips 60° with cohesion $c_j = 135$ psi and friction angle $\phi_j = 35^\circ$. For convenience, fix the x-axis due east parallel with S_h . Determine joint safety factors before excavation, after excavation and the improvement that a bolting plan could achieve.

Solution: Joint normal and shear stresses for this purpose may be obtained by rotation of the reference axes. Thus, before excavation

$$\sigma_j = \left(\frac{1,200 + 1,350}{2} \right) - \left(\frac{1,200 - 1,350}{2} \right) \cos(120^\circ) = 1,313 \text{ psi}$$

$$\tau_j = - \left(\frac{1,200 - 1,350}{2} \right) \sin(120^\circ) = 65 \text{ psi}$$

where the presence of the joint is assumed not to affect the given stress field. The joint safety factor is

$$FS_j(\text{before}) = \frac{1,313 \tan(35^\circ) + 135}{65} = 16.2$$

so joint slip or separation is not impending. After excavation

$$\sigma_j = \left(\frac{0 + 1,350}{2} \right) - \left(\frac{0 - 1,350}{2} \right) \cos(120^\circ) = 1,013 \text{ psi}$$

$$\tau_j = - \left(\frac{0 - 1,350}{2} \right) \sin(120^\circ) = 585 \text{ psi}$$

which shows some normal stress reduction but a significant increase in shear stress; both tend to reduce safety. The joint safety factor is now

$$FS_j(\text{after}) = \frac{1,013 \tan(35^\circ) + 135}{585} = 1.44$$

which is greatly reduced and probably too low for shaft wall safety.

If bolts are installed horizontally on a square pattern of three foot centers and tensioned to 18,500 pounds force, the safety factor improvement is

$$\Delta FS_j = \frac{18,500 [\sin(60^\circ - 0) \tan(35^\circ) + \cos(60^\circ - 0)]}{(585)(3)(3) \cos(0) / \sin(60^\circ)} = 0.023$$

or less than 2% over the post-excavation joint safety factor. The small improvement in joint safety factor obtained by bolting suggests that bolting may not be an effective support method. This suggestion is definitely *not* the case. The example results do clearly indicate that stresses induced by bolting are quite small relative to stresses *in situ*. Indeed, bolts are effective when they prevent rock blocks from falling into an excavation and are essential

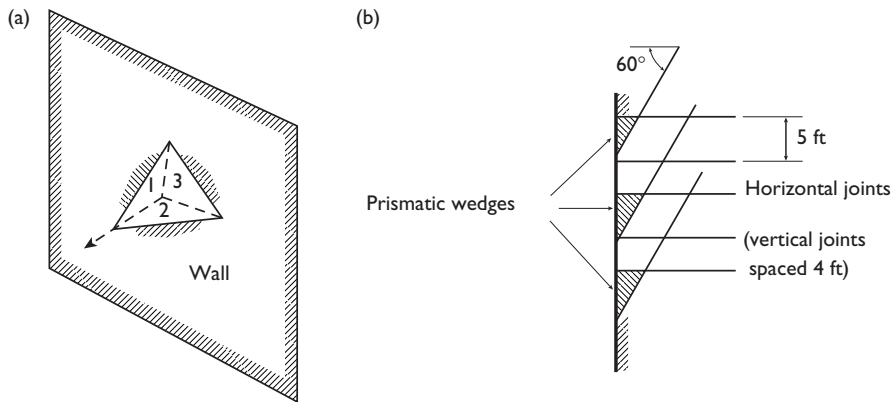


Figure 3.14 Potential wedge failures in a shaft wall: (a) tetrahedral wedge and (b) vertical section looking north.

to shaft safety. In this regard, slip at a point on a joint is not synonymous with rock block fall.

If a joint is the base of a rock block defined by other intersecting joints, then shaft safety requires the potential failure to be eliminated. However, not all rock blocks defined by intersecting joints, fractures, cracks, and bedding planes, pose a threat to safety. Only those blocks that are kinematically disposed to move into the excavation are of concern. For example, tetrahedral blocks formed by three intersecting joints that have a baseline of intersection dipping away from an excavation are safe. If the line of intersection dips into the excavation, as shown in Figure 3.14(a), then a safety analysis and possibly bolting reinforcement are indicated. A variety of block shapes may be formed by joints of different orientations and spacings and the excavation wall proper.

Example 3.16 Consider a wedge formed in a vertical shaft excavated where one joint set strikes due north and dips 60° west with uncertain cohesion c_j and friction angle $\phi_j = 35^\circ$, a second set is vertical with spacing of four feet and dip direction due north, no cohesion and friction angle $\phi_j = 35^\circ$ and a third is horizontal with spacing of five feet, no cohesion and nil friction angle. Fix the x -axis due east perpendicular to strike of the first joint, as shown in Figure 3.14(b). Determine whether the block will be a threat to excavation safety and, if so, the second question is then what bolting plan could eliminate the threat by improving a suitable factor of safety to some acceptable value.

Solution: A rough first analysis may be based on a model of a block sliding down an inclined plane. This model neglects sides and top and because the inclination of the base is greater than the friction angle, sliding would occur under the weight of the block if cohesion were also neglected. Thus, further analysis is required. In view of the uncertain joint cohesion, one may determine what cohesion is necessary to achieve an acceptable safety factor, say, $FS_j = 3.5$. A force safety factor against sliding is

$$\begin{aligned}
FS_j &= \frac{\text{resisting force}}{\text{driving force}} \\
&= \frac{R}{D} \\
&= \frac{W \cos(\delta) \tan(\phi_j) + c_j A_j}{W \sin(\delta)} \\
3.5 &= \frac{(4,500) \cos(60^\circ) \tan(35^\circ) + c_j (23)(144)}{(4,500) \sin(60^\circ)}
\end{aligned}$$

where W is block weight and A_j is area of the block base (Volume = $(4)(5)(5 \tan(30)/2) = 28.87$ cubic feet). Solving this last equation gives $c_j = 3.6$ psi, assuming a specific weight of rock of 156 pcf. Thus, a small amount of joint cohesion may be important.

But whether a reliable determination can be made is another question. Without cohesion, the safety factor is 0.40, and as a practical matter, bolting would probably be specified. For example, a horizontal bolt centered on the block face and is tensioned to 18,500 pounds improves the force safety factor against sliding by

$$\Delta FS_j = \frac{18,500 [\sin(60^\circ) \tan(\phi) + \cos(60^\circ)]}{4,500 \sin(60^\circ)} = 5.3$$

This result for a bolted rock block provides an entirely different view of joint reinforcement by bolting. Even if the rock block weight were tripled, the improvement in joint safety factor would, say, for the horizontal bolting plan, be 1.8.

In essence, a *dead weight* load model was adopted for the alternative joint reinforcement analysis, an approach that is conservative because of the neglect of physical features that would otherwise increase safety. In blocky ground formed by a multiplicity of joint intersections, block size and kinematics pose a formidable design challenge that may be met with specialized “key block” computer programs.

Consideration of gravity sliding of an individual block aides understanding how rock bolts reinforce a jointed rock mass, but when blocks are numerous and variable, a systematic bolting plan is indicated. An approach to understanding how pattern bolting may be effective is based on the influence of confining pressure on rock strength and how bolting generates a confining pressure at a shaft wall.

Compressive strength C_p under confining pressure p according to the Mohr–Coulomb strength criterion is

$$C_p = C_o + \left(\frac{C_o}{T_o} \right) p \quad (3.18)$$

The ratio of unconfined compressive to tensile strength for intact rock tested in the laboratory is usually in the range of 10–20, so a small confining pressure is greatly amplified and may contribute substantially to compressive strength. Intact, laboratory-scale strengths may be reduced or scaled to characterize a field-scale rock mass, perhaps by an order of magnitude. A 10,000 psi laboratory strength would be 1,000 psi in the field. Causes of such a reduction or scaling are the natural joints in the rock mass and cracks and fractures induced by excavation which are certainly formed by

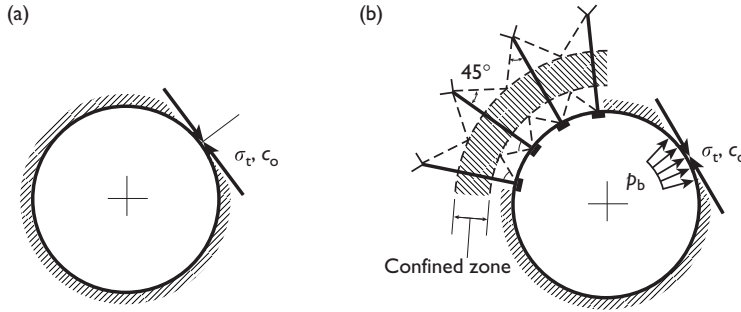


Figure 3.15 Radial shaft wall bolting plan: (a) before bolting and (b) after bolting.

blasting. However, the strength ratio would remain constant, provided that both strengths were reduced by the same scale factor.

Shaft wall safety factors (with respect to compression) before and after application of a confining pressure by bolting and the improvement achieved by bolting are therefore

$$FS_o = \frac{C_o}{\sigma_c}, \quad FS = \frac{C_o + (C_o/T_o)p_b}{\sigma_c}, \quad \Delta FS = \frac{(C_o/T_o)p_b}{\sigma_c} \quad (3.19)$$

where the confining pressure p_b is specifically attributed to bolt forces. Implicit in (3.19) is the assumption that the post-excavation stress σ_c concentrated at the shaft wall is unaffected by bolting. Bolting would tend to decrease this compression, so the assumption is conservative in the sense that the safety factor would be somewhat higher if the effect were easily quantified and taken into account. Figure 3.15 shows stress concentration in a circular shaft wall before bolting and a ring of bolts installed in the wall after excavation. Bolting pressure is a statically equivalent distributed normal stress given by $p_b = F_b/A_r$ where A_r is the rock area influenced by a bolt. This area is simply defined by bolt spacing, that is, $A_r = l_h l_v$, the product of horizontal by vertical bolt spacing on a shaft wall. A modification to the equivalent distributed load view is to increase the area of distribution gradually from the collar and anchor points of force application. A 45° slope is often used for this purpose, as shown in Figure 3.15. The confining pressure in the clamped ring of rock is the same in either case.

Example 3.17 Consider bolts tensioned to 18,500 pounds force on three foot centers horizontally and four foot centers vertically, as shown in Figure 3.15, and suppose the tangential stress concentration at the shaft wall is 3,000 psi. Bolting pressure is about 15 psi. If intact rock unconfined compressive strength $C_o = 12,000$ psi and tensile strength $T_o = 1,200$ psi and both are scaled by a 1/4, then rock mass $C_o = 3,000$ psi. Determine safety factors before and after bolting and show the safety factor improvement achieved by bolting is about 0.05 (5%).

Solution: By definition

$$FS_o = \frac{C_o}{\sigma_c} = \frac{3,000}{3,000} = 1.0$$

$$FS = \frac{C_o + (C_o/T_o)p_b}{\sigma_c} = \frac{3,000 + (10)(18,500/9)(1/144)}{3,000} = 1.05$$

$$\Delta FS\% = (1.05 - 1.00)100 = 5\%$$

Again, in Example 3.17, the effects of bolting forces are small relative to rock mass stresses and thus bring into question whether the concept of bolt functioning by increasing strength with confining pressure is valid. The suggestion is that bolts may function according to a different physical mechanism, for example, by restraining rock blocks from falling and thus by preventing raveling of ground into a major failure. Indeed, bolts are generally considered temporary support of excavations designed for a long service life. They function mainly by preventing small block falls developing into large failures. Bolts may be used in combination with metal strapping, wire mesh, chain link, plastic webbing, or even membrane coatings for this purpose.

Bolts with distributed anchorage, bolts cemented into a hole, and friction bolts, function in much the same way as point anchored bolts that are tensioned during installation. Pretensioned bolts are *active* because they provide immediate resistance to rock mass displacement at the time of installation. Distributed anchorage bolts are ordinarily not tensioned during installation and are *passive* because they provide support only to the extent that the rock mass tensions bolts after installation. Both types support the rock mass by resisting displacement along the bolt hole. In addition, distributed anchorage bolts provide resistance to shearing motion across bolt holes because the hole is filled. This resistance is *dowel* action. Point anchored bolts have an empty annulus between bolt and hole that allows for easy displacement across bolt holes and thus are much less effective against shearing action. *Both types of bolts should be oriented to be stretched in tension by anticipated rock mass displacements.* Cable bolts in particular must be tensioned to be effective.

Circular shaft liners

Shaft liners are usually concrete that is formed to the shaft wall. Concrete may be used in combination with steel liner plate or steel rings which may be embedded in the concrete. Shaft walls may also be sprayed with cement (shotcrete, gunite) after bolting and screening. Liner plate and light segment corrugated steel may also be used in shafts. Concrete and shotcrete are brittle materials that may be reinforced by steel rebar (concrete) or by steel or glass fibers (shotcrete) to improve strength and resistance to fracture.

There are two systems for installing concrete shaft liners. For shafts of moderate depth, the liner is installed after the shaft is sunk. Concrete is poured from the bottom up using slip forms. In this case, shaft wall motion induced by mining is complete before the liner is installed assuming the rock mass responds elastically to excavation. Economics of deep shafts dictates installation of the liner concurrently with sinking. By keeping the last pour several shaft “diameters” above the shaft bottom, motion of the shaft walls after the liner is installed is slight and induces little stress in the liner as sinking continues, again, assuming the

rock mass response is elastic. In either case, the liner experiences negligible loading from contact with the rock mass.

If the rock mass exhibits a pronounced time-dependent component of deformation and thus creeps significantly or behaves like a highly viscous fluid, then high stresses may eventually be induced in the liner. Indeed, as time passes, the full premining stress may come to bear on the liner. Under such circumstances, a rigid liner is a poor support choice. Support that accommodates yielding or flow of the rock mass would be a better choice.

Water is an important source of shaft liner load and represents an all around pressure on the liner after installation. During shaft sinking, a grout curtain is often used to dam the flow of water to the unlined shaft, but in time the curtain may deteriorate, fracture, and allow seepage to the liner. Liners should be designed for this eventuality.

A model for a *circular* shaft liner loaded with water pressure may be found in the solution to the problem of a hollow (thick-walled) cylinder loaded by radial “pressures” p_a and p_b acting on the inside of the liner at a radius a and at the outside of the liner at a radius b (unlined shaft diameter $D_o = 2b$). The cylinder is considered elastic with Young’s modulus E and Poisson’s ratio ν and under plane strain conditions ($\varepsilon_{zz} = 0$). The problem is also one of axial symmetry. Stresses, strains, and displacements are therefore independent of z and θ . Under these conditions there is only the radial displacement u to be considered. In fact, $u = Ar + B/r$ (see, e.g. Love, 1944) and in cylindrical coordinates ($r\theta z$), the radial and circumferential strains are

$$\varepsilon_{rr} = A - \frac{B}{r^2}$$

$$\varepsilon_{\theta\theta} = A + \frac{B}{r^2}$$

According to Hooke’s law under plane strain conditions, the radial and circumferential stresses are related to strains by

$$\sigma_{rr} = \frac{E}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_{rr} + \nu\varepsilon_{\theta\theta}]$$

$$\sigma_{\theta\theta} = \frac{E}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_{\theta\theta} + \nu\varepsilon_{rr}]$$

Hence,

$$\sigma_{rr} = \frac{E}{(1+\nu)(1-2\nu)} \left[A - (1-2\nu) \frac{B}{r^2} \right]$$

$$\sigma_{\theta\theta} = \frac{E}{(1+\nu)(1-2\nu)} \left[A + (1-2\nu) \frac{B}{r^2} \right]$$

Use of the boundary conditions on the radial stress at $r = a$ and $r = b$ allows for the determination of the constants A and B . Thus,

$$\frac{EA}{(1+\nu)(1-2\nu)} = \left(\frac{p_a a^2 - p_b b^2}{a^2 - b^2} \right)$$

$$\frac{EB}{(1+\nu)} = \left(\frac{p_a - p_b}{a^2 - b^2} \right) (a^2 b^2)$$

Hence,

$$\begin{aligned}\sigma_{rr} &= \left(\frac{p_a a^2 - p_b b^2}{a^2 - b^2} \right) - \left(\frac{p_a - p_b}{a^2 - b^2} \right) \left(\frac{a^2 b^2}{r^2} \right) \\ \sigma_{\theta\theta} &= \left(\frac{p_a a^2 - p_b b^2}{a^2 - b^2} \right) + \left(\frac{p_a - p_b}{a^2 - b^2} \right) \left(\frac{a^2 b^2}{r^2} \right) \\ u &= \left(\frac{1 + \nu}{E} \right) \left[(1 - 2\nu) \left(\frac{p_a a^2 - p_b b^2}{a^2 - b^2} \right) r + \left(\frac{p_a - p_b}{a^2 - b^2} \right) \left(\frac{a^2 b^2}{r} \right) \right]\end{aligned}$$

where, if compression is positive, then a positive radial displacement is inward.

In the usual case, there is no internal pressure p_a acting on the liner, so

$$\begin{aligned}\sigma_{rr} &= \left(\frac{p_b}{1 - (a/b)^2} \right) \left(1 - \frac{a^2}{r^2} \right) \\ \sigma_{\theta\theta} &= \left(\frac{p_b}{1 - (a/b)^2} \right) \left(1 + \frac{a^2}{r^2} \right) \\ u &= \left(\frac{1 + \nu}{E} \right) \left[(1 - 2\nu) \left(\frac{p_b}{1 - (a/b)^2} \right) r + \left(\frac{p_b}{1 - (a/b)^2} \right) \left(\frac{a^2}{r} \right) \right]\end{aligned}$$

A Mohr–Coulomb failure criterion is appropriate to concrete and includes steel as a special case. A convenient form in terms of principal stresses is

$$\left(\frac{\sigma_1 - \sigma_3}{2} \right) = \left(\frac{\sigma_1 + \sigma_3}{2} \right) \sin(\phi) + (c) \cos(\phi)$$

where c and ϕ are cohesion and angle of internal friction, respectively. In the hollow cylinder model of a shaft liner, the radial and circumferential stresses are principal stresses. Hence,

$$\left(\frac{p_b}{1 - (a/b)^2} \right) \left(\frac{a^2}{r^2} \right) = \left(\frac{p_b}{1 - (a/b)^2} \right) \sin(\phi) + (c) \cos(\phi)$$

that is first satisfied where r is the least, that is, at $r = a$. At greater r the left side of the failure criterion is too small to satisfy the equality. Thus, liner failure initiates at the inside of the liner when

$$p_b = \left[\frac{(c) \cos(\phi)}{1 - \sin(\phi)} \right] \left[1 - (a/b)^2 \right]$$

The circumferential stress on the inside of the liner ($r = a$) is maximum. Thus,

$$\sigma_{\theta\theta}(a) = \frac{2p_b}{1 - (a/b)^2}$$

where the radial stress is zero. As b becomes indefinitely large, the stress approaches two times the applied stress which agrees with the case of a circular hole under hydrostatic stress. If p_b is a reference stress and K is a liner stress concentration factor, then

$$\sigma_{\theta\theta} = Kp_b, \quad K = \left[\frac{2}{1 - (a/b)^2} \right]$$

The safety factor for the liner is given by

$$FS_c = \frac{C_o}{\sigma_c} = \frac{C_o}{Kp_b} = C_o \left[\frac{1 - (a/b)^2}{2p_b} \right] \quad (3.20)$$

If the liner thickness h is specified, so $h = b - a$, then the factor of safety may be calculated, given the liner strength C_o and applied stress p_b .

Alternatively, a liner thickness may be calculated for a given safety factor or, equivalently, for a given maximum allowable stress (C_o/FS). Two formulas are possible, one in terms of the inner liner radius a , the other in terms of the outer liner radius b which is also the radius of the unlined shaft wall. Thus, after solving the liner safety factor equation,

$$\begin{aligned} h &= b \left[1 - \left(\frac{C_o - 2p_b FS}{C_o} \right)^{(1/2)} \right] \\ h &= a \left[\left(\frac{C_o}{C_o - 2p_b FS} \right)^{(1/2)} - 1 \right] \end{aligned} \quad (3.21)$$

Either may be used; both follow directly from the liner safety factor formula. In consideration of the square root operation, the liner load can in no case exceed one-half the unconfined compressive strength of the liner material. Liner loads from water pressure are an order of magnitude less, although there are notable exceptions. In exceptional cases, a concrete liner alone may not suffice; combinations of steel and concrete may be needed to support very high water pressure.

Example 3.18 A circular concrete liner in a vertical shaft is subject to an external pressure of 90 psi. Concrete strength is 5,000 psi. If the inside diameter of the liner must be 18 ft, determine the liner thickness necessary to achieve a liner safety factor of 3.5.

Solution: According the second equation of (3.21)

$$\begin{aligned} h &= \left(\frac{18}{2} \right) \left\{ \left[\frac{5,000}{5,000 - (2)(9)(3.5)} \right]^{1/2} - 1 \right\} \\ &= 0.627 \text{ ft} \\ h &= 7.52 \text{ in.} \end{aligned}$$

This thickness is rather small and as a practical matter the actual liner would probably be one foot thick.

Example 3.19 A circular concrete liner in a vertical shaft is subject to an unknown external pressure that cause the liner to fail in compression on the inside of the liner wall where traces of vertical shear cracks are observed. Concrete strength is 5,000 psi. The inside diameter of the liner is 18 ft and the thickness is 7.52 in. Determine the liner load at failure.

Solution: At failure, the liner safety factor is 1.0; according to (3.20), this safety factor is

$$\begin{aligned} \text{FS}_c &= C_o \left[\frac{1 - (a/b)^2}{2p_b} \right] \\ \therefore p_b &= C_o \left[\frac{1 - (a/b)^2}{2\text{FS}_c} \right] \\ &= (5,000) \left\{ \frac{1 - [9/(9 + 7.52/12.0)^2]}{(2)(1)} \right\} \\ p_b &= 315 \text{ psi} \end{aligned}$$

The small thickness leads to a relatively small failure load. A one foot thick liner would require a pressure of 475 psi, about 50% greater, to cause failure.

Example 3.20 A 3 ft thick circular concrete shaft liner is designed for an allowable stress of 3,500 psi in compression. Outside diameter is 24 ft. Determine the thickness of a steel (A36) liner with a safety factor of 1.5 that would support the same load but allow for a greater inside diameter and therefore a greater useful cross-sectional area of the shaft.

Solution: The load the concrete liner is designed to support is

$$p_b = C_o \left[\frac{1 - (a/b)^2}{2\text{FS}_c} \right] = (3,500) \left[\frac{1 - (9/12)^2}{2} \right] = 766 \text{ psi}$$

According to the first equation of (3.21)

$$\begin{aligned} h &= (12) \left[1 - \left(\frac{36,000 - 2(766)(1.5)}{36,000} \right)^{1/2} \right] \\ &= 0.389 \text{ ft} \\ h &= 4.67 \text{ in.} \end{aligned}$$

The inside shaft diameter would increase from 21 ft to 23 ft. Whether the increase is warranted would require a cost analysis.

The *average* circumferential stress acting through the thickness of the liner may be computed by integrating the circumferential stress through the thickness of the liner and then dividing by the thickness. The result is

$$\sigma_{\text{ave}} = \frac{p_b b}{h}$$

which may also be obtained by substituting $a = b - h$ in the safety factor formula, then expanding the square and neglecting $(h/b)^2$ of the result on the grounds that the thickness is small compared with the radius, that is, $(b - a)/(b + a) \ll 1$. Figure 3.16 shows how the ratio of maximum to average circumferential stress varies with the ratio of thickness to outer radius. The average stress is within about 10% of the maximum stress up to a thickness of about 0.2 the outer radius (unlined shaft radius). A 20 ft diameter shaft lined with concrete 1 ft

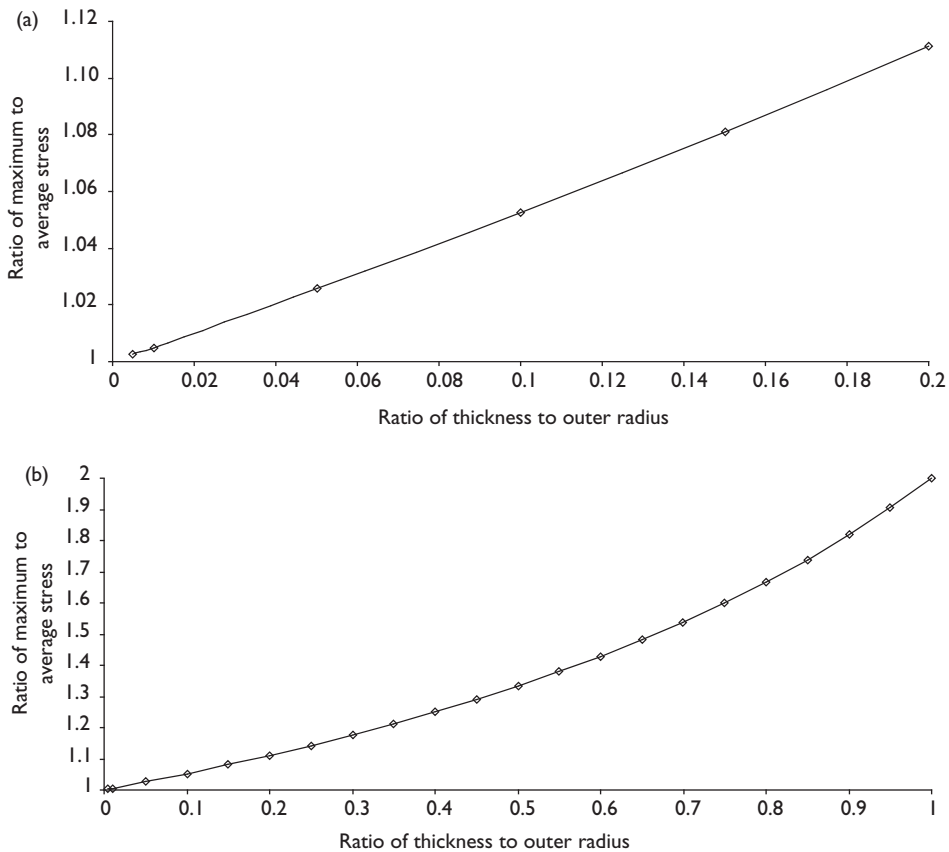


Figure 3.16 Ratio of maximum to average liner stress under external pressure as a function of the ratio of liner thickness to outer radius. (a) Small range and (b) large range.

thick could be designed using the thick-walled cylinder formula (exact solution) or using the simpler average stress. The maximum stress in this example is $10.52 p_b$ while the average is $10.00 p_b$, a difference of about 5%.

Example 3.21 Suppose a 20 ft diameter shaft is sunk through an aquifer where the fluid pressure is 180 psi and a safety factor of 4.8 is desired using concrete with $C_o = 4,800$ psi. The required liner thickness according to (3.21) is

$$h = (20/2) \left[1 - \left(\frac{4,800 - (2)(180)(4.8)}{4,800} \right)^{1/2} \right] = 2.0 \text{ ft}$$

assuming the diameter given is the outside diameter. The finished inside shaft diameter would be 16 ft.

If the finished shaft diameter needed were 20 ft, then the liner thickness required would be

$$h = (20/2) \left[\left(\frac{4,800}{4,800 - (2)(180)(4.8)} \right)^{1/2} - 1 \right] = 2.5 \text{ ft}$$

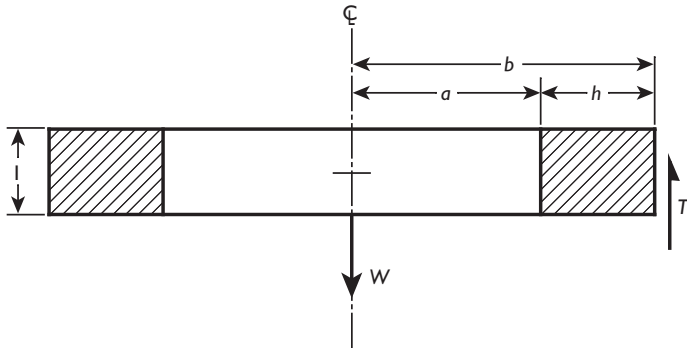


Figure 3.17 Shaft ring weight equilibrated.

If the liner were steel instead of concrete, and $C_o = 36,000$ psi, then by the same formula that leads to a 2.5 ft thickness of concrete, $h = 0.25$ ft or a 3.0 in. thickness of steel. The useful area of a shaft is, of course, defined by the inside diameter (20 ft).

In this example, the unlined shaft would need to be excavated to a diameter of 25 ft to allow for the required concrete liner thickness and finished diameter of 20 ft where the shaft route encounters the high water pressure.

Weight of a concrete liner also loads the liner, so some account of this load is of interest. The weight of a 20 ft diameter shaft liner, 1 ft thick, is the product of concrete specific weight and ring volume. With reference to Figure 3.17 that shows a ring in equilibrium with a shearing force acting on the perimeter of the ring at the concrete-rock contact with no change in vertical force from top to ring bottom, the shear force required for equilibrium is just

$$W = T$$

$$\gamma\pi(b^2 - a^2) = 2\pi b\tau$$

where τ is the average shear stress acting over a one foot run of liner. Solution for τ gives

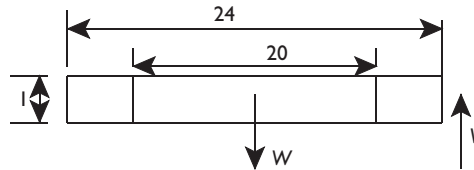
$$\tau = \frac{\gamma}{2b}(b - a)(b + a)$$

$$\tau \approx \frac{\gamma}{2b}(h)(2b) = \gamma h$$

Thus, shear stress required to support the liner is about equal to the product of liner specific weight γ and liner thickness h . This product is roughly 1 psi per foot of liner thickness. A one foot thick liner would require about 1 psi of supporting shear stress at the liner-rock contact. In consideration of the roughness of the unlined shaft wall and the adhesion of concrete to rock, a shear stress of a few psi is easily supported. For this reason, the weight of a liner does not accumulate with depth. The load is distributed over the perimeter of the liner and does not develop a large vertical force over the bottom of the liner.

Example 3.22 A circular concrete shaft liner has an inside diameter of 20 ft and an outside diameter of 24 ft. Determine the bond strength between concrete and rock such that liner weight does not accumulate. Assume specific weight of concrete is 140 pcf.

Solution: A free body diagram of a section of the liner is shown in the sketch.



For weight not to accumulate, each vertical foot of liner must be supported in shear at the concrete–rock interface. Thus,

$$W = V = (140)(\pi/4)(24^2 - 20^2)(1) = \tau(\pi)(24)(1)$$

$\therefore$

$$\tau = 257 \text{ psf (1.9 psi)}$$

However, if a liner is decoupled from the rock and thus moves freely in the vertical direction, then liner weight does accumulate and must be supported by reactions at the bottom of the liner. Decoupling may be desirable in the presence of shaft wall motion or subsidence induced by mining or other activity such as drainage of an aquifer or fault.

If indeed the rock mass does not load a liner because post-excavation displacement is negligible, then liner thickness should be independent of shaft depth. Empirical evidence for this inference is shown in Figure 3.18 (Cornish, 1967) where depth is plotted as a function of the ratio of liner thickness to diameter (in./ft) for 43 shafts. Depth ranges to over 6,000 ft, but there is clearly no trend line that would indicate a correlation between depth and liner thickness to diameter ratio.

Figure 3.19 illustrates a computer simulation of concurrent shaft sinking and concrete lining using the popular finite element method (Pariseau, 1977). The shaft is circular and the analysis is axially symmetric with a preexcavation principal stress ratio of one ($M = 1$). Unlined shaft diameter was 22 ft and liner thickness was 1 ft; finished diameter was 20 ft. Results in Figure 3.19 show that total shaft wall displacement increases linearly with depth as does stress. However, liner displacement is small and independent of depth. Most of the shaft wall displacement occurred simultaneously with sinking rounds that were equal to 20 ft initially and then for computation efficiency increased to 40 ft. The liner was never installed closer than one shaft diameter to the shaft bottom and was never more than 60 ft from the shaft bottom. Keeping support at least one diameter from the shaft bottom not only defends support against residual wall displacement but also reduces damage from blasting.

Example 3.23 A 1-ft thick circular concrete shaft liner has an unconfined compressive strength of 3,500 psi, a Young's modulus of $3.5(10^6)$ psi, a Poisson's ratio of 0.20, and a specific weight of 140 pcf. Inside diameter is 20 ft. Determine the change in inside diameter when the load is sufficient to cause liner failure.

Solution: The radial displacement of the liner is given by

$$u(r) = \left(\frac{1+\nu}{E} \right) \left[(1-2\nu) \left(\frac{p_b}{1-(a/b)^2} \right) r + \left(\frac{p_b}{1-(a/b)^2} \right) \left(\frac{a^2}{r} \right) \right]$$

$$\therefore u(a) = \left(\frac{1+0.20}{3.5(10^6)} \right) \left[(1-2(0.20)) \left(\frac{p_b}{1-(9/10)^2} \right) 9 + \left(\frac{p_b}{1-(9/10)^2} \right) \left(\frac{9^2}{9} \right) \right]$$

$$= 2.6(10^{-5})p_b$$

At failure, the liner load is

$$p_b = (C_o/2)[1 - (a/b)^2] = (3,500/2)[1 - (9/10)^2] = 333 \text{ psi}$$

$$\therefore u(a) = 2.6(10^{-5})(333)(12) = 0.104 \text{ in.}$$

The change in diameter is just twice the radial displacement, so $\Delta D = 0.208$ in. Although, seemingly small, accurate measurement during a monitoring program would not be difficult with a calibrated tape extensometer or similar device.

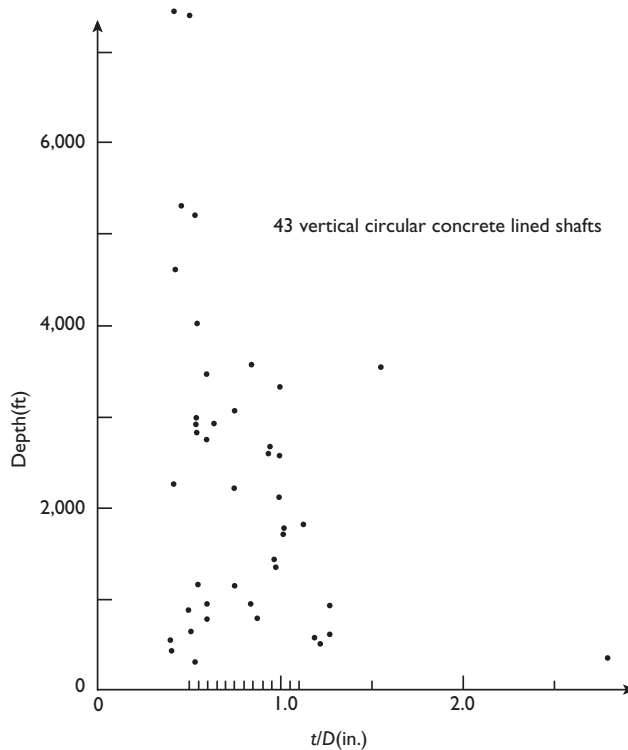


Figure 3.18 Shaft depth versus liner thickness to diameter ratio (after Cornish, 1967).

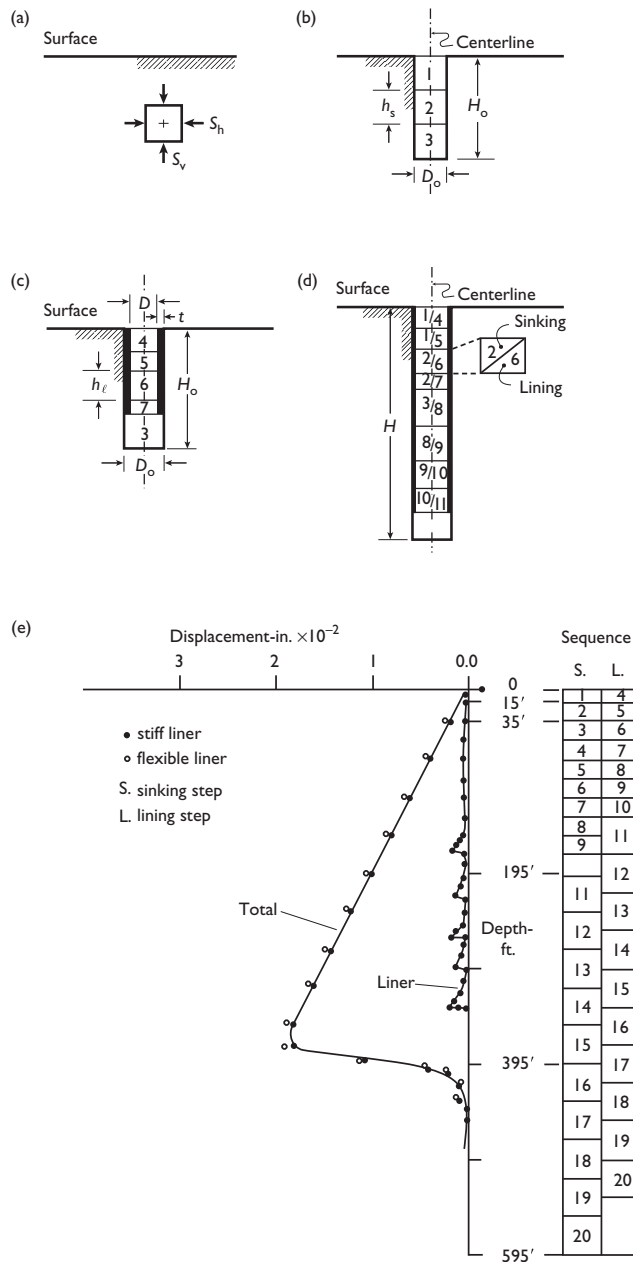


Figure 3.19 Schematic representation of shaft sinking and lining operations performed by the computer. Numbers identify the sequence of operations. (a) Generation of the premining stress field, (b) collaring of shaft and initial sinking to depth H_o in lifts of height h_s , (c) first installation of the liner in lifts of height h_l , (d) shaft deepening to depth H with concurrent sinking and lining operation, and (e) shaft sinking and lining sequence. Also total shaft and liner wall displacements at shaft depth of 395 ft.

Note

Actual curves are based on more than 100 data points.

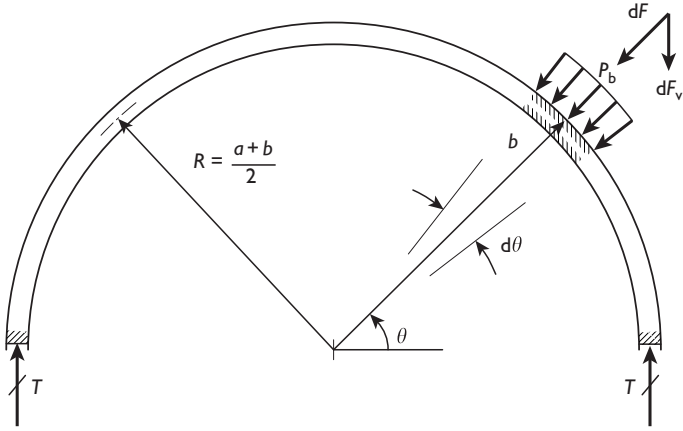


Figure 3.20 Thin, continuous ring equilibrium.

Circular steel rings

Integration of radial pressure over a semicircular shaft ring allows for estimation of the net horizontal force applied to the ring. This force must be equilibrated by internal reactions at the semi-ring ends as shown in Figure 3.20. The equilibrium requirement allows for calculation of the average stress acting through the ring thickness. Thus,

$$\begin{aligned}
 F_v &= 2T \\
 &= \int_0^\pi dF_v \\
 &= \int_0^\pi \sin(\theta) b p_b S d\theta \\
 F_v &= 2b S p_b
 \end{aligned}$$

where S is distance along the ring, which is assumed to be continuous and has cross section Sh . Hence, $T = b S p_b$ and the average stress is just $T/Sh = b p_b/h$ in this continuous liner ring.

If the ring is not continuous but discrete, a steel set, for example, then each ring supports the load one half the distance to the next set. Some load may be transmitted to a ring by lagging or lacing (wood, steel) between sets, but the largest load would be a line load P distributed over the outside of the ring with magnitude $p_b S$. In this case, the support load or pressure p_b is likely to be a rock pressure rather than a water load. Regardless of the identity of the load, if the spacing of ring sets is S , then ring support reaction T is

$$T = p_b R S \quad (3.22)$$

where the radius is now to the center of the ring which is of acceptable accuracy in consideration of ring radius and thickness. Clearly if the liner load were an internal pressure tending to expand the ring, the support reaction T would be a tension given by the same formula within the accuracy of practical geometry (thickness small compared with radius, say, a few inches compared with several feet).

A safety factor for steel ring sets in vertical shafts may be defined with respect to forces acting and resisting or simply as a ratio of strength to stress. Thus,

$$FS = \frac{C_s A_s}{T} = \frac{C_s}{T/A_s} = \frac{C_s}{\sigma_s} \quad (3.23)$$

where C_s and σ_s are steel compressive strength and stress, respectively.

Equations (3.22) and (3.23) allow one to estimate steel size (cross-sectional area) needed to support an anticipated rock pressure p_b in an unlined shaft of radius R (or b). Steel rings are curved beams shaped and sized to design specifications and are usually A36 steel that has an initial yield strength of about 36,000 psi. Steel area and ring spacing interact, so there is no unique solution to support design.

Example 3.24 Consider an unlined shaft 20 ft in diameter where the rock pressure is estimated to be 15 psi, a steel safety factor of 2.5 is specified, and the maximum allowable stress in a ring is 36,000/2.5 or 14,400 psi. Determine the steel area A_s required.

Solution: Although the spacing is not uniquely determined, an expression relating steel area to spacing may be obtained. Thus,

$$A_s = \frac{(15)(20/2)(144)S}{14,400} = 1.5S$$

where A_s is in square inches and spacing S is in feet.

A wide spacing clearly implies a heavier ring. A W or wide flange shape with an 8 in. flange width spaced on 6 ft centers with an area of about 9.0 square inches would weigh 31 pounds ($W8 \times 31$) per lineal foot. Increasing spacing by 33% to 8 ft would increase weight proportionally and call for a $W8 \times 40$ shape (*Manual of Steel Construction* (7th ed.) AISC, NY, 1970). This calculation points out once again, that seemingly small support loads, in this case, 15 psi, lead to sizeable support requirements and significant costs that must be considered in the overall design.

A wide spacing tends to lower costs by reducing the number of ring sets installed over a given length of excavation, while increasing cost and installation time because of greater difficulty in handling heavier steel. An important consideration in this regard is any tendency of the rock mass to ravel. Even falls of small rock pieces cannot be tolerated in a shaft, so lagging between rings is essential. Lagging may be wood, galvanized corrugated sheet metal, or some other structural material and may be placed between flanges or on top of the outside flange. Lagging is thin relative to length and experiences beam action under rock pressure. A common 3 in. thick piece of wood lagging 6 ft long under a distributed load of 15 psi experiences a tensile stress of 6480 psi which is well above design value for No. 1 Douglas Fir

parallel to the grain. (*National Design Specification for Wood Construction Supplement* (1997 ed.) American Forest & Paper Association). Double lagging (6 in.) would reduce the bending stress by 1/4 at the very most, to 1,620 psi which is marginal. Smaller sets closer together may be required to allow thick lagging to safely restrain small rock pieces detached from the rock mass.

3.3 Multiple naturally supported shafts

Often several shafts are needed to meet requirements for ventilation, personnel, and materials handling. Efficient placement of surface and underground facilities favors closely spaced shafts, while wide spacing favors lower stress concentration caused by interaction between excavations. Multiple shafts also raise an important rock mechanics or safety question concerning the most favorable *shaft row orientation*. Two new rules of thumb provide guidance to these design questions.

The first new rule states that *excavations should be separated by at least one “diameter” to reduce interaction to an acceptable amount*. Separation refers to wall to wall distance. In case of circular shafts “diameter” is obviously the unlined shaft diameter. An acceptable amount of interaction occurs when stress concentration is reduced to about 10% of that associated with a single excavation. For example, by separating circular shafts by one diameter, “1-D,” one hopes to reduce peak stress concentrations associated with a row of circular shafts to that of a single opening in the same preshaft stress field. In case of a single vertical shaft of circular cross section excavated in a stress field caused by gravity alone, circumferential stress concentration $K = 1 + (a/r)^2$. At 1-D away from the shaft wall $r = 3a$ and $K = 1.11$, so the stress concentration has decreased to within 11% of the preshaft value. In case of elliptical and rectangular shafts, “diameter” refers to a characteristic linear dimension of the section that reduces interaction to an acceptable amount. As a practical matter, the characteristic linear dimension of an ellipse or rectangular section for application of the 1-D rule is the *long* dimension of the opening.

The second new rule of thumb states that *the most favorable orientation of a row of identical excavations is with the row axis parallel to the major principal preexcavation compression*. Thus, the most favorable row orientation is the same as the most favorable orientation of a single opening.

Results from the mathematical theory of elasticity supply stress distributions and concentrations about multiple circular holes in a row. Results are in a variety of forms (analytical, numerical, computer code) and often require considerable additional work for practical assessment. Three reasonably assessable cases are of interest here: (1) two identical circular holes; (2) an infinite row of identical circular holes, and (3) two circular holes of different diameters.

Circular shafts in a row

Results from solutions to the problem of two circular holes side by side and an infinite row of circular holes allow one to evaluate the influence of one hole upon the other and to further assess the “1-D” rule of thumb for treating multiple identical openings as isolated, single openings. Figure 3.21 shows two identical circular holes side by side centered on the x -axis. Points of interest are holes sides (A and B), and top and bottom (C) where the stresses are at or near peak compressive and tensile values. Stress concentration at these points depends on the

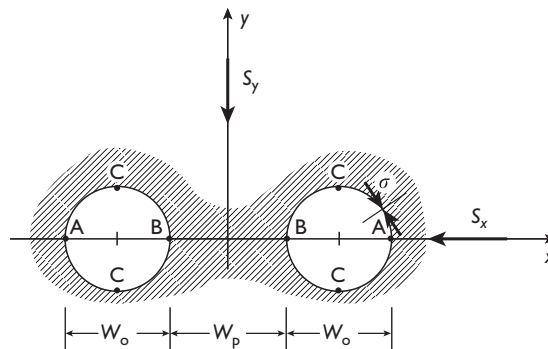


Figure 3.21 Two identical circular openings.

applied stresses (S_x , x = east and S_y , y = north) and on hole separation $W_o + W_p$ relative to hole size where W_o = width of opening (diameter) and W_p = width of pillar.

A plot of stress concentration at points A, B, and C as a function of the ratio W_p/W_o when a compression is applied *parallel* to the x -axis is shown in Figure 3.22(a). The reference stress $S_x = S_1$ and is the major principal stress (compression positive). When the holes are far apart, the stress concentration factors are -1.0 at points A and B and $+3.0$ at points C. When the holes are separated by one diameter (pillar width equals opening width, $W_p/W_o = 1$), the tension outside at A is reduced only slightly to about -0.9 , while the tension inside at B is reduced to about -0.6 . The compression at the top and bottom, point C, is reduced about 10% to 2.7 from 3.0. These results are in keeping with the rule of thumb stating holes more than one diameter apart experience only a slight interaction. As hole separation is reduced further from $W_p/W_o = 1$ and the holes nearly touch ($W_p/W_o = 0.2$), compression at C is reduced further. This effect occurs because one hole is in the “shadow” of the other when the major compression acts parallel to the axis of hole centers. The peak tension on the outside at A changes very little from -1.0 regardless of separation, while tension on the inside at B is reduced as separation is reduced.

Figure 3.22(b) shows the results when the major compression is *perpendicular* to the axis of hole centers. In this case $S_y = S_1$ and is the reference stress. The peak tension in this case occurs at the hole tops and bottoms, points C, and changes very little with hole separation. The peak compression occurs at B inside the holes and increases very rapidly from $+3.0$ as hole separation decreases below one diameter. Clearly, the most favorable orientation of the axis of hole centers in the case of two identical circular holes is parallel to the major compression when the choice is between parallel or perpendicular. The material between holes is shielded, shadowed, by the holes when the compression is parallel, but directly exposed when the applied compression is perpendicular to the axis of hole centers.

The two cases of uniaxial compression shown in Figure 3.22 may be combined under the principle of superposition. Table 3.6 presents the stress concentration factors for the two uniaxial cases. Thus in the hydrostatic compression case ($S_x = 1$, $S_y = 1$) with $W_p/W_o = 0.5$, stress concentration at points A, B, and C are 2.25, 2.89, and 1.74, respectively. Either applied

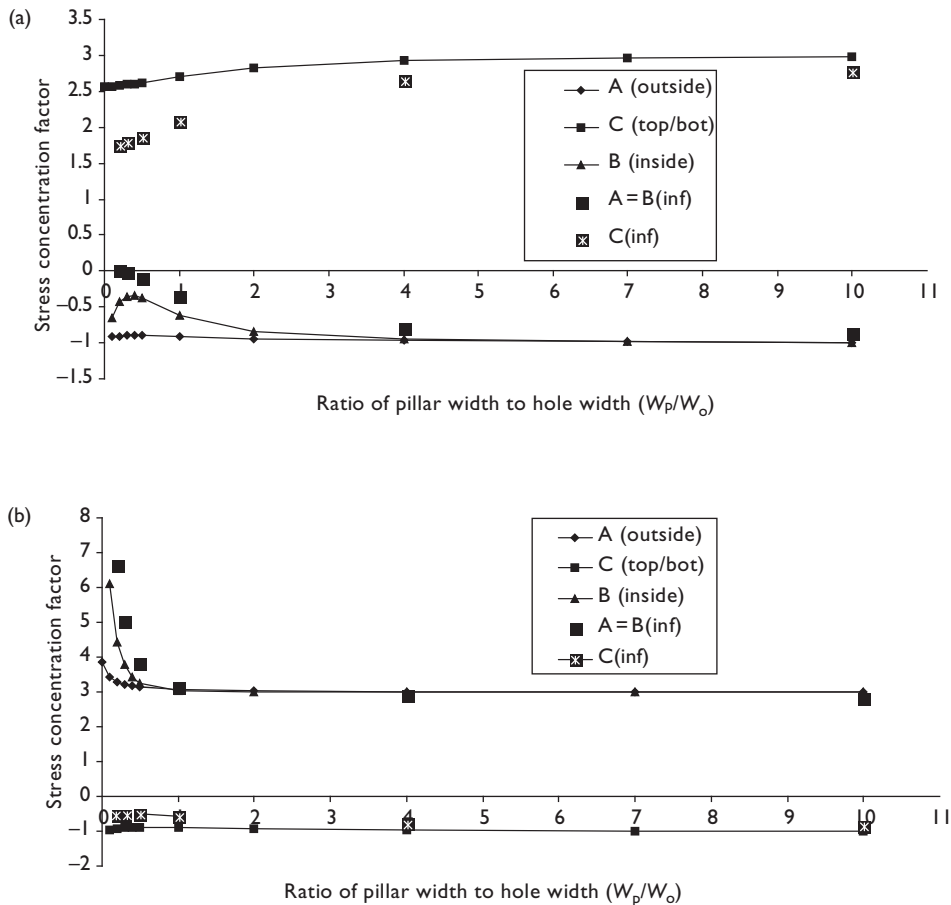


Figure 3.22 Stress concentration about two circular holes loading: (a) compression parallel to the row axis; $S_x = 1, S_y = 0$ and (b) compression perpendicular to row axis; $S_x = 0, S_y = 1$.

stress may be used as a reference stress. A single circular hole would experience a stress concentration of 2.0 at all points on the hole wall.

As another example of superposition, in case of a relatively low S_y [$S_y = (1/3)S_x$] and $W_p/W_o = 0.5$, the stress concentrations at hole sides, top and bottom (points A, B, and C) are, 0.15, 2.33, and 0.71, respectively (reference stress is S_x). When S_y is high relative to S_x [$S_x = (1/3)S_y$] and $W_p/W_o = 0.5$ as before, then stress concentrations at A, B, and C are 2.85, 3.14, and -0.01 , respectively (reference stress is S_y). If one of the applied stresses is a tension, for example, S_y and $S_y = -S_x$, then stress concentrations for $W_p/W_o = 0.5$ at A, B, and C are -4.05 , -3.64 , and 3.51 , respectively (S_x is the reference stress). If S_x is the tension, then the signs of the stress concentrations are reversed. If both of the applied stresses are tension, the stress concentration factors are the same as if both were compression.

Table 3.6 Stress concentration about two identical circular holes

W_p/W_o	S_x, S_y					
	$S_x = 1.0 \ S_y = 0.0$			$S_x = 0.0 \ S_y = 1.0$		
	A	C	B	A	C	B
0.0		2.569		3.869		
0.1	-0.918	2.568	-0.651	3.414	-0.958	6.106
0.2	-0.905	2.580	-0.423	3.289	-0.924	4.423
0.3	-0.899	2.593	-0.351	3.231	-0.905	3.768
0.4	-0.896	2.608	-0.346	3.185	-0.892	3.443
0.5	-0.896	2.623	-0.377	3.151	-0.884	3.264
1.0	-0.908	2.703	-0.609	3.066	-0.881	3.020
2.0	-0.94	2.825	-0.838	3.020	-0.92	2.992
4.0	-0.927	2.927	-0.948	3.004	-0.964	2.997
7.0	-0.987	2.970	-0.981	3.001	-0.985	2.998
10.0	-0.993	2.984	-0.991	3.000	-0.992	3.000

Example 3.25 Consider two vertical, circular shafts of the same diameter (7 m) in a preexcavation stress field defined by

$$S_{xx} = 2.5 \quad S_{yy} = 3.5 \quad S_{zz} = 10.0 \quad T_{yz} = 0.0 \quad T_{zx} = 0.0 \quad T_{xy} = -1.5$$

where units are MPa, x = east, y = north, z = up. Determine the best orientation of the shaft pair and the stress concentrations in plan view at points A, B, and C defined in Figure 3.21. Assume the shafts are separated by 1.4 m.

Solution: According to rule of thumb, the row axis of multiple openings should be parallel to the major preexcavation compression. The direction of the preexcavation principal stress may be found by the solution of

$$\begin{aligned} \tan(2\alpha) &= \frac{\tau_{xy}}{(1/2)(\sigma_{xx} - \sigma_{yy})} \\ &= \frac{T_{xy}}{(1/2)(S_{xx} - S_{yy})} \\ &= \frac{-1.5}{(1/2)(2.5 - 3.5)} \end{aligned}$$

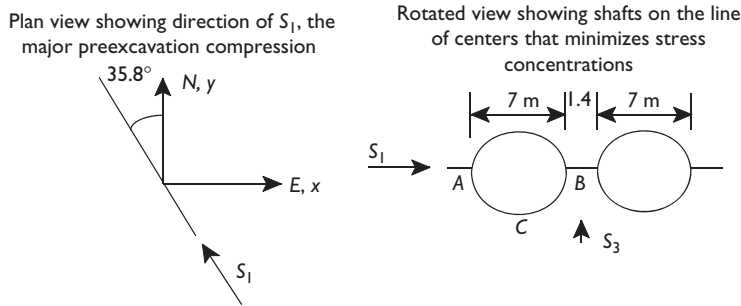
$$\tan(2\alpha) = 3.0$$

$\therefore$

$$2\alpha = 71.6^\circ, 251.6^\circ$$

$$\alpha = 35.8^\circ, 125.8^\circ$$

The proper choice of the two solutions may be made by measuring from the axis of the largest preexcavation normal stress, in this example, the y -axis, and observing the sign convention that positive angles are measured counterclockwise, as shown in the sketch.



Numerical evaluation of stress concentrations at points A, B, and C shown in the sketch requires magnitudes of the preexcavation principal stresses. These are

$$\left. \begin{matrix} S_1 \\ S_3 \end{matrix} \right\} = \left(\frac{S_{xx} + S_{yy}}{2} \right) \pm \sqrt{\left(\frac{S_{xx} + S_{yy}}{2} \right)^2 + T_{xy}^2}$$

$$= \left(\frac{2.5 + 3.5}{2} \right) \pm \sqrt{\left(\frac{2.5 + 3.5}{2} \right)^2 + (-1.5)^2}$$

$$\left. \begin{matrix} S_1 \\ S_3 \end{matrix} \right\} = 3.0 \pm 1.58$$

$\therefore$

$$S_1 = 4.58$$

$$S_3 = 1.42$$

Accordingly, the ratio $M = 1.42/4.58 = 0.31$. The pillar width to opening width ratio $W_p/W_o = 1.4/7.0 = 0.2$. With S_1 and S_3 in the roles of S_x and S_y in Table 3.6, the stress concentrations are

$$K_A = -0.905 + 0.31(3.289) = 0.11$$

$$K_B = -0.423 + 0.31(4.423) = 0.95$$

$$K_C = 2.580 + 0.31(-0.924) = 2.29$$

There is no tension and the peak compressive stress concentration is 2.29. A single shaft would have a peak stress concentration of 2.69, about 14% higher in compression, and a tensile stress concentration of -0.07 .

Example 3.26 Consider the data in Example 3.25 and suppose the major compression is perpendicular to the shaft pair axis rather than parallel. Determine the stress concentrations in this orientation.

Solution: As before, the ratio $M = 1.42/4.58 = 0.31$. The pillar width to opening width ratio $W_p/W_o = 1.4/7.0 = 0.2$. With S_1 and S_3 in now in the roles of S_y and S_x in Table 3.6, the stress concentrations are

$$K_A = 3.289 + 0.31(-0.905) = 3.01$$

$$K_B = 4.423 + 0.31(-0.423) = 4.29$$

$$K_C = -0.924 + 0.31(2.58) = -0.124$$

Table 3.7 Stress concentration about an infinite row of identical circular holes

W_p/W_o	S_x, S_y			
	$S_x = 1.0$	$S_y = 0.0$	$S_x = 0.0$	$S_y = 1.0$
	A = B	C	A = B	C
0.2	-0.000	1.753	6.642	-0.535
0.3	-0.014	1.788	5.036	-0.518
0.5	-0.101	1.855	3.824	-0.495
1.0	-0.365	2.077	3.159	-0.555
4.0	-0.762	2.648	2.876	-0.795
10.0	-0.852	2.777	2.827	-0.851
∞	-1.000	3.000	3.000	-1.000

The tensile stress concentration factor has magnitude 0.124, which was absent in example 3.25. The peak compressive stress concentration is 4.29 which is 87% greater than when the shaft row is oriented parallel to the preexcavation principal stress (compression positive).

As more holes are added to the two-hole row, stress concentrations change. Eventually, the addition of one more identical circular hole results in a negligible change to holes in the interior of the row. When the row is of indefinite (infinite) extent, stress concentrations about each hole are repeated from hole to hole. Because of the periodicity of the array, stress concentration at points A and B are the same. Stress concentrations at the hole sides (A, B), top and bottom (C) are given in Table 3.7.

Comparison of two-hole row data with the infinite row data points in Figure 3.22 shows that loading *parallel* to the row axis leads to *lower* compressive and tensile stress concentrations in the infinite row case. This result indicates that the shadow effect is increased as holes are added to the row. In the case of loading *normal* to the row axis, compressive stress concentrations are generally *higher* while tensile stress concentrations are lower in the infinite row case. This effect is accentuated as the distance between holes is decreased. The generally lower stress concentrations under parallel loading lends further support to the rule of thumb that states the most favorable row orientation is parallel to the major principal stress (compression).

Superposition applies, so the results in Table 3.7 may be used to determine stress concentrations for combined loadings. For example, under hydrostatic compression ($S_x = S_y$) at $W_p/W_o = 1.0$, stress concentrations at A (and B) and C are 2.794 and 1.552, respectively. If $S_y = (1/3)S_x$, then stress concentration at A (and B) and C are both compressive, 3.037 and 0.137, respectively.

As the distance between holes ($W_o + W_p$) is decreased in the case of an infinite row of circular holes loaded *perpendicularly* to the line of hole centers, not only does the compressive stress concentration increase at the hole sides, but the average stress increases as well. Figure 3.23 shows the distribution of σ_{xx} (east stress) and σ_{yy} (north stress) along the x -axis between holes. Even though there is no applied stress in the east or x -direction, a stress develops that increases from zero towards the interior of the material between holes. The interior portion of the material between holes is confined by this x -direction normal stress buildup.

The average north (y -direction) pillar stress S_p may be calculated exactly from equilibrium of forces for holes in an infinite row. Thus, $S_p W_p = S_n (W_o + W_p)$, so $S_p = S_n / (1 - R)$ where S_n is

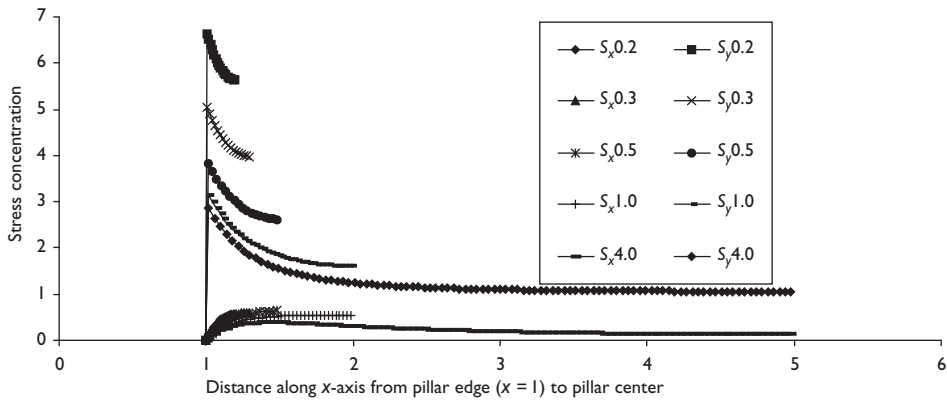


Figure 3.23 Stress concentration between two circular holes loaded perpendicularly to the line of centers (x = east = parallel to line of centers, y = north = perpendicular to line of centers). Plots start at pillar edges and end at pillar centers. Pillar to opening width ratio W_p/W_o ranges from 0.2 to 4.0.

the preexcavation normal stress acting in the north direction and $R = W_o/(W_o + W_p)$, ratio of opening width to hole spacing. In the cases shown in Figure 3.23, $W_p/W_o = 4, 1, 0.5, 0.3$, and 0.2 , $S_p/S_n = 1.25, 2.00, 3.00, 4.33$, and 6.00 , respectively. These averages may be compared with the corresponding peaks: 2.88, 3.16, 3.82, 5.03, and 6.64, respectively (Table 3.7). As the distance between holes is decreased both peak and average stress increase, while the difference decreases. At $W_p/W_o = 0.2$, the difference between peak and average stress is only about 10%.

Shaft pillar safety

A shaft pillar factor of safety based on *average pillar stress* S_p rather than the peak stress is easy to compute and if too low indicates the need for more detailed analysis including determination of peak stresses. There is another type of shaft pillar about mine shafts. This second type of shaft pillar protects shafts from damage that may be induced by extracting ore too close to the shafts. Concern here is with pillars formed by multiple shafts. These pillars should be dimensioned to prevent damaging interaction between shafts. As always, geologic structure, “joints,” need to be taken into account. The rock mass between shafts of equal size in an infinite row of shafts forms pillars that experience an average stress $S_p = S_n(W_o + W_p)/W_p$ where S_n is the horizontal preshaft stress acting normal to the line of centers and S_p is horizontal post-shaft stress (average). An area extraction ratio R may be defined as the ratio of the area excavated to a total area of excavation and pillar, that is, $R = W_o/(W_o + W_p)$, measured along the line of centers (horizontally) and one unit of distance vertically. The desired safety factor based on average compressive stress is then

$$FS_{ave} = \frac{C_p}{S_p} = \frac{C_o(1 - R)}{S_n} \quad (3.24)$$

where C_p is shaft wall compressive strength and C_o is rock mass unconfined compressive strength. If a size effect on strength is considered important, then C_o must include such an effect. Equation (3.24) may be used to evaluate a safety factor based on average pillar stress or to evaluate pillar size associated with a specified safety factor.

Example 3.27 Suppose a safety factor of 3.0 is desired in a rock mass with unconfined compressive strength of 4,800 psi where the preshaft stress is hydrostatic and 1,200 psi. A row of 25 ft diameter circular shafts is planned. Determine the shaft separation needed to meet the required safety factor.

Solution: By definition

$$FS_p = \frac{C_p}{S_p} = \frac{C_o}{S_n/(1-R)}$$

where no size effect on strength is assumed. This equation may be solved for $1-R$,

$$\begin{aligned} 1-R &= \frac{S_n(FS_p)}{C_o} \\ &= \frac{(1,200)(3)}{4,800} \end{aligned}$$

$$1-R = 0.75$$

The ratio R may also be computed from the shaft geometry, that is,

$$\begin{aligned} 1-R &= \frac{W_p}{W_o + W_p} \\ \therefore 0.75W_o &= W_p(1-0.75) \\ \therefore W_p &= 3W_o = (3)(25) = 75 \text{ ft} \end{aligned}$$

This pillar size separates the shafts of three diameters, so the shafts are effectively isolated and spaced on 100 ft centers. Single shafts would have a stress concentration of 2.0 in compression and tension would be absent. A safety factor based on peak stress at the shaft walls would be $FS = C_o/K_c S_1 = 4,800/(2)(1,200) = 2.0$. Thus, a design layout based on averages, although seemingly conservative at $FS = 3.0$, masks a danger of potentially unsafe peak stress conditions at excavation walls. In this case, the shaft wall safety factor is 2.0 and thus indicates a safe condition with respect to rock mass failure.

Because the safety factor based on average stress separates the shafts by more than one diameter, the spacing could be reduced to one diameter without significantly increasing peak stress. At a one diameter pillar width, $W_p = 25$ ft, $R = 0.5$, and $FS_{ave} = (4,800)(0.5)/1,200 = 2.0$. The safety factor at the shaft wall would be only slightly higher.

Example 3.28 According to the analysis summarized in Figure 3.23, the difference between peak and average stress concentration about two identical circular shafts is about 10% when $W_p/W_o = 0.2$. At this ratio, the peak stress concentration is 6.64; the average stress

concentration is approximately 6.00. Estimate the rock mass strength needed under these circumstances to achieve a safety factor of 3 in a 1,200 psi hydrostatic stress field.

Solution: Because of the closeness of average to peak stress concentration, one may use either, therefore

$$\begin{aligned} \text{FS}_c &= \frac{C_o}{\sigma_c} \\ &= \frac{C_o}{K_c S_1} \\ 3 &= \frac{C_o}{(6.64)(1,200)} \\ \therefore \\ C_o &= 23,900 \text{ psi} \end{aligned}$$

Use of unconfined compressive strength tends to underestimate the safety factor based on average stress. The reason is the buildup of horizontal stress in the pillars, as shown in Figure 3.23. At a pillar width equal to 1-D, W_o , the central portion of a pillar is confined by a stress almost equal to the preexcavation horizontal stress. If one adopts a Mohr–Coulomb rock mass failure criterion, then $C_p = C_o + (C_o/T_o)p$ where p = horizontal confining stress or “pressure.” For example, even if p were only one-half the preexcavation stress in a hydrostatic stress field of 1,200 psi, the rock mass compressive strength $C_p = 4,800 + (10)(600) = 10,800$ psi where a reasonable ratio of unconfined compressive to tensile strength of 10 is assumed and the unconfined compressive strength is 4,800 psi. In this case, other factors being equal, the pillar core has a safety factor that is more than double an estimate based on unconfined compressive strength.

A technically sound and economically viable shaft pillar defense strategy for circular shafts in a row allows for relatively close spacing (pillar width equal to shaft diameter), based on the extraction ratio Formula (3.24), with provision of shaft wall safety against peak stress failure. The latter provision almost certainly requires rock mass reinforcement that prevents falls of small rock and the start of progressive failure that could lead to pillar collapse. As in the case of single circular shafts, if the shaft wall is safe, then the rock mass beyond is also safe.

Formula (3.24) does not depend on shaft shape and may therefore be applied to other shaft sections in an infinite row of identical vertical shafts. Figure 3.24 illustrates before and after stress distributions normal to the line of centers of rectangular shafts. The x and y axes coincide with preshaft principal directions. Of course, the most favorable orientation would align the shaft section axis and row axis parallel to the major principal stress before excavation. In any case, the preshaft normal stress $S_n = S_y$ which is a principal stress. The post-shaft row pillar stress $S_p = S_n(1 - R)$ where R is the area extraction ratio $W_o/(W_o + W_p)$ and W_o is the section dimension *parallel* to the x -axis. Opening dimension in the y -direction, “height” of the opening is H_o which is also “height” of the pillar H_p . Of course, these distinctions were unnecessary in the case of circular shafts in a row.

Possibilities of failure along faults and major joints of considerable continuity in shaft pillars also need to be considered. Numerous small joints and fractures are considered in estimation of rock mass strength and corresponding safety factor calculations. Analysis of

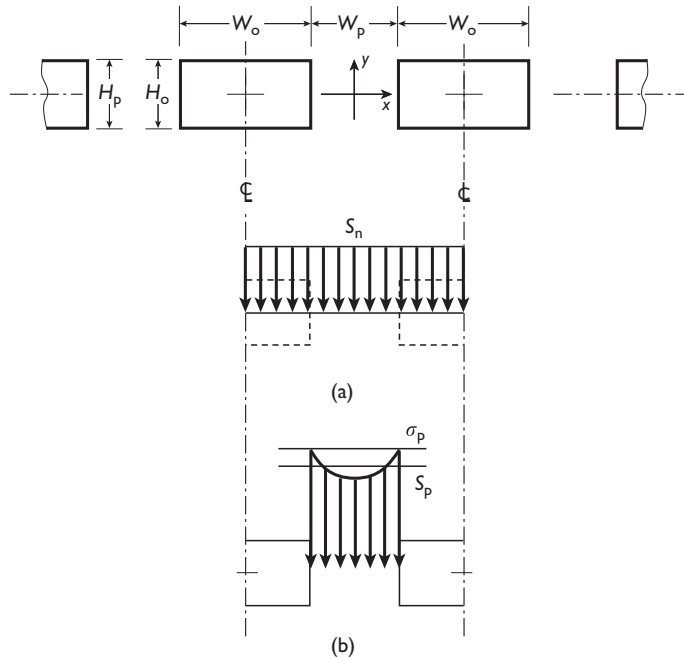


Figure 3.24 Pillar stress distribution normal to the line of centers before and after excavation: (a) before and (b) after.

pillar joint safety based on average stresses proceeds in much the same manner as wall joint safety discussed in the section on *Shaft Wall Strengths*. Here, “compass” coordinates, xyz are aligned with preshaft principal directions and with the understanding that the shaft row axis or line of centers is parallel to one of these principal directions. The best choice is with x parallel to S_1 . Stresses with respect to xyz are: $S_{xx} = 0$, $S_{yy} = S_p$, and $S_{zz} = S_v$; shear stresses $T_{xy} = T_{yz} = T_{zx} = 0$, and the notation implies average stress. The next step is to compute the direction cosines of the normal to a joint plane of interest (n_x , n_y , n_z). Components of the stress vector (T_x , T_y , T_z) can then be calculated and finally the normal and shear stresses (S_j , T_j) acting on the joint plane may be obtained. The pillar joint safety factor, assuming Mohr–Coulomb failure, is then

$$FS_j = \frac{\sigma_j \tan(\phi_j) + c_j}{\tau_j} \quad (3.25)$$

Example 3.29 Consider a row of 18 ft diameter vertical shafts spaced on 45 ft centers at a depth of 3,200 ft where the vertical preexcavation stress is 3,600 psi and the horizontal stress is 1,800 psi. A small fault with dip direction 30° and dip 30° transects a potential shaft pillar. The gouge in the fault zone has an angle of internal friction of 28° and a cohesion of 20 psi. If the shaft excavation plan is realized, is there a danger of fault slip, that is, what is the post-excavation fault safety factor?

Solution: With the x -axis along the line of centers, the post-shaft stresses may be estimated as $S_{xx} = 0$, $S_{yy} = S_p$, $S_{zz} = S_v$, and $T_{xy} = T_{yz} = T_{zx} = 0$. The area extraction ratio $R = W_o/(W_o + W_p) = 18/45 = 0.4$, and $S_p = S_v/(1-R) = 1,800/(1 - 0.4) = 3,000$ psi. Direction cosines of the fault plane normal are:

$$\begin{aligned}n_x &= \sin(\delta)\sin(\alpha) = \sin(60^\circ)\sin(30^\circ) = (\sqrt{3}/2)(1/2) = \sqrt{3}/4 \\n_y &= \sin(\delta)\cos(\alpha) = \sin(60^\circ)\cos(30^\circ) = (\sqrt{3}/2)(\sqrt{3}/2) = 3/4 \\n_z &= \cos(\delta) = \cos(60^\circ) = 1/2\end{aligned}$$

Tractions on the fault plane are then

$$\begin{aligned}T_x &= (0)(\sqrt{3}/4) + (0)(3/4) + (0)(1/2) = 0 \\T_y &= (0)(\sqrt{3}/4) + (3,000)(3/4) + (0)(1/2) = 2,250 \\T_z &= (0)(\sqrt{3}/4) + (0)(3/4) + (3,600)(1/2) = 1,800\end{aligned}$$

The normal stress on the fault plane is

$$N_j = (0)\left(\sqrt{3}/4\right) + (2,250)(3/4) + (1,800)(1/2) = 2,588$$

and the shear stress is

$$S_j = [(0)^2 + (2,250)^2 + (1,800)^2 - (2,588)^2]^{1/2} = 1,267$$

The fault plane factor of safety according to a Mohr-Coulomb strength criterion is then

$$FS_j = \frac{(2,588)\tan(28^\circ) + 20}{1,267} = 1.10$$

which is close to failure and would be cause for concern.

Of interest is the fault plane safety factor before excavation. From a similar analysis, $FS_j = 1.56$, which indicates excavation would tend to induce fault plane shearing.

However, the pillar is 27 ft wide, more than one diameter, so consideration of confining pressure effects is reasonable. There are three effects, one that increases the normal stress after excavation, one that reduces the post-excavation shear stress, and one that increases strength. The strength effect increases compressive strength under confining pressure, but does not affect cohesion or angle of internal friction. If a cautious view of the confining pressure effects is taken by assuming the post-excavation normal stress in the x -direction in the pillar core is 900 psi, just one-half of the preexcavation horizontal stress, the effect is to increase the post-excavation normal stress to 2,334 psi, while reducing the shear stress to 889 psi. The fault plane safety factor after taking confining pressure into account is 1.42, which is somewhat less than the preexcavation safety factor but significantly greater than the unconfined safety factor calculation that lead to an estimate of 1.10. These results indicate that the core of the pillar would be safe with respect to fault plane slip. Slip at the shaft walls would still need to be considered in conjunction with loading by the circumferential stress concentrated at the shaft wall and the vertical stress, as in the case of a single shaft.

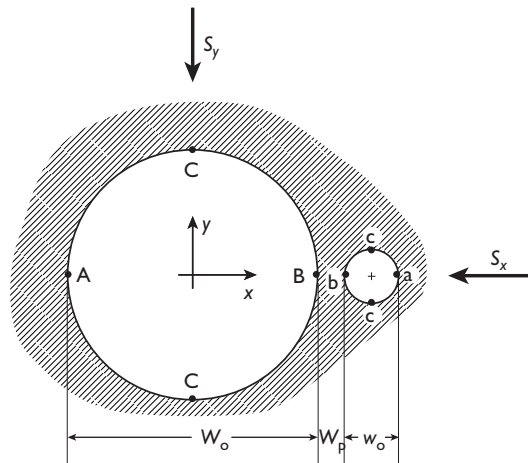


Figure 3.25 Circular excavations of different diameters.

Two circular shafts of different diameter

When two circular holes or shafts of *different* diameters are excavated in the same neighborhood, one may suppose that the large hole will significantly influence stress concentration near the small hole, while the small hole will have little influence on stress about the big hole. A practical example would involve a small diameter borehole for utilities excavated adjacent to a larger service shaft. Figure 3.25 illustrates the situation where W_o is width (diameter) of the large opening, w_o is the small opening width (diameter), and W_p is the width of the pillar between.

Peak stress concentration data for two diameter ratios W_o/w_o of 5 and 10 and for different pillar width to large hole width (diameter) W_p/W_o ratios are shown in Figure 3.26. The lower case legend with solid symbols pertains to the small hole; the upper case legend with open symbols pertain to the large hole. Loading is *parallel* to the line of hole centers in Figure 3.26 (a) and *perpendicular* in Figure 3.26(b). If either hole were isolated, then under compressive stress loading, peak compressive stress concentrations of 3.0 would occur at points C and c. Peak tensions of -1.0 would occur at A and B, and a and b (Figure 3.25). When loading is normal to the line of hole centers, the peak compression and tension would occur at points A, B, a, b, and at points C, c, respectively, provided the holes were isolated from each other. Data in Figure 3.26 show that the large diameter hole is largely unaffected by the small hole even when the separation or pillar width is less than the small hole diameter (about $W_p/W_o < 0.1$ in Figure 3.25). The large hole tends to shadow the small hole and reduces the peak small hole compression in the vicinity of points c with reduction in W_p/W_o . However, peak tension on the inside of the small hole at b tends to increase rapidly over -1.0 with close approach ($W_p/W_o < 0.1$) of the small hole to the big hole. High tensile stress in the pillar could pose a serious threat to stability of both excavations.

Data for the case when the load is applied *perpendicular* to the line of hole centers is shown in Figure 3.26(b). Peak tensile stress concentrations about the big hole occur at points C and are much the same as if the hole were isolated, that is, $K_t = -1.0$. Peak compressive

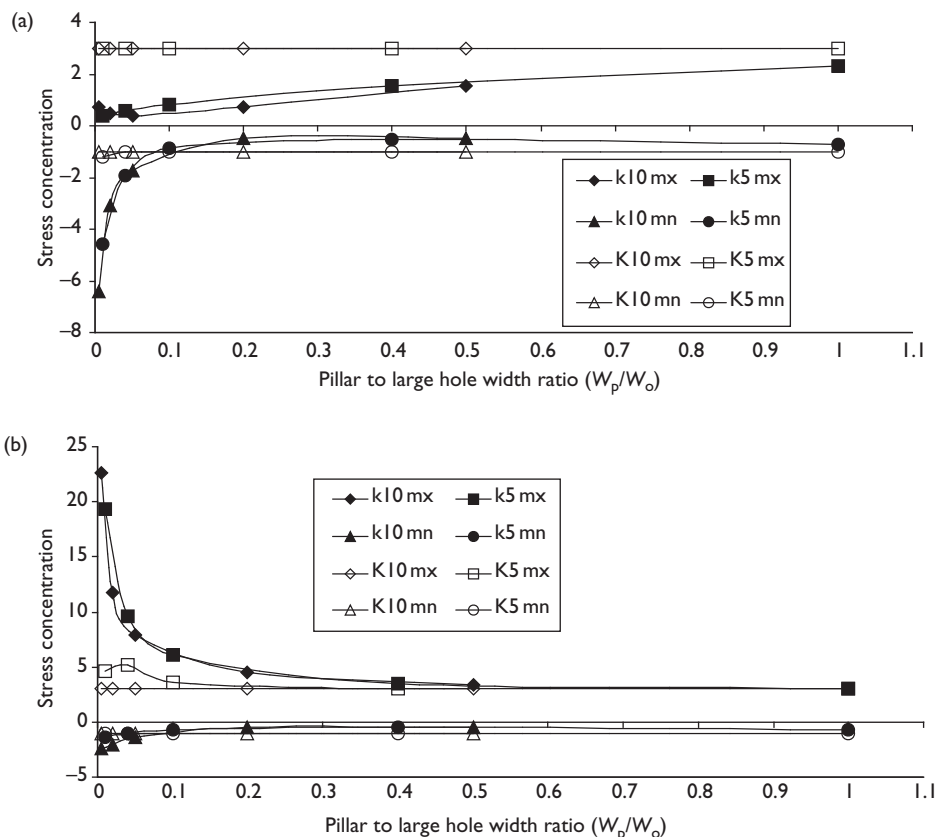


Figure 3.26 Maximum and minimum stress concentration about two circular holes of unequal diameters of 5 and 10. Lower case solid symbols pertain to the small hole. Upper case open symbols pertain to the large hole. Large hole diameter W_o is 5 and 10 times small hole diameter w_o . Rock distance between holes is W_p (after Haddon, 1967). (a) Parallel and (b) perpendicular to the line of centers.

stress concentration K_c is also largely unaffected by the small hole until the holes are quite close, $W_p/W_o < 1.0$. Peak compression at the wall of the large hole occurs on the inside at B. There is a hole size effect in that the case of the smaller big hole ($W_o/w_o = 5$), compressive stress increases to about 5.0 from 3.0 and then decreases with even closer approach of the small hole. This effect would be amplified as the big hole diameter is decreased. Indeed, when hole diameters are equal and $W_p/W_o = 0.1$, the compressive stress concentration is 6.11 (Table 3.6). Peak tension and compression at the wall of the small hole increase with decreasing separation. The increase accelerates with close approach as W_p/W_o decreases below 0.1, as seen in Figure 3.26(b).

The data for two circular holes of unequal diameter show that stress concentrations about the large hole are not significantly affected by the small hole, while stress concentration about the small hole is definitely influenced by the large hole. Intuitively, one might suppose that the

1-D rule applies using the large hole diameter and, in fact, if the two holes were separated by a pillar width $W_p = W_o$, then the data in Figure 3.26 make clear that the two holes are effectively isolated with negligible interaction. When isolated, maximum and minimum stress concentrations are 3.0 and -1.0 . However, such separation is too conservative; a lesser separation is possible. Superposition applies and in practice, peak stress concentrations will be less than the uniaxial load case shown in Figure 3.26. Using the small hole diameter for the 1-D rule would be risky. A reasonable compromise and guide would be to separate circular holes of unequal diameter by $W_p = 0.5 W_o$, when the large hole is more than five times the diameter of the small hole. Otherwise the 1-D rule should be applied using the large hole diameter in case of long life openings such as shafts. The rule of thumb for the most favorable orientation with the line of hole centers parallel to the row axis still applies as the data in Figure 3.26 indicate.

Consider a 25 ft diameter vertical shaft and 5 ft diameter borehole excavated in a hydrostatic stress field ($M = 1$) at a depth of 3,600 ft in a preexcavation stress field caused by gravity alone and where the horizontal stress is 1,200 psi and rock mass strength is 4,800 psi. If the holes are effectively isolated according to the 1-D rule (25 ft apart, $W_p = 25$) then $W_p/W_o = 1.0$. Tension is then absent and the peak compressive stress concentration is 2.0. The safety factor with respect to compression is then 2.0. Superposition of stress concentration data from Figure 3.26 allows for closer spacing. At $W_p/W_o = 0.4$, tension is still absent, while the peak compression stress concentration appears at the inside of the small hole and is almost 3.0. The holes are 10 ft apart ($W_p = 10$) and the associated safety factor is 1.33.

Elliptical shafts in a row

When the holes in a row are not circular, two orientation questions arise: (1) the best orientation for the individual holes and (2) the best orientation for the row of holes. The first question may be tentatively answered by applying a rule of thumb for isolated, single openings that states the best orientation is to align the long axis of the hole with the direction of the major principal compressive stress. Whether this applies for multiple noncircular holes in a row when interaction occurs is an open question. The second question calls into play the rule of thumb that states the most favorable row of holes orientation is with the row axis, the line of centers, also parallel to the major compression. Whether this rule applies to noncircular holes is also an open question. Analytical or closed form solution to this problem is not known. However, a purely numerical approach allows for quantitative evaluation.

A comparison of the analytical and numerical results in the case of two identical circular holes excavated in a hydrostatic stress field is presented in Figure 3.27. Numerical data in Figure 3.27 were obtained using the finite element method. The agreement between the two methods is excellent and indicates reliability of the finite element results. Data in Figure 3.27 may also be computed using superposition and Table 3.6 data for points A and B outside and inside the two holes.

Figure 3.28 shows four combinations of a pair of elliptical holes under uniaxial loading. The ratio of the major to the minor axes of the ellipses is two; the ratio of hole separation to hole width (W_p/W_o), both measured along the x -axis is 0.5, so interaction is expected. The results are summarized in Table 3.8. These limited results support the applicability of the

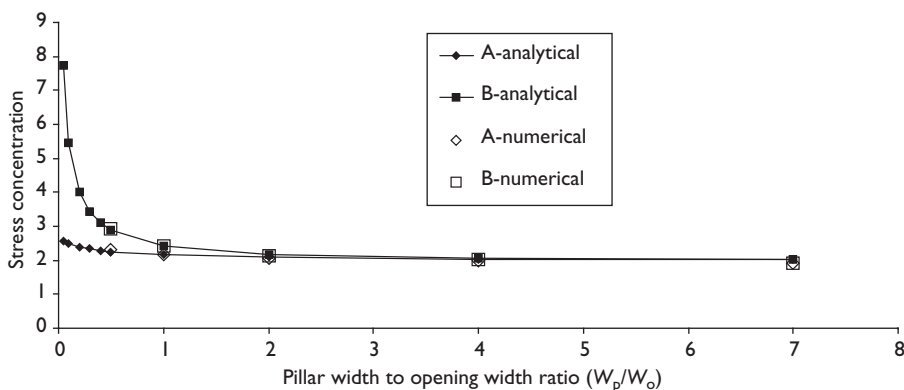


Figure 3.27 Comparison of analytical and numerical results and stress concentration in the case of two identical circular holes excavated in a hydrostatic stress field ($M = 1$).

combined rules of thumb for single openings and rows of openings that state the long axis of the hole and the row should be parallel to the major compression.

Inspection of Table 3.8 shows that the greatest compression (5.56) occurs inside ellipses (B) when the applied compressive stress is *normal* to the hole and row axes – combination (c) in Figure 3.28. The tension in this case (at C) is also very nearly equal to the greatest tension (–1.03). The smallest tension and compression (–0.64, 1.91, respectively) occur when the applied stress is *parallel* to the hole and row axes – combination (a) – at points B and C, respectively. For comparison, if the ellipses were widely separated and effectively isolated then loading parallel to the hole axis leads to compressive stresses of 2.0 at C and tensile stress of –1.0 at A and B. The noticeable reduction in tension at points B between the ellipse pair indicates a significant shadow effect in combination (a) of Table 3.8.

A single ellipse with an axis ratio of two under an applied stress acting *normal* to the long axis of the hole would experience stress concentrations of –1.0 at the hole top and bottom (C) and 5.0 at the hole sides. By comparison, the ellipse pair in case (c) shows a tension of –1.02 and compressions 5.16 and 5.56 outside and inside the holes (points A and B), respectively, and thus an interaction amounting to about a 10% increase in peak compression occurs in this unfavorable case.

Superposition applies, so one may use the data in Table 3.8 for combined S_x and S_y loading and $W_p/W_o = 0.5$. For example in the hydrostatic case when $S_x = S_y$, and the ellipse axes are parallel to the x -axis, stress concentrations at A, B, and C are 4.33, 4.92, and 0.89, respectively. When the ellipse axes are parallel to the y -axis, stress concentrations at A, B, and C are 1.19, 2.00, and 3.23, respectively. In either case tensions are absent under compressive hydrostatic loading.

Table 3.9 shows stress concentrations for the four combinations of hole and infinite row axis orientation relative to the applied compression used in the two-ellipse cases (Table 3.8). The main trends are similar to the infinite circular hole row trends: (1) lower tensile and

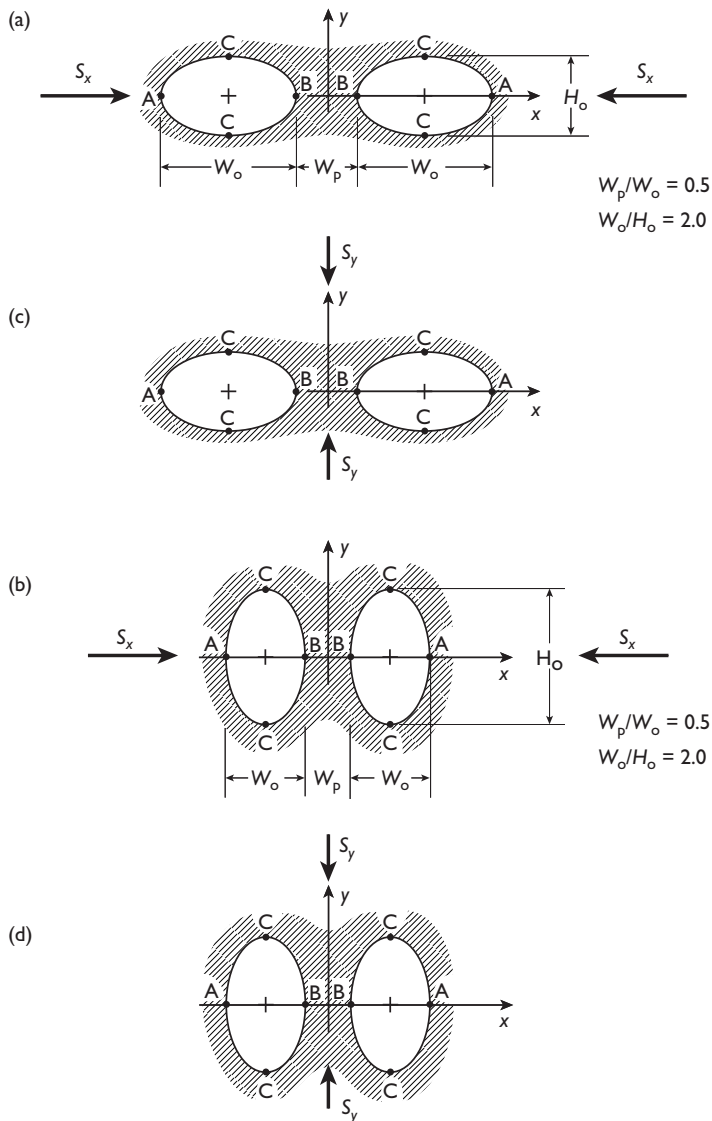


Figure 3.28 Four configurations of two elliptical excavations.

compressive stress concentration when the load is parallel to the row axis, and (2) lower tensile, but higher compressive stress concentrations when the load is normal to the row axis. Again the most favorable combination is with hole and row axes parallel to the major compression; the least favorable combination is with hole and row axes normal to the major compression.

Table 3.8 Stress concentration about two identical elliptical holes ($W_p/W_o = 0.5$)

Combination	S_x, S_y					
	$S_x = 1.0 \ S_y = 0.0$			$S_x = 0.0 \ S_y = 1.0$		
	A	C	B	A	C	B
(a) I-I	-0.83	1.91	-0.64			
(b) 0-I	-1.03	4.09	-0.55			
(c) 0-0				5.16	-1.02	5.56
(d) I-0				2.22	-0.86	2.55

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal.

Table 3.9 Stress concentration about an infinite row of identical elliptical holes

Combination	S_x, S_y			
	$S_x = 1.0$	$S_y = 0.0$	$S_x = 0.0$	$S_y = 1.0$
	A = B	C	A = B	C
(a) I-I	-0.34	1.55		
(b) 0-I	-0.00	2.19		
(c) 0-0			5.58	-0.66
(d) I-0			3.81	-0.31

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal.

Rectangular shafts in a row

An elliptical section is useful for illustrating the concept of stress concentration about single and multiple holes and transition from analytical to numerical solutions. However, a rectangular section is a much more practical shape. Again, two related questions arise. One concerns orientation of individual holes; the second concerns orientation of a row of holes. Both orientations are with respect to the direction of the major preexcavation compression S_1 . Four combinations of identical rectangular holes are shown in Figure 3.29; these combinations are the same used in analyses of elliptical holes. The long dimension of all holes is always twice the short dimension and the ratio of hole separation to hole width (W_p/W_o) is always 0.5 as measured along the x -axis. Thus, the openings are relatively close together and interact significantly, as was the case for elliptical holes.

Table 3.10 shows stress concentration factors for two cases of uniaxial loading of just *two* rectangular holes with different hole and row axis alignment relative to the load direction. Peak stress concentration occurs at corners in all cases. When loading is parallel to the x -axis (cases *a* and *b*), peak stress occurs on the outside corners (D_o); when loading is parallel to the y -axis (cases *c* and *d*), peak stress occurs on the inside corners (D_i). The most favorable case with respect to peak compressive stress concentration (under compressive loading) is case (*a*)

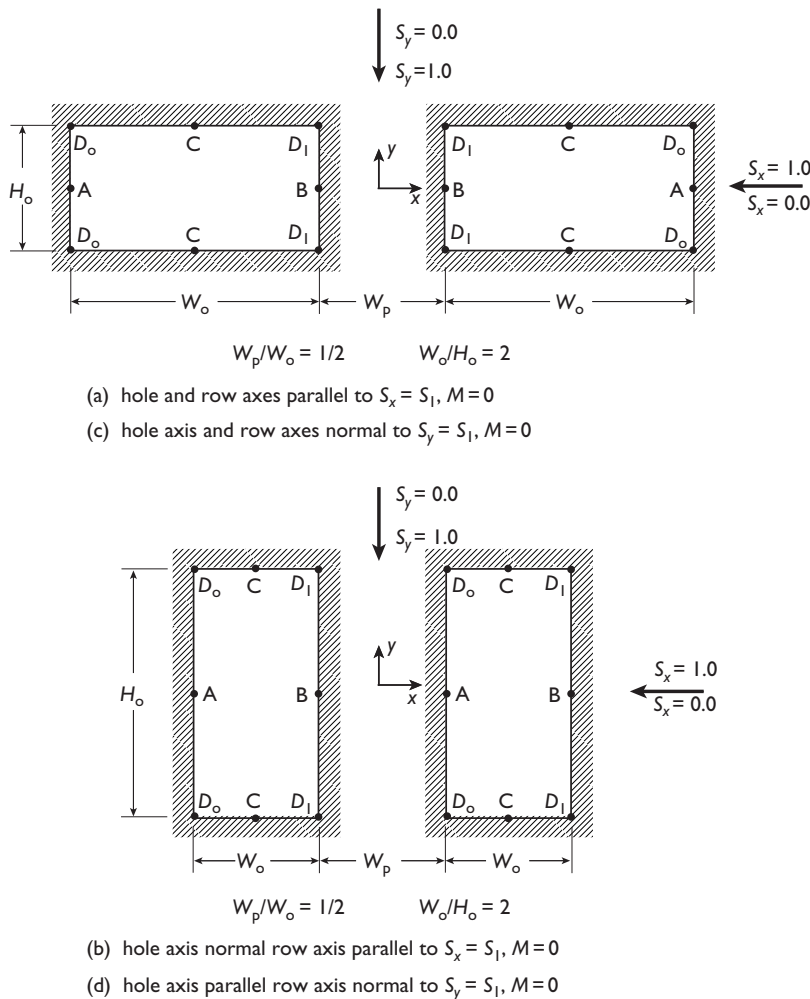


Figure 3.29 Four configurations of two identical rectangular excavations.

with hole and row axes parallel to the direction of the major preexcavation compression S_1 (S_x). The least favorable case is (c) with both axes normal to the applied stress. The same conclusions are obtained relative to tension: case (a) is most favorable; case (c) is least favorable.

Table 3.11 shows stress concentrations at the ends of hole semi-axes (A , B , C) and at hole corners (D_o , D_i) in four cases of an *infinite* row of identical rectangular holes. As in cases of two rectangular holes, peak compressive stress concentration always occurs at the corners of a rectangular section. Tension, when present, occurs “under the load,” that is, at the ends of the semi-axis that is parallel to the major compression. Data in Table 3.11 show that loading *normal* to the row axis or line of centers (c) is quite unfavorable because of high compressive stress concentration at corners (D_o , D_i) and high tensile stress at midsides (C). Stress concentration at corners is greatly reduced when the row axis is oriented *parallel* to the major preexcavation compression; tension is also reduced.

Table 3.10 Stress concentration about two identical rectangular holes

Case	S_x, S_y									
	$S_x = 1.0 \ S_y = 0.0$					$S_x = 0.0 \ S_y = 1.0$				
	A	C	B	D_0	D_1	A	C	B	D_0	D_1
(a) 1-1	-0.81	1.36	-0.42	2.84	2.01					
(b) 0-1	-1.02	2.04	-0.39	4.81	2.27					
(c) 0-0						2.50	-1.05	2.85	5.58	6.66
(d) 1-0						1.64	-0.90	2.07	3.32	3.40

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal.

Table 3.11 Stress concentration about an infinite row of identical rectangular holes

Combination	S_x, S_y					
	$S_x = 1.0 \ S_y = 0.0$			$S_x = 0.0 \ S_y = 1.0$		
	A = B	C	$D_0 = D_1$	A = B	C	$D_0 = D_1$
(a) 1-1	-0.23	1.06	1.81			
(b) 0-1	0.00	1.07	1.57			
(c) 0-0				2.94	-0.84	7.30
(d) 1-0				3.00	-0.77	5.13

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal.

The main trends as holes are added to the two rectangular hole cases are similar to those for elliptical holes: (1) lower tensile and compressive stress concentration when the load is *parallel* to the row axis (cases *a* and *b*) because of increased “shadow” effects and (2) lower tensile, but higher compressive stress concentrations when the load is *normal* to the row axis (cases *c* and *d*).

Interestingly, the condition of uniaxial loading (*b*), with the hole axis normal to the load direction and the row axis parallel to the load direction, is slightly more favorable than (*a*), which was more favorable for elliptical holes. In consideration of a more realistic *in situ* stress state than uniaxial ($M = 0$), for example $M = 1/3$ or $M = 1$, case (*a*) with both axes (hole and row) parallel to S_1 , (*b*) is still more favorable. This trend continues at larger separations (greater W_p). To see these results, one may use superposition of data from Table 3.11 for midside points *A*, *B*, and *C* and by numerical calculation for corners (D_0 and D_1). In case $M = 1/3$ with hole and row axes *parallel* to $S_1 = S_x$ and $S_3 = S_y = (1/3)S_1$, stress concentrations at *C* and *A(B)* and are 0.78 and 0.75, respectively. Tension is absent. Peak compressive stress concentration still occurs at the corners (D_0 , D_1). However, direction and location of peak corner compression varies somewhat from case to case, so superposition does not apply at corners. A separate series of calculations leads to the data in Table 3.12 for $M = 1/3$. Results for hydrostatic case, $M = 1$, are given in Table 3.13.

Table 3.12 Stress concentration about an infinite row of identical rectangular holes

Combination	S_x, S_y					
	$S_x = 1 \ S_y = 1/3$			$S_x = 1/3 \ S_y = 1$		
	$A = B$	C	$D_0 = D_1$	$A = B$	C	$D_0 = D_1$
(a) 1-1	0.75	0.78	3.49			
(b) 0-1	1.00	0.81	2.69			
(c) 0-0				2.86	-0.49	7.62
(d) 1-0				3.00	-0.42	5.24

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal. $W_p/W_o = 0.5$

Table 3.13 Stress concentration about an infinite row of identical rectangular holes

Combination	S_x, S_y					
	$S_x = 1 \ S_y = 1$			$S_x = 1 \ S_y = 1$		
	$A = B$	C	$D_0 = D_1$	$A = B$	C	$D_0 = D_1$
(a) + (c)	2.71	0.22	8.20			
(b) + (d)				3.00	0.30	5.46

Notes

(a) hole axis and row axis parallel to S_1 ; (b) hole axis normal, row axis parallel; (c) hole axis and row axis normal; and (d) hole axis parallel, row axis normal. $W_p/W_o = 0.5$

Stress concentration data for an infinite row of rectangular holes thus indicates that (b) is more favorable than (a) with respect to peak compressive stress concentration at corners. However, the reverse was true with just two rectangular holes; (a) was most favorable. In the case, say, of five rectangular holes, stress concentration about the outside holes and associated pillars would be similar to stress concentration about just two holes. Stress concentration about the inside holes and pillars would more likely be similar to the infinite row of holes data. Because stress concentrations are higher in the two hole cases, especially at the outside corners, than in the infinite row of holes cases, the choice of configuration (a) or (b) is logically decided by the two hole case in favor of (a). Thus, alignment of hole and row axes parallel to the major compression is the best choice, when the number of holes in the row is few, say, about five.

Example 3.30 Consider a relatively close spaced row of vertical, rectangular shafts such that $W_p = 0.5W_o$ with hole and row axis parallel. Further suppose the opening dimensions are 12×24 ft and therefore $H_o/W_o = 1/2$. These data correspond to the geometry of Figure 3.29 (a). The shaft route penetrates a weak zone at 3,600 ft where rock mass compressive and

tensile strengths are 15,000 and 1,500 psi, respectively. If the preexcavation stress field is caused by gravity alone, estimate the shaft wall safety factors with respect to tension and compression.

Solution: In plan view, the preexcavation gravity field shows equal stress, so $M = 1$. Either the results in Table 3.11 or Table 3.13 may be used for stress concentration values. From Table 3.13, a peak compressive stress concentration of 8.20 occurs at the corners (D); no tension is indicated. With respect to compression

$$FS_c = \frac{C_o}{K_c S_1} = \frac{15,000}{(8.20)(1,200)} = 1.52$$

which is low for shaft wall stability. In comparison, the safety factor at midside of the short edges (A, B) is 4.61. The midside points of the long edges (C) have even higher safety factors. Danger of large-scale failure is therefore not indicated.

Example 3.31 Stresses at a shaft wall are composed of two parts, a preshaft part and a change part induced by excavation of the shaft. Thus, $\sigma = \sigma^o + \Delta\sigma$, where σ , σ^o , and $\Delta\sigma$ post-shaft wall stress, preshaft stress, and stress change. In case of a vertical, circular shaft, the post-shaft wall circumferential stress $\sigma_t = (S_1 + S_3) - 2(S_1 - S_3) \cos(2\theta)$ as given by equation (3.3) and illustrated in Figure 3.1. Suppose the preshaft stress field with respect to compass coordinates (x =east, y =north, z =up) is $S_x = S_H = S_2$, $S_y = S_h = S_3$, and $S_z = S_v = S_1$. Shear stresses are nil. Compression is positive and the principal stresses in three dimensions are ordered such that $S_1 \geq S_2 \geq S_3$. Determine the stress changes induced by shaft excavation as seen in a plan view of the shaft.

Solution: The normal stress perpendicular to the shaft wall (σ_r) is nil, so the stress change is equal in magnitude but opposite in sign to the preshaft stress. Similarly the shear stress in plan view ($\tau_{r\theta}$), also referred to cylindrical coordinates ($r\theta z$), is equal but opposite in sign to the preshaft stress. The circumferential stress change is the final stress less the preshaft stress. Thus,

$$\Delta\sigma_t = \Delta\sigma_\theta = (S_1 + S_3) - 2(S_1 - S_3)\cos(2\theta) - [(1/2)(S_1 + S_3) - (1/2)(S_1 - S_3)\cos(2\theta)]$$

where in plan view $S_1 = S_H$ and $S_3 = S_h$ and the term in [...] is the preshaft circumferential stress. The preshaft stress is obtained by rotation of reference axes about the given, vertical, z -axis. Hence, $\Delta\sigma_t = \Delta\sigma_\theta = (1/2)(S_1 + S_3) - (3/2)(S_1 - S_3) \cos(2\theta)$. Summarizing, the induced stress changes are

$$\begin{aligned}\Delta\sigma_r &= -[(1/2)(S_H + S_h) + (1/2)(S_H - S_h)\cos(2\theta)] \\ \Delta\sigma_\theta &= (1/2)(S_H + S_h) - (3/2)(S_H - S_h)\cos(2\theta) \\ \Delta\tau_{r\theta} &= (1/2)(S_H - S_h)\sin(2\theta)\end{aligned}$$

Example 3.32 Consider a circular shaft of radius a excavated in a preshaft stress field relative to compass coordinates (x =east, y =north, z =up), so $S_x = S_H = S_2$, $S_y = S_h = S_3$, and $S_z = S_v = S_1$. Shear stresses are nil as in Example 3.31. Further suppose the shaft axis is inclined with direction cosines $n_x = \cos(x, c) = \cos(\alpha)$, $n_y = \cos(y, c) = \cos(\beta)$, $n_z = \cos(z, c) = \cos(\gamma)$ where the direction angles of the shaft axis c are (α, β, γ) . Find

formulas for the preshaft stresses relative to a rotated set of axes (abc) where a is dip of a plane normal to the shaft axis c , and b is horizontal.

Solution: A plane normal to the shaft axis is analogous to a “joint” plane, so a table of direction cosines between axes (xyz) and (abc) is just that given in the text, that is, Table 2.1. Figure 2.13 shows the relationship between axes. Stresses in the shaft system (abc) may be computed from formulas given in Appendix D. Thus, from (D.4), $[\sigma(abc)] = [R][\sigma(xyz)][R]^t$ where the matrix $[R]$ is given by (D.6) which is essentially Table 2.1. The multiplication is lengthy but straightforward. The result in a compact form using the symbols in (D.2) for (D.6) and the given stress state is

$$\begin{aligned}\sigma_{aa} &= S_H l_1^2 + S_h m_1^2 + S_v n_1^2 \\ \sigma_{bb} &= S_H l_2^2 + S_h m_2^2 + S_v n_2^2 \\ \sigma_{cc} &= S_H l_3^2 + S_h m_3^2 + S_v n_3^2 \\ \tau_{ab} &= S_H l_1 l_2 + S_h m_1 m_2 + S_v n_1 n_2 \\ \tau_{bc} &= S_H l_2 l_3 + S_h m_2 m_3 + S_v n_2 n_3 \\ \tau_{ca} &= S_H l_3 l_1 + S_h m_3 m_1 + S_v n_3 n_1\end{aligned}$$

Note that the (xyz) directions are actually principal directions. For this reason no shear stresses are present in these formulas.

Example 3.33 Suppose the shaft axes in Example 3.32 has dip direction due north and dip of 60 degrees. Further suppose the stress state before shaft sinking is caused by gravity alone. Estimate the preshaft stresses at a depth of 300 m relative to an (abc) set of axes with a down the dip of the cross-section, b on strike, and c the shaft axis.

Solution: A reasonable estimate of the preshaft vertical stress is 25 kPa/m. The horizontal stresses are equal in a gravity field and are some fraction of the vertical stress, say, 1/3. Thus,

$$S_v = 7.5 \text{ MPa, and } S_H = S_h = 2.5 \text{ MPa.}$$

The cross-section of the shaft is in the role of a “joint” plane that is normal to the shaft axis. The dip of the cross-section is 30 degrees. A table of direction cosines is needed. Thus, with $\alpha = 0$ and $\delta = 30$ degrees,

system	x	y	z
a (dip)	$l_1 = \cos(30)\sin(0)$ $l_1 = 0$	$m_1 = \cos(30)\cos(0)$ $m_1 = \sqrt{3}/2$	$n_1 = -\sin(30)$ $n_1 = -1/2$
b (strike)	$l_2 = -\cos(0)$ $l_2 = -1$	$m_2 = \sin(0)$ $m_2 = 0$	$n_2 = 0$
c (axis)	$l_3 = \sin(30)\sin(0)$ $l_3 = 0$	$m_3 = \sin(30)\cos(0)$ $m_3 = 1/2$	$n_3 = \cos(30)$ $n_3 = \sqrt{3}/2$

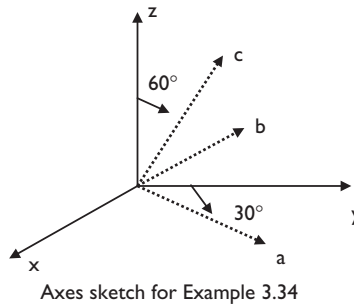
Using the formulas from the previous example:

$$\begin{aligned}
 \sigma_{aa} &= [(1/3)(0) + (1/3)(3/4) + (1/4)](7.5) \\
 \sigma_{bb} &= [(1/3)(1) + (1/3)(0) + (0)](7.5) \\
 \sigma_{cc} &= [(1/3)(0) + (1/3)(1/4) + (3/4)](7.5) \\
 \tau_{ab} &= [(1/3)(0)(-1) + (1/3)(\sqrt{3}/2)(0) + (-1/2)(0)](7.5) \\
 \tau_{bc} &= [(1/3)(-1)(0) + (1/3)(0)(1/2) + (0)(\sqrt{3}/2)](7.5) \\
 \tau_{ca} &= [(1/3)(0)(0) + (1/3)(1/2)(\sqrt{3}/2) + (\sqrt{3}/2)(-1/2)](7.5)
 \end{aligned}$$

These evaluate to:

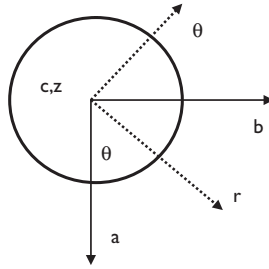
$$\begin{aligned}
 \sigma_{aa} &= 3.75 \\
 \sigma_{bb} &= 2.50 \\
 \sigma_{cc} &= 6.25 \\
 \tau_{ab} &= 0.00 \\
 \tau_{bc} &= 0.00 \\
 \tau_{ca} &= -2.17
 \end{aligned} \quad MPa$$

Example 3.34 Consider the results from Example 3.33 that give the preshaft stresses relative to a rectangle coordinate system (abc) after rotation from compass coordinates (xyz). The axis c is parallel to the shaft axis that dips 60 degrees with dip direction due north, b is horizontal, and a is inclined. Sketch the relationship between the axes and then compute algebraically the preshaft stresses relative to a cylindrical coordinate system attached to the shaft.



Solution: The sketch with x =east, y =north, and z =up is looking down the shaft axes and gives a plan view of the circular shaft and the relationship between (abc) and cylindrical coordinates ($r\theta z$). Thus, Stresses in cylindrical coordinates are obtained by a rotation of reference axes from (abc) to ($r\theta z$) about the shaft axes. Thus, from (D.1) in Appendix D, with the necessary change in subscripts,

$$\begin{aligned}
 \sigma_{rr} &= (1/2)(\sigma_{aa} + \sigma_{bb}) + (1/2)(\sigma_{aa} - \sigma_{bb})\cos(2\theta) + \tau_{ab}\sin(2\theta) \\
 \sigma_{\theta\theta} &= (1/2)(\sigma_{aa} + \sigma_{bb}) - (1/2)(\sigma_{aa} - \sigma_{bb})\cos(2\theta) - \tau_{ab}\sin(2\theta) \\
 \tau_{r\theta} &= (-1/2)(\sigma_{aa} - \sigma_{bb})\sin(2\theta) + \tau_{ab}\cos(2\theta)
 \end{aligned}$$



Shaft coordinates, rectangular and cylindrical.

$$\sigma_{zz} = \sigma_{cc}$$

$$\tau_{rz} = \tau_{ac}\cos(\theta) + \tau_{bc}\sin(\theta)$$

$$\tau_{\theta z} = -\tau_{ac}\sin(\theta) + \tau_{bc}\cos(\theta)$$

These are the preshaft stresses relative to shaft coordinates.

Example 3.35 Analysis of an inclined circular shaft shows that the stress changes induced by mining at the shaft wall are given by:

$$\Delta\sigma_r = -[(1/2)(S_r + S_\theta) + (1/2)(S_r - S_\theta)]$$

$$\Delta\sigma_\theta = [(1/2)(S_r + S_\theta) - (3/2)(S_r - S_\theta)]$$

$$\Delta\tau_{r\theta} = -T_{r\theta}$$

$$\Delta\sigma_{zz} = -(2\nu)(S_r - S_\theta)$$

$$\Delta\tau_{rz} = -T_{rz}$$

$$\Delta\tau_{\theta z} = T_{\theta z}$$

where the capital letters indicate preshaft stresses relative to the shaft coordinate system ($r\theta z$). These are the stresses given in Example 3.34 with a simple change in notation, e.g., $S_r = \sigma_{rr}$ and $T_{rz} = \tau_{rz}$ and so on. Determine the post-shaft wall stresses in terms of the preshaft stresses.

Note: The stress change formulas are computed on the assumption that the shaft diameter is small compared with shaft depth and the section of interest is beyond the influence of the surface where the shaft is collared, say, several diameters.

Solution: The r -stress changes are equal but opposite in sign to the preshaft stresses. Consequently, the total post-shaft stresses $\sigma_r = S_r + \Delta\sigma_r$, $\tau_{rz} = T_{rz} + \Delta\tau_{rz}$, and $\tau_{r\theta} = T_{r\theta} + \Delta\tau_{r\theta}$ are zero as, indeed, they should be in consideration of the traction-free shaft wall that is generated through excavation.

The remaining three post-shaft wall stresses by similar logic are

$$\sigma_\theta = S_\theta + [(1/2)(S_r + S_\theta) - (3/2)(S_r - S_\theta)]$$

$$\sigma_{zz} = S_z - (2\nu)(S_r - S_\theta)$$

$$\tau_{\theta z} = T_{\theta z} + T_{\theta z}$$

Alternatively, after collecting terms

$$\begin{aligned}\sigma_\theta &= -S_r + 3S_\theta \\ \sigma_{zz} &= S_z - (2\nu)(S_r - S_\theta) \\ \tau_{\theta z} &= 2T_{\theta z}\end{aligned}$$

Example 3.36 Consider a circular shaft excavated in a stress field caused by gravity only. Inclination of the shaft axis with respect to the vertical varies. Derive a procedure to evaluate the maximum and minimum shaft wall stress concentration factors for circumferential stress, $K(\max)$ and $K(\min)$ at any orientation from the vertical to the horizontal. In this case, use the vertical preshaft stress S_v as the reference stress.

Solution: The preshaft stress field is caused by gravity alone. Hence, the vertical and horizontal preshaft stresses have the form $S_v = \gamma H$, $S_h = S_H = K_o S_v$ where γ , H , and K_o are specific weight of rock, depth to a point of interest, and the ratio of horizontal to vertical preshaft stress (some number between 0 and 1, perhaps 1/3).

By definition, stress concentration $K = (\text{stress})/(\text{reference stress})$ and $K(\max) = \sigma(\max)/S_1$, $K(\min) = \sigma(\min)/S_1$ where ordinarily S_1 is the major principal stress *in the considered view*. However, in this example, the reference stress is given simply as S_v . The reason is S_1 in the cross-section view varies continuously with inclination while S_v is constant and in fact is the preshaft major principal stress in three dimensions.

From Example 3.35, the post-shaft wall stresses in terms of preshaft stresses are S_r , S_θ , S_z , T_{rz} , $T_{r\theta}$, $T_{\theta z}$ where S is a normal stress and T is a shear stress. These stresses are referred to a cylindrical coordinate system attached to the shaft ($r\theta z$). Stress changes induced by shaft sinking are $\Delta\sigma_r$, $\Delta\sigma_\theta$, $\Delta\sigma_z$, $\Delta\tau_{rz}$, $\Delta\tau_{r\theta}$, $\Delta\tau_{\theta z}$. Thus,

$$\begin{aligned}\sigma_r &= S_r + \Delta\sigma_r = 0 \\ \tau_{rz} &= T_{rz} + \Delta\tau_{rz} = 0 \\ \tau_{r\theta} &= T_{r\theta} + \Delta\tau_{r\theta} = 0\end{aligned}$$

in consideration of the traction-free shaft wall that is generated through excavation. The remaining three post-shaft wall stresses from Example 3.35 are

$$\begin{aligned}\sigma_\theta &= S_\theta + [(1/2)(S_r + S_\theta) - (3/2)(S_r - S_\theta)] \\ \sigma_{zz} &= S_z - (2\nu)(S_r - S_\theta) \\ \tau_{\theta z} &= T_{\theta z} + T_{\theta z}\end{aligned}$$

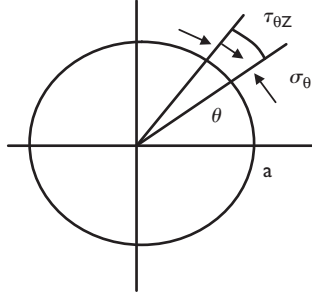
Alternatively, after collecting terms

$$\begin{aligned}\sigma_\theta &= -S_r + 3S_\theta \\ \sigma_{zz} &= S_z - (2\nu)(S_r - S_\theta) \\ \tau_{\theta z} &= 2T_{\theta z}\end{aligned}\tag{3.26}$$

The stresses seen in cross-section are the circumferential stress σ_θ and shear stress $\tau_{\theta z}$. These are shown in the sketch. The post-shaft shear stress is given as twice the preshaft shear stress. The preshaft shear stress is

$$T_{\theta z} = -\tau_{ac}\sin(\theta) + \tau_{bc}\cos(\theta) \quad (3.27)$$

where the shear stresses on the right are referred to a rectangular set of axes (abc) that are rotated from compass coordinates (xyz) . The angle θ is measure from the a -axis.



Sketch for Example 3.36 looking down the shaft axis.

To implement the rotation from Example 3.32, the direction cosines are needed. These are:

$$\begin{aligned} l_1 &= 0, & m_1 &= \cos(\delta), & n_1 &= -\sin(\delta) \\ l_2 &= -1, & m_2 &= 0, & n_2 &= 0 \\ l_3 &= 0, & m_3 &= \sin(\delta), & n_3 &= \cos(\delta) \end{aligned}$$

where δ is the *dip of the shaft cross-section* which is 90-shaft axis inclination. For example, when the shaft is vertical, $\delta=0$. The direction cosines are obtained from Table 2.1 in the text with the dip direction $\alpha=0$.

The shear stresses needed are then

$$\begin{aligned} \tau_{ac} &= [K_o \sin(\delta) \cos(\delta) - \sin(\delta) \cos(\delta)] S_v \\ \tau_{bc} &= 0, \quad \tau_{ab} = 0 \end{aligned} \quad (3.28)$$

When the section is horizontal (shaft) or vertical (tunnel), the shear stresses are nil.

The radial, circumferential, and axial preshaft normal stresses may be computed from the usual equations of transformation (rotation about axis c). Thus,

$$\begin{aligned} S_r &= (1/2)(\sigma_{aa} + \sigma_{bb}) + (1/2)(\sigma_{aa} - \sigma_{bb})\cos(2\theta) + \tau_{ab}\sin(2\theta) \\ S_\theta &= (1/2)(\sigma_{aa} + \sigma_{bb}) - (1/2)(\sigma_{aa} - \sigma_{bb})\cos(2\theta) + \tau_{ab}\sin(2\theta) \\ S_{zz} &= \sigma_{cc} \end{aligned} \quad (3.29)$$

Again the preshaft stresses relative to (abc) on the right are obtained by rotation of axes from compass coordinates (xyz) as in Example 3.2. Thus,

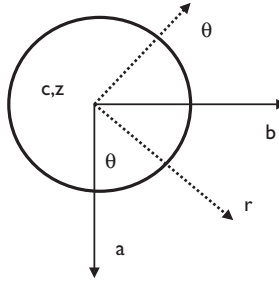
$$\begin{aligned} \sigma_{aa} &= [K_o \cos^2(\delta) + \sin^2(\delta)] S_v = K_1 S_v \\ \sigma_{bb} &= K_o S_v \\ \sigma_{cc} &= [K_o \sin^2(\delta) + \cos^2(\delta)] S_v = K_2 S_v \end{aligned} \quad (3.30)$$

When the shaft is vertical and $\delta = 0$, $\sigma_{aa} = \sigma_{xx} = K_o S_v$, $\sigma_{bb} = \sigma_{yy} = K_o S_v$, $\sigma_{cc} = \sigma_{zz} = S_v$ which checks. When the shaft is horizontal and $\delta = 90^\circ$, $\sigma_{aa} = \sigma_{zz} = S_v$, $\sigma_{bb} = \sigma_{yy} = K_o S_v$, $\sigma_{cc} = \sigma_{xx} = K_o S_v$ which also checks.

Equations (3.29) can now be written in terms of the preshaft stresses in compass coordinates. Thus,

$$\begin{aligned} S_r &= (S_v/2)(K_1 + K_o) + (S_v/2)(K_1 - K_o)\cos(2\theta) \\ S_\theta &= (S_v/2)(K_1 + K_o) - (S_v/2)(K_1 - K_o)\cos(2\theta) \\ S_{zz} &= K_2 S_v \\ T_{\theta z} &= K_3 S_v \end{aligned} \quad (3.31)$$

The last equation is added from (3.27) and (3.28) for completeness. A sketch of the shaft coordinates is helpful. From (3.26) and (3.31)



Shaft coordinate sketch looking down the c-axis.

$$\begin{aligned} \sigma_\theta &= S_r[(K_1 + K_o) - 2(K_1 - K_o)\cos(2\theta)] \\ \sigma_{zz} &= S_v[K_2 - 2v(K_1 - K_o)\cos(2\theta)] \\ \tau_{\theta z} &= 2K_3 S_v \end{aligned} \quad (3.32)$$

The constant K's are

$$\begin{aligned} K_1 &= [K_o \cos^2(\delta) + \sin^2(\delta)] \\ K_2 &= [K_o \sin^2(\delta) + \cos^2(\delta)] \\ K_3 &= [K_o \sin^2(\delta)\cos(\delta) - \sin(\delta)\cos(\delta)] \end{aligned}$$

As a check on (3.32), in the vertical case $\delta=0$ and

$$\begin{aligned} K_1 &= K_o & \sigma_\theta &= 2S_r K_o = 2S_v(S_h)/S_v = 2S_1 \text{ (} S_1 \text{ in the view plane)} \\ K_2 &= 1 & \text{and } \sigma_{zz} &= S_v \\ K_3 &= 0 & \tau_{\theta z} &= 0 \end{aligned}$$

Again, these results are for a gravity only preexcavation stress field.

From (3.32), the stress concentrations are easily obtained. Thus,

$$\begin{aligned} K_\theta &= \sigma_\theta/S_r = [(K_1 + K_o) - 2(K_1 - K_o)\cos(2\theta)] \\ K_z &= \sigma_z/S_v = [K_2 - 2v(K_1 - K_o)\cos(2\theta)] \\ K_\tau &= \tau_{\theta z}/S_v = 2K_3 \end{aligned}$$

The required maximum and minimum circumferential stress concentration factors are obtained from (3.32). Stationary values of the first derivatives occur at $\theta=0, \pi/2$ (measured from the α -axis that points down dip. Thus, with addition of subscript θ for clarity,

$$\begin{aligned} K_{\theta}(\max) &= 3K_1 - K_o \\ K_{\theta}(\min) &= -K_1 + 3K_o \end{aligned} \quad (3.33)$$

These expressions have the same structure as those discussed in the text following equation (3.4). Indeed, they reduce to the maximum and minimums given there in case of a vertical shaft excavated in the gravity field.

This example illustrates the great utility of axis rotation. Although lengthy, the derivation is a straightforward application of fundamentals to achieve a useful design goal. The same formulas apply in case of equal but high horizontal stress. In this case, $K_o \geq 1$. A similar procedure can be used to examine stress concentration about shafts of arbitrary orientation excavated in any preshaft stress field. An outline of the solution including displacements as well as stresses and a demonstration of correctness are given by Pariseau (1987).

Comment: There are yet other factors of safety that take into account the biaxial nature of stress at the wall of a naturally supported shaft. These safety factors depend on the failure criterion adopted for design analyses. An example is a factor of safety using the well-known Mohr-Coulomb (MC) failure criterion that is discussed in Appendix B. The MC criterion may be put into a form that indicates the maximum shear stress the material can sustain. Thus,

$$\tau(\text{strength}) = \left(\frac{\sigma_1 + \sigma_3}{2} \right) \sin(\phi) + (c) \cos(\phi)$$

where σ_1 , σ_3 , ϕ , and c are major principal stress, minor principal stress, angle of internal friction, and cohesion, respectively. Compression is considered positive in this expression. The intermediate principal stress σ_2 is assumed to have no effect on failure.

The stress normal to the shaft wall (radial stress) is zero, of course. However, the circumferential stress acting tangent to the shaft wall and the vertical stress are not zero. Moreover, there may be a shear stress present at the shaft wall as well as the two normal stresses. Thus, to apply the MC criterion, one must first determine the principal stress at the shaft wall. The radial stress is a principal stress and the normal (radial) direction to the shaft wall is a principal direction because the shaft wall surface is shear free. Thus, a rotation about a radius will lead to the required principal stresses. If these two principal stresses bracket the zero radial stress, then the radial stress is indeed the intermediate principal stress. This bracketing requires one principal stress to be greater than zero (compression) and one to be less than zero (tension). If both are compression, then the minor principal stress is the zero radial stress. If in the very unlikely event that both are tension, then the intermediate principal stress is the tension of smallest magnitude.

The most likely case is for the radial stress to be the minor principal stress. In the absent of a shear stress at the shaft wall, either the vertical or circumferential stress is the major principal stress. In either case, MC assumes the form

$$\begin{aligned} \tau(\text{strength}) &= \left(\frac{\sigma_1}{2} \right) \sin(\phi) + (c) \cos(\phi) \\ \left(\frac{\sigma_1}{2} \right) &= \left(\frac{\sigma_1}{2} \right) \sin(\phi) + (c) \cos(\phi) \end{aligned}$$

When this expression is solved for the major principal stress at failure, the result is simply

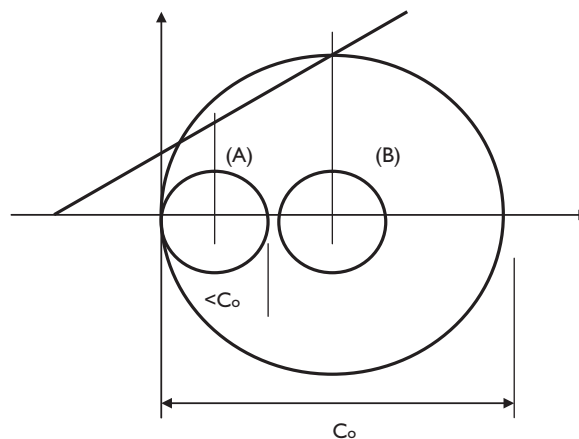
$$\sigma_1 = C_o.$$

Thus, the MC criterion leads to a safety factor with respect to compression given in (3.1). There are two such factors of safety, one respect to the circumferential stress and one with respect to the vertical stress. Naturally, the lowest factor of safety guides the design.

The case when the maximum compression is zero and the least principal stress is a tension leads to $\sigma_3 = -T_o$ and a safety factor with respect to tension. As a reminder, only the numerical values of strength and stress are used in factory of safety calculations.

Graphically, the safety factor using the Mohr-Coulomb criterion is simply the ratio of the maximum radius of a Mohr circle representing the state of stress at failure to the actual radius of the Mohr circle representing the given stress state. A subtlety that enters all factor of safety concepts is how the stresses would increase from the current state to a failure state. The process is unknown, so an assumption is needed. One assumption is to suppose that the mean stress remains constant, so the center of the Mohr circle is fixed. The radius of the failure circle is then readily computed from the Mohr criterion. An alternative assumption is to suppose the minor principal stress is fixed, while the major principal stress increases. This situation occurs during conventional triaxial strength testing that fixes confining pressure and increases axial load to failure of a test cylinder.

MC strength is graphically seen in the sketch as the radius of the Mohr circle representing failure **at the given mean normal stress** where the circle is centered. These are the vertical lines extending from the circle centers to the sloping failure line. A maximum radius divided by the radius of the circle representing the actual state of stress is the MC safety factor. Mohr circles representing the actual state of stress have the same radii but different centers and therefore different associated strengths and safety factors.



Sketch showing alternative safety factors with respect to Mohr-Coulomb strength.

The circle (A) occurs at a naturally supported excavation wall in consideration of the zero normal stress perpendicular to the wall and in consideration of the remaining two principal stresses at the wall that are assumed compressive. In this common situation, the minor principal stress is zero and the largest of the remaining principal stresses is the major principal

stress. In a gravity field, the vertical shaft wall stress increases linearly with depth while the radial stress remains zero. The corresponding Mohr circle increases in diameter as the major principal stress moves to the right while the minor principal stress remains fixed at the origin. At failure $S_v = \sigma_1 = C_o$.

Circle (B) is a stress state where the mean normal stress is just one-half of the unconfined compressive strength. If the mean normal stress is fixed, while the radius increases, then at failure $\sigma_3 = 0$ and $\sigma_1 = C_o$. The safety factor associated with (A) is less than that associated with (B).

3.4 Problems

Single, naturally supported shafts

- 3.1 A 20 foot diameter circular shaft is planned in a massive rock. Laboratory tests on core from exploration drilling show that

$$\begin{aligned} C_o &= 22,000 \text{ psi} & T_o &= 1,200 \text{ psi} \\ \gamma &= 144 \text{ pcf} & E &= 5 \times 10^6 \text{ psi} \\ G &= 2.0 \times 10^6 \text{ psi} \end{aligned}$$

Depth is 3,000 ft. Site measurements show that no tectonic stresses are present. Determine the factors of safety with respect to failure in compression and tension.

- 3.2 If the opening in Problem 3.1 is in an *in situ* stress field such that $\sigma_h = \sigma_v$, what are the safety factors?
- 3.3 If the opening in Problem 3.1 is in an *in situ* stress field such that $\sigma_h = 0.01\sigma_v$, is there a possibility of failure? Justify your answer.
- 3.4 A 6 m diameter circular shaft is planned in massive rock. Laboratory tests on core from exploration drilling show that

$$\begin{aligned} C_o &= 152 \text{ MPa} & T_o &= 8.3 \text{ MPa} \\ \gamma &= 23 \text{ kN/m}^3 & E &= 34.5 \text{ GPa} \\ G &= 13.8 \text{ GPa} \end{aligned}$$

Depth is 915 m. Site measurements show that no tectonic stresses are present. Determine the factors of safety with respect to failure in compression and tension.

- 3.5 If the opening in Problem 3.4 is in an *in situ* stress field such that $\sigma_h = \sigma_v$, what are the safety factors?
- 3.6 If the opening in Problem 3.4 is in an *in situ* stress field such that $\sigma_h = 0.01\sigma_v$, is there a possibility of failure? Justify your answer.
- 3.7 Which is preferable from the rock mechanics viewpoint: an elliptical, ovaloidal, or rectangular opening of W_o/H_o of 2 for a stress field $M = 1/3$? $M = 0$? Justify your choices.
- 3.8 A rectangular shaft 10 ft wide by 20 ft long is sunk vertically in ground where the state of premining stress is

$$\sigma_{xx} = 1,141, \quad \sigma_{yy} = 2,059, \quad \sigma_{zz} = 1,600, \quad \tau_{xy} = 221, \quad \tau_{yz} = 0, \quad \tau_{zx} = 0,$$

with compression positive, x = east, y = north, and z = up, in psi. Rock properties are:

$$E = 4.5 \times 10^6 \text{ psi}, \quad \nu = 0.20, \quad C_o = 15,000 \text{ psi}, \quad T_o = 900 \text{ psi}.$$

Find (a) the most favorable orientation of the shaft and (b) the safety factors in this orientation. Indicate by sketch where the peak stresses occur in cross section.

- 3.9 A rectangular shaft 3 m wide by 6 m long is sunk vertically in ground where the state of premining stress is

$$\sigma_{xx} = 7.9, \quad \sigma_{yy} = 14.2, \quad \sigma_{zz} = 11.0, \quad \tau_{xy} = 1.5, \quad \tau_{yz} = 0, \quad \tau_{zx} = 0,$$

with compression positive, x = east, y = north, and z = up, in MPa. Rock properties are:

$$E = 31.0 \text{ GPa}, \quad \nu = 0.20, \quad C_o = 103 \text{ MPa}, \quad T_o = 6.2 \text{ MPa}.$$

Find (a) the most favorable orientation of the shaft and (b) the safety factors in this orientation. Indicate by sketch where the peak stresses occur in cross section.

- 3.10 A 12 ft by 24 ft rectangular shaft is sunk to a depth of 3,000 ft in ground where the premining stress field is given by formulas

$$S_v = 1.2h, \quad S_h = 120 + 0.5h, \quad S_H = 3,240 + 0.3h$$

where h is depth in feet and the stresses are in psi. Show by sketch, the best orientation of the shaft relative to the directions of h and H . Is this the best orientation at all depths to 3,000 ft?

- 3.11 A 3.66 m by 7.32 m rectangular shaft is sunk to a depth of 914 m in ground where the premining stress field is given by formulas

$$S_v = 27.2h, \quad S_h = 826 + 11.3h, \quad S_H = 22,345 + 6.8h$$

where h is depth in m and the stresses are in kPa. Show by sketch, the best orientation of the shaft relative to the directions of h and H . Is this the best orientation at all depths?

- 3.12 Thin overburden is scrapped to expose fresh bedrock at the site of a planned shaft. A 0-45-90 strain gauge rosette is bonded to the bedrock after a five inch square portion of the surface is ground smooth. The 0-gauge is oriented N30E. A 6-inch diameter coring bit is then used to relieve the *in situ* strains by overcoring the gauge. Final readings after overcoring are: $\epsilon(0) = 1,000 \mu \text{ in./in.}$, $\epsilon(45) = 500 \mu \text{ in./in.}$, $\epsilon(90) = 3,000 \mu \text{ in./in.}$ where *the strain meter reads tension as positive*.

- If Young's modulus is 2.4 million psi and Poisson's ratio is 0.20, what is the state of strain and the state of stress at the gauge site relative to compass coordinates? Note: The *in situ* strains are opposite in sign to the relieved strains.
- What is the best orientation of a 10 × 20 ft rectangular shaft at this site?
- What unconfined compressive and tensile strengths (C_o , T_o) are needed to ensure safety factors of 2 and 4 in compression and tension, respectively?

- 3.13 With reference to Problem 3.12, if the state of stress at the surface continues to depth in addition to gravity load, and the specific weight of rock is 28 kN/m^3 , what strengths are needed at a depth of 1,234 m for the same safety factors?
- 3.14 Thin overburden is scrapped to expose fresh bedrock at the site of a planned shaft. A0–45–90 strain gauge rosette is bonded to the bedrock after a 12.7 cm square portion of the surface is ground smooth. The 0-gauge is oriented N30E. A 15.2 cm diameter coring bit is then used to relieve the *in situ* strains by overcoring the gauge. Final readings after overcoring are: $\varepsilon(0) = 1,000 \mu \text{ in./in.}$, $\varepsilon(45) = 500 \mu \text{ in./in.}$, $\varepsilon(90) = 3,000 \mu \text{ in./in.}$ where the strain meter reads tension as positive.
- If Young's modulus is 16.55 GPa and Poisson's ratio is 0.20, what is the state of strain and the state of stress at the gauge site relative to compass coordinates? Note: The *in situ* strains are opposite in sign to the relieved strains.
 - What is the best orientation of a $3 \times 6 \text{ m}$ rectangular shaft at this site?
 - What unconfined compressive and tensile strengths (C_o , T_o) are needed to ensure safety factors of 2 and 4 in compression and tension, respectively?
- 3.15 With reference to Problem 3.14, if the state of stress at the surface continues to depth in addition to gravity load, and the specific weight of rock is 28 kN/m^3 , what strengths are needed at a depth of 1,234 m for the same safety factors?
- 3.16 A rectangular shaft 10 ft by 20 ft with the long axis parallel to the N-S line exists at a depth of 950 ft. The mining plan calls for deepening the shaft to 1,800 ft. Estimate safety factors for the shaft wall at a depth of 1,750 ft. Rock properties are:

$$C_o = 23,700 \text{ psi}, \quad T_o = 1,480 \text{ psi}, \quad E = 5.29(10^6 \text{ psi}), \quad \nu = 0.27, \\ \gamma = 162 \text{ pcf}$$

The premining stress state relative to compass coordinates is:

$$S_E = 350 + 0.2h, \quad S_N = 420 + 0.35h, \quad S_V = 1.12h$$

where stresses are in psi, h = depth in ft, E , N , V refer to compass coordinates (x = east, y = north, z = up) and compression is positive. Premining shear stresses are nil relative to compass coordinates.

- 3.17 If the premining stress state in Problem 3.16 were hydrostatic (3D), what shape would be more favorable, an ellipse, rectangle, or ovaloid (with semi-axes ratio of 2)?
- 3.18 A rectangular shaft $3 \times 6 \text{ m}$ with the long axis parallel to the N–S line exists at a depth of 290 m. The mining plan calls for deepening the shaft to 549 m. Estimate safety factors for the shaft wall at a depth of 533 m. Rock properties are:

$$C_o = 164 \text{ MPa}, \quad T_o = 10.2 \text{ MPa}, \quad E = 36.5 \text{ GPa}, \quad \nu = 0.27, \\ \gamma = 26.6 \text{ kN/m}^3$$

The premining stress state relative to compass coordinates is:

$$S_E = 2,414 + 4.5h, \quad S_N = 2,897 + 7.9h, \quad S_V = 25.3h$$

where stresses are in kPa, h = depth in m, E , N , V refer to compass coordinates (x = east, y = north, z = up) and compression is positive. Premining shear stresses are nil relative to compass coordinates.

- 3.19 If the premining stress state in Problem 3.18 where hydrostatic (3D), what shape would be more favorable, an ellipse, rectangle, or ovaloid (with semi-axes ratio of 2)?
- 3.20 A rectangular shaft 18 ft by 24 ft is planned for a depth of 4,800 ft where the premining stresses relative to compass coordinates (x = east, y = north, z = up) are given by:

$$\sigma_{xx} = 250 + 0.5h, \quad \sigma_{yy} = 800 + 0.2h, \quad \sigma_{zz} = 1.1h,$$

where h is depth and compression is positive. Find the best shaft orientation (explain), then determine if shaft wall support is needed at some depth. Wall rock properties are: $E = 6.2$ million psi, $\nu = 0.33$, c (cohesion) = 5,600 psi, $\phi = 52^\circ$, $\gamma = 158$ pcf.

- 3.21 Considering the data and results from Problem 3.20, determine the shaft wall safety factors in tension and compression.
- 3.22 Given the stresses relative to compass coordinates (x = east, y = north, z = up),

$$\begin{aligned} \sigma_{xx} &= 2,155, \quad \sigma_{yy} = 3,045, \quad \sigma_{zz} = 4,200, \quad \tau_{yx} = -1,222, \\ \tau_{xz} &= 0, \quad \tau_{yz} = 0 \end{aligned}$$

in psi with compression positive, find the principal stresses σ_1 , σ_2 , σ_3 (magnitude and orientation). Show with a suitable sketch. Then consider an unlined shaft that will be 13 ft by 26 ft in section, choose the best opening shape (ellipse, ovaloid, or rectangle) and best orientation from the rock mechanics perspective, assuming the given stress state is critical to the design. Sketch.

- 3.23 A rectangular shaft 6×8 m is planned for a depth of 1,500 m where the premining stresses relative to compass coordinates (x = east, y = north, z = up) are given by:

$$\sigma_{xx} = 1,724 + 11.3h, \quad \sigma_{yy} = 5,517 + 4.5h, \quad \sigma_{zz} = 2.49h,$$

where h is depth in meters and stress is in kPa; compression is positive. Find the best shaft orientation (explain), then determine if shaft wall support is needed at some depth. Wall rock properties are: $E = 42.8$ GPa, $\nu = 0.33$, c (cohesion) = 38.6 MPa, $\phi = 52^\circ$, $\gamma = 25.0$ kN/m³.

- 3.24 Considering the data and results from Problem 3.23, determine the shaft wall safety factors in tension and compression.
- 3.25 Given the stresses relative to compass coordinates (x = east, y = north, z = up),

$$\sigma_{xx} = 14.9, \quad \sigma_{yy} = 21.0, \quad \sigma_{zz} = 29.0, \quad \tau_{yx} = -8.4, \quad \tau_{xz} = 0, \quad \tau_{yz} = 0$$

in MPa with compression positive, find the principal stresses σ_1 , σ_2 , σ_3 (magnitude and orientation). Show with a suitable sketch. Then consider an unlined shaft that will be 4×8 m in section, choose the best opening shape (ellipse, ovaloid, or rectangle) and best orientation from the rock mechanics perspective, assuming the given stress state is critical to the design. Sketch.

- 3.26 A large vertical shaft is planned for an underground hard rock mine. Laboratory tests on core from exploration drilling show that

$$C_o = 21,500 \text{ psi}, \quad T_o = 1,530 \text{ psi}, \quad E = 6.25(10^6) \text{ psi},$$

$$G = 2.5(10^6) \text{ psi}, \quad \gamma = 144 \text{ pcf}$$

while premining stress measurements can be fit to the formulas

$$S_E = 200 + 0.3h, \quad S_N = 600 + 0.9h, \quad S_V = 1.1h$$

where E, N, V refer to compass coordinates ($x = \text{east}, y = \text{north}, z = \text{up}$) and compression is positive. Premining shear stresses are nil relative to compass coordinates. Find: (a) the most preferable shape of shaft (elliptical, rectangular, or ovaloidal) when the cross section is 14×28 ft and (b) the resulting factors of safety with respect to compression and tension at a depth of 2,800 ft.

- 3.27 A large vertical shaft is planned for an underground hard rock mine. Laboratory tests on core from exploration drilling show that

$$C_o = 148.3, \quad T_o = 10.6, \quad E = 43.10 \text{ GPa}, \quad G = 17.24 \text{ GPa},$$

$$\gamma = 22.78 \text{ kN/m}^3$$

while premining stress measurements can be fit to the formulas

$$S_E = 1,379 + 6.8h, \quad S_N = 4,138 + 20.4h, \quad S_V = 24.9h$$

where E, N, V refer to compass coordinates ($x = \text{east}, y = \text{north}, z = \text{up}$), stress is in kPa, depth h is in m, and compression is positive. Premining shear stresses are nil relative to compass coordinates. Find: (a) the most preferable shape of shaft (elliptical, rectangular, or ovaloidal) when the cross section is 4.5×9.0 m and (b) the resulting factors of safety with respect to compression and tension at a depth of 854 m.

- 3.28 Laboratory tests on core from exploration drilling show that

$$C_o = 21,500 \text{ psi}, \quad T_o = 1,530, \quad E = 6.25(10^6) \text{ psi},$$

$$G = 2.5(10^6) \text{ psi}, \quad \gamma = 144 \text{ pcf}$$

while premining stress measurements can be fit to the formulas

$$S_E = 200 + 0.3h, \quad S_N = 600 + 0.9h, \quad S_V = 1.1h$$

where E, N, V refer to compass coordinates ($x = \text{east}, y = \text{north}, z = \text{up}$), stress is in psi, depth h is in ft, and compression is positive. Premining shear stresses are nil relative to compass coordinates. Suppose a 12×24 ft rectangular shaft is selected and for some reason is oriented with the long axis N30W. Find the resulting factors of safety at 2,150 ft.

- 3.29 A rectangular shaft 12 ft by 24 ft is planned for deepening from a depth of 3,200 to 4,800 ft in ground where the preshaft stress state relative to compass coordinates (x = east, y = north, z = up) is given by

$$S_{xx} = 2,000 + 1.1d, \quad S_{yy} = 50 + 0.9d, \quad S_{zz} = 1.15d, \quad T_{xy} = -350 - 0.3d, \\ T_{yz} = 0.0, \quad T_{zx} = 0.0$$

where compression is positive and d is depth below surface.

- Find the best orientation of the shaft at a planned 3,850 shaft station and show by sketch.
 - Also show on the sketch where the peak stress concentrations are likely to occur.
 - Finally, estimate the peak stress concentrations.
- 3.30 Laboratory tests on core from exploration drilling show that

$$C_o = 148.3 \text{ MPa}, \quad T_o = 10.55 \text{ MPa}, \quad E = 43.10 \text{ GPa}, \\ G = 17.24 \text{ GPa}, \quad \gamma = 22.78 \text{ kN/m}^3$$

while premining stress measurements can be fit to the formulas

$$S_E = 1,379 + 6.8h, \quad S_N = 4,138 + 20.4h, \quad S_V = 24.9h$$

where E, N, V refer to compass coordinates (x = east, y = north, z = up), stress is in kPa, depth h is in m, and compression is positive. Premining shear stresses are nil relative to compass coordinates. Suppose a 3.7×7.4 m rectangular shaft is selected and for some reason is oriented with the long axis N30W. Find the resulting factors of safety at 655 m.

- 3.31 A rectangular shaft 3.7×7.4 m is planned for deepening from a depth of 975 to 1,463 m in ground where the preshaft stress state relative to compass coordinates (x = east, y = north, z = up) is given (in kPa) by

$$S_{xx} = 13,793 + 24.9d, \quad S_{yy} = 345 + 20.4d, \quad S_{zz} = 26.0d, \\ T_{xy} = -2,414 - 6.8d, \quad T_{yz} = 0.0, \quad T_{zx} = 0.0$$

where compression is positive and d is depth in m below surface.

- Find the best orientation of the shaft at a planned 1,175 m deep shaft station and show by sketch.
 - Also show on the sketch where the peak stress concentrations are likely to occur.
 - Finally, estimate the peak stress concentrations.
- 3.32 A vertical circular shaft will pass through a water bearing stratum at a depth of 2,780 ft where the water pressure is estimated to be 210 psi. Finished shaft diameter must be 18 ft. The preshaft stress state is attributable to gravity alone. Rock properties are: $C_o = 6,750$ psi, $T_o = 675$ psi, $E = 2.4$ million psi, $\nu = 0.25$, $\gamma = 156$ pcf. Estimate the unlined shaft wall safety factors.
- 3.33 A three-compartment rectangular shaft 12×24 ft in cross section is planned for a depth of 3,000 ft. The presinking stress field is assumed to be caused by gravity only.

The weakest rock along the proposed shaft route has an unconfined compressive strength of 8,000 psi and a tensile strength of 750 psi. Young's modulus and Poisson's ratio are estimated to be 4.5 million psi and 0.25. Specific weight of rock is 162 pcf. Determine the safety factors, with respect to tension and compression. What is the optimum orientation of the shaft?

- 3.34 The presinking stress field assumed to be caused by gravity alone (Problem 3.33) turns out to be wrong. The actual stress field has a tectonic component that adds a constant 1,250 psi to the east–west horizontal stress attributable to gravity and twice that amount to the north–south stress attributable to gravity. Orient the shaft in the optimum direction and then determine the unlined shaft wall safety factors.
- 3.35 A circular shaft 18 ft in diameter is decided upon rather than the proposed rectangular shaft. Determine the unlined shaft wall safety factors. Note: The stress field from Problem 3.34 applies.
- 3.36 The *in situ* stress field for Problems 3.34 and 3.35 changes between 3,000 and 3,500 ft to one described by the formulas

$$\sigma_v = 1.125h, \quad \sigma_H = 3,500 + 0.33h, \quad \sigma_h = 3,500 + 0.33h$$

where the stresses are in psi and the depth h is in ft. The stresses σ_v , σ_H , σ_h are principal stresses in the vertical and horizontal directions. Find the safety factor at 4,500 ft for an unlined, circular shaft wall. Assume the same rock properties.

- 3.37 A three-compartment rectangular shaft 3.7×7.4 m in cross section is planned for a depth of 914 m. The presinking stress field is assumed to be caused by gravity only. The weakest rock along the proposed shaft route has an unconfined compressive strength of 55.17 MPa and a tensile strength of 5.17 MPa. Young's modulus and Poisson's ratio are estimated to be 31.0 GPa and 0.25. Specific weight of rock is 25.6 kN/m³. Determine the safety factors, with respect to tension and compression. What is the optimum orientation of the shaft?
- 3.38 The presinking stress field assumed to be caused by gravity alone (Problem 3.37) turns out to be wrong. The actual stress field has a tectonic component that adds a constant 8.62 MPa to the east–west horizontal stress attributable to gravity and twice that amount to the north–south stress attributable to gravity. Orient the shaft in the optimum direction and then determine the unlined shaft wall safety factors.
- 3.39 A circular shaft 5.5 m in diameter is decided upon rather than the proposed rectangular shaft. Determine the unlined shaft wall safety factors. Note: The stress field from Problem 3.38 applies.
- 3.40 The *in situ* stress field for Problems 3.38 and 3.39 changes between 914 and 1,067 m to one described by the formulas

$$\sigma_v = 25.45h, \quad \sigma_H = 24,138 + 7.47h, \quad \sigma_h = 24,138 + 7.47h$$

where the stresses are in kPa and the depth h is in m. The stresses σ_v , σ_H , σ_h are principal stresses in the vertical and horizontal directions. Find the safety factor at 1,372 m for an unlined, circular shaft wall. Assume the same rock properties.

- 3.41 A circular shaft liner is sunk to a depth of 3,750 ft (1,143 m). If the premining stress field is caused by gravity alone, what unconfined compressive rock strength in psi (MPa) is needed for a rock factor of safety of 3.0?

Supported shafts, liners, bolts, rings

- 3.42 A circular vertical shaft is planned to have a finished, inside diameter of 19 ft in an underground hard rock mine. Rock properties are:

$$C_o = 23,700 \text{ psi}, \quad T_o = 1,480 \text{ psi}, \quad E = 5.29(10^6 \text{ psi}), \\ \nu = 0.27, \quad \gamma = 162 \text{ pcf}$$

The premining stress state relative to compass coordinates is:

$$S_E = 350 + 0.2h, \quad S_N = 420 + 0.35h, \quad S_V = 1.12h$$

where stresses are in psi, h = depth in ft, E , N , V refer to compass coordinates (x = east, y = north, z = up) and compression is positive. Premining shear stresses are nil relative to compass coordinates.

Properties of concrete are:

Compressive strength = 5,740 psi, tensile strength = 425 psi, Young's modulus = $4.75(10^6)$ psi, Poisson's ratio = 0.25.

Find:

- (a) the unlined shaft wall safety factors in tension and compression at a depth of 4,250 ft,
 - (b) concrete liner thickness in the vicinity of a water-bearing formation where the pressure in the undisturbed ground is 90 psi. Note: The maximum allowable compression in the concrete is 3,500 psi.
- 3.43 With reference to the previous problem data, if changes in the inside diameter of the lined shaft are monitored for safety, what "reading" in inches would indicate impending failure of the liner?
- 3.44 A circular vertical shaft is planned to have a finished, inside diameter of 5.8 m in an underground hard rock mine. Rock properties are:

$$C_o = 163.5 \text{ MPa}, \quad T_o = 10.21 \text{ MPa}, \quad E = 36.48 \text{ GPa}, \\ \nu = 0.27, \quad \gamma = 25.63 \text{ kN/m}^3$$

The premining stress state relative to compass coordinates is:

$$S_E = 2,414 + 4.53h, \quad S_N = 2,897 + 7.92h, \quad S_V = 25.34h$$

where stresses are in kPa, h = depth in m, E , N , V refer to compass coordinates (x = east, y = north, z = up) and compression is positive. Premining shear stresses are nil relative to compass coordinates.

Properties of concrete are:

Compressive strength = 39.6 MPa, tensile strength = 2.93 MPa, Young's modulus = 32.76 GPa, Poisson's ratio = 0.25.

Find:

- (a) the unlined shaft wall safety factors in tension and compression at a depth of 1,295 m,

- (b) concrete liner thickness in the vicinity of a water-bearing formation where the pressure in the undisturbed ground is 0.621 MPa. Note: The maximum allowable compression in the concrete is 24.14 MPa.
- 3.45 With reference to the previous problem data, if changes in the inside diameter of the lined shaft are monitored for safety, what “reading” in centimeters would indicate impending failure of the liner?
- 3.46 A circular vertical shaft is planned to have a finished, inside diameter of 26 ft in an underground hard rock mine. Rock properties are:

$$C_o = 27,400 \text{ psi}, \quad T_o = 1,840 \text{ psi}, \quad E = 6.19(10^6 \text{ psi}), \\ \nu = 0.22, \quad \gamma = 159 \text{ pcf}$$

The premining stress state relative to compass coordinates is:

$$S_E = 600 + 0.3h, \quad S_N = 200 + 0.4h, \quad S_V = 1.12h$$

where E, N, V refer to compass coordinates ($x = \text{east}, y = \text{north}, z = \text{up}$), h is depth in ft, stress is in psi, and compression is positive. Premining shear stresses are nil relative to compass coordinates. Properties of concrete are: Compressive strength = 5,500 psi, tensile strength = 550 psi, Young’s modulus = $5.5(10^6)$ psi, Poisson’s ratio = 0.25. Find: The concrete liner thickness in the vicinity of a water-bearing cavernous dolomite where the pressure in the undisturbed ground is 210 psi. Note: The maximum allowable compression in the concrete is 3,500 psi.

- 3.47 With reference to the previous problem data, if changes in the *inside* diameter of the lined shaft are monitored for safety, what “reading” in inches would indicate impending failure of the liner?
- 3.48 A circular vertical shaft is planned to have a finished, inside diameter of 8 m in an underground hard rock mine. Rock properties are:

$$C_o = 189.0 \text{ MPa}, \quad T_o = 12.69 \text{ MPa}, \quad E = 42.69 \text{ GPa}, \\ \nu = 0.22, \quad \gamma = 25.15 \text{ kN/m}^3$$

The premining stress state relative to compass coordinates is:

$$S_E = 4,138 + 6.79h, \quad S_N = 1,379 + 9.05h, \quad S_V = 25.34h$$

where E, N, V refer to compass coordinates ($x = \text{east}, y = \text{north}, z = \text{up}$), h is depth in m, stress is in kPa and compression is positive. Premining shear stresses are nil relative to compass coordinates. Properties of concrete are: compressive strength = 37.93 MPa, tensile strength = 3.79 MPa, Young’s modulus = 37.93 GPa, Poisson’s ratio = 0.25. Find: the concrete liner thickness in the vicinity of a water-bearing cavernous dolomite where the pressure in the undisturbed ground is 1.45 MPa. Note: the maximum allowable compression in the concrete is 24.14 MPa.

- 3.49 With reference to the previous problem data, if changes in the *inside* diameter of the lined shaft are monitored for safety, what “reading” in cm would indicate impending failure of the liner?

- 3.50 A vertical unlined 22 ft diameter circular shaft is fitted with a concrete shaft liner to withstand water pressure of 123 psi in a massive sandstone aquifer at a depth of 2,890 ft. Liner properties are: $E = 3.5$ million psi, $\nu = 0.30$, $C_o = 4,500$ psi, $T_o = 450$ psi, $\gamma = 152$ pcf. A liner safety factor of 3.85 is required in compression. Find the liner thickness (in.) and finished shaft diameter (ft).
- 3.51 Show why the weight of a concrete shaft liner that is poured to the walls of the unlined shaft is not important to liner stress.
- 3.52 A large vertical finished (inside) shaft diameter of 32 ft is required for hoisting capacity in a planned high volume underground oil shale mine. An aquifer is encountered at a depth of 1,270 ft where the water pressure is 240 psi.
- Determine the thickness of a concrete liner needed for a liner safety factor of 2.5 when the concrete compressive strength (C_o) is 3,500 psi, Young's modulus is $5.6(10^6)$ psi, Poisson's ratio is 0.27. Explain, defend your answer.
 - If changes in the inside diameter of the shaft are monitored for safety, what "reading" in inches would indicate impending failure of the liner?
- 3.53 A vertical circular shaft will pass through a water-bearing stratum at a depth of 2,780 ft where the water pressure is estimated to be 210 psi. Finished shaft diameter must be 18 ft. The preshaft stress state is attributable to gravity alone. Rock properties are: $C_o = 6,750$ psi, $T_o = 675$ psi, $E = 2.4$ million psi, $\nu = 0.25$, $\gamma = 156$ pcf. What concrete liner thickness is indicated, assuming the maximum allowable concrete liner stress is 3,500 psi?
- 3.54 A large vertical finished (inside) shaft diameter of 9.75m is required for hoisting capacity in a planned high volume underground oil shale mine. An aquifer is encountered at a depth of 387 m where the water pressure is 1.66 MPa.
- Determine the thickness of a concrete liner needed for a liner safety factor of 2.5 when the concrete compressive strength (C_o) is 24.2 MPa, Young's modulus is 38.62 GPa, Poisson's ratio is 0.27. Explain, defend your answer.
 - If changes in the inside diameter of the shaft are monitored for safety, what "reading" in cm would indicate impending failure of the liner?
- 3.55 A vertical circular shaft will pass through a water-bearing stratum at a depth of 847 m where the water pressure is estimated to be 1.45 MPa. Finished shaft diameter must be 5.5 m. The preshaft stress state is attributable to gravity alone. Rock properties are: $C_o = 46.6$ MPa, $T_o = 4.66$ MPa, $E = 16.6$ GPa, $\nu = 0.25$, $\gamma = 24.7$ kN/m³. What concrete liner thickness is indicated, assuming the maximum allowable concrete liner stress is 24.14 MPa?
- 3.56 An *in situ* stress field between 3,000 ft and 3,500 ft is fit to the formulas

$$\sigma_v = 1.125h, \quad \sigma_H = 3,500 + 0.33h, \quad \sigma_h = 3,500 + 0.33h$$

where the stresses are in psi and the depth h is in ft. Stresses σ_v , σ_H , σ_h are principal stresses in the vertical and horizontal directions. The weakest rock along the proposed shaft route has an unconfined compressive strength of 8,000 psi and a tensile strength of 750 psi. A circular, concrete liner is planned below 3,500 ft for a shaft with an outside diameter of 18 ft. If the liner is one foot thick and the concrete properties are: $E = 5.0$ million psi, $\nu = 0.20$, $\gamma = 156$ pcf, $C_o = 4,500$ psi and $T_o = 450$ psi, determine

what uniform radial stress would just cause the liner to fail, assuming the concrete follows a Mohr–Coulomb criterion.

- 3.57 With reference to the previous problem, determine the reduction in diameter of the liner when the liner first fails.
- 3.58 With reference to Problem 3.56, determine the radial displacement of the interface between the liner and shaft wall when the liner first fails.
- 3.59 With reference to Problem 3.56 data, water pressure of 80 psi is anticipated at 4,500 ft. What liner thickness is indicated, if a liner safety factor of 2.5 is required. Note: Minimum thickness is 1 ft.
- 3.60 If a steel liner is used in Problem 3.59 instead of concrete and the steel strength is 36,000 psi (compressive and tensile strengths are equal), what is the corresponding steel liner thickness?
- 3.61 An *in situ* stress field between 914 and 1,067 m is fit to the formulas

$$\sigma_v = 25.45h, \quad \sigma_H = 24,138 + 7.47h, \quad \sigma_h = 24,138 + 7.47h$$

where the stresses are in kPa and the depth h is in m. Stresses σ_v , σ_H , σ_h are principal stresses in the vertical and horizontal directions. The weakest rock along the proposed shaft route has an unconfined compressive strength of 55.2 MPa and a tensile strength of 5.17 MPa. A circular, concrete liner is planned below 1,067 m for a shaft with an outside diameter of 5.5 m. If the liner is 0.3 m thick and the concrete properties are: $E = 34.48$ GPa, $\nu = 0.20$, $\gamma = 24.7$ kN/m³, $C_o = 31.0$ MPa, and $T_o = 3.10$ MPa, determine what uniform radial stress would just cause the liner to fail, assuming the concrete follows a Mohr–Coulomb criterion.

- 3.62 With reference to the previous problem, determine the reduction in diameter of the liner when the liner first fails.
- 3.63 With reference to Problem 3.61, determine the radial displacement of the interface between the liner and shaft wall when the liner first fails.
- 3.64 With reference to Problem 3.61 data, water pressure of 0.552 MPa is anticipated at 1,372 m. What liner thickness is indicated, if a liner safety factor of 2.5 is required. Note: Minimum thickness is 0.3 m.
- 3.65 If a steel liner is used in Problem 3.64 instead of concrete and the steel strength is 248 MPa (compressive and tensile strengths are equal), what is the corresponding steel liner thickness?
- 3.66 A circular concrete shaft liner with a Young's modulus of 3.4 million psi, Poisson's ratio of 0.25, unconfined compressive strength 3,500 psi and tensile strength 350 psi is considered for control of water pressure (190 psi) at a depth of 3,750 ft. Inside shaft diameter after lining must be 18 ft. Find the liner thickness necessary to achieve a safety factor of 2.5. What outside shaft diameter is indicated?
- 3.67 A circular concrete shaft liner with Young's modulus of 3.4 million psi, Poisson's ratio of 0.25, unconfined compressive strength 3,500 psi and tensile strength 350 psi is loaded to the verge of failure in dry ground. If the inside shaft diameter is 22 ft and the liner is one foot thick, what is the contact pressure between rock and liner, assuming it is uniform (psi)?
- 3.68 With reference to Problem 3.67, measurements are made between points on the inside of the liner on opposite ends of a diametral line. What change in diameter from the no-load condition is indicated at the verge of liner failure (inches)?

- 3.69 A circular concrete shaft liner with a Young's modulus of 23.45 GPa, Poisson's ratio of 0.25, unconfined compressive strength 24.1 MPa, and tensile strength 2.41 MPa is considered for control of water pressure (1.31 MPa) at a depth of 1,143 m. Inside shaft diameter after lining must be 5.5 m. Find the liner thickness necessary to achieve a safety factor of 2.5. What outside shaft diameter is indicated?
- 3.70 A circular concrete shaft liner with Young's modulus of 23.45 GPa, Poisson's ratio of 0.25, unconfined compressive strength 24.1 MPa, and tensile strength 2.41 MPa is loaded to the verge of failure in dry ground. If the inside shaft diameter is 6.7 m and the liner is 0.3 m thick, what is the contact pressure between rock and liner, assuming it is uniform (in kPa)?
- 3.71 With reference to Problem 3.70, measurements are made between points on the inside of the liner on opposite ends of a diametral line. What change in diameter from the no-load condition is indicated at the verge of liner failure (in cm)?

Multiple shafts

- 3.72 A 12 ft by 24 ft rectangular shaft is sunk to a depth of 3,000 ft in ground where the premining stress field is given by formulas

$$S_v = 1.2h, \quad S_h = 120 + 0.5h, \quad S_H = 3,240 + 0.3h$$

where h is depth in feet and the stresses are in psi. Show by sketch, the best orientation of the shaft relative to the directions of h and H . Is this the best orientation at all depths to 3,000 ft?

If mine expansion plans call for two additional shafts of the same size, show by sketch the position and orientation of these additional shafts. Explain.

- 3.73 A 3.7×7.4 m rectangular shaft is sunk to a depth of 914 m in ground where the premining stress field is given by formulas

$$S_v = 27.2h, \quad S_h = 282 + 11.3h, \quad S_H = 22,345 + 6.8h$$

where h is depth in m and the stresses are in kPa. Show by sketch, the best orientation of the shaft relative to the directions of h and H . Is this the best orientation at all depths to 914 m?

If mine expansion plans call for two additional shafts of the same size, show by sketch the position and orientation of these additional shafts. Explain.

- 3.74 A rectangular shaft 10 ft by 20 ft with the long axis parallel to the N–S line exists at a depth of 950 ft. The mining plan calls for deepening the shaft to 1,800 ft. The premining stress state relative to compass coordinates is:

$$S_E = 350 + 0.2h, \quad S_N = 420 + 0.35h, \quad S_V = 1.12h$$

where stresses are in psi, h = depth in ft, E , N , V refer to compass coordinates (x = east, y = north, z = up) and compression is positive. Premining shear stresses are nil relative to compass coordinates. Rock properties are:

$$C_o = 23,700 \text{ psi}, \quad T_o = 1,480 \text{ psi}, \quad E = 5.29(10^6 \text{ psi}),$$

$$\nu = 0.27, \quad \gamma = 162 \text{ pcf}$$

A second shaft of identical shape is also planned to the new depth of 1,800 ft. Show a location and orientation (plan view) of the two shafts that is most favorable with respect to stress concentration and explain the basis of your choice.

- 3.75 Given the stresses relative to compass coordinates ($x = \text{east}$, $y = \text{north}$, $z = \text{up}$),

$$\sigma_{xx} = 2,155, \quad \sigma_{yy} = 3,045, \quad \sigma_{zz} = 4,200, \quad \tau_{yx} = -1,222, \\ \tau_{xz} = 0, \quad \tau_{yx} = 0$$

in psi with compression positive, find the principal stresses $\sigma_1, \sigma_2, \sigma_3$ (magnitude and orientation). Show with a suitable sketch. Then consider an unlined shaft that will be 13 ft by 26 ft in section and (a) choose the best opening shape (ellipse, ovaloidal, or rectangle) and best orientation from the rock mechanics perspective, assuming the given stress state is critical to the design, (b) suppose an identical second shaft is planned near the first. Show the orientation and location of the second shaft relative to the first such that interactions are likely to be acceptable and not unduly high. Sketch. Justify your answer.

- 3.76 A rectangular shaft 3 m $\times$ 6 m with the long axis parallel to the N–S line exists at a depth of 290 m. The mining plan calls for deepening the shaft to 550 m. The premining stress state relative to compass coordinates is:

$$S_E = 3,414 + 4.5h, \quad S_N = 2,897 + 7.9h, \quad S_V = 25.3h$$

where stresses are in kPa, $h = \text{depth in m}$, E, N, V refer to compass coordinates ($x = \text{east}$, $y = \text{north}$, $z = \text{up}$) and compression is positive. Premining shear stresses are nil relative to compass coordinates. Rock properties are:

$$C_o = 163.5 \text{ MPa}, \quad T_o = 10.2 \text{ MPa}, \quad E = 36.48 \text{ GPa}, \\ \nu = 0.27, \quad \gamma = 25.6 \text{ kN/m}^3$$

A second shaft of identical shape is also planned to the new depth of 550 m. Show a location and orientation (plan view) of the two shafts that is most favorable with respect to stress concentration and explain the basis of your choice.

- 3.77 Given the stresses relative to compass coordinates ($x = \text{east}$, $y = \text{north}$, $z = \text{up}$),

$$\sigma_{xx} = 14.86, \quad \sigma_{yy} = 21.00, \quad \sigma_{zz} = 29.00, \quad \tau_{yx} = -8.43, \quad \tau_{xz} = 0, \quad \tau_{yx} = 0$$

in MPa with compression positive, find the principal stresses $\sigma_1, \sigma_2, \sigma_3$ (magnitude and orientation). Show with a suitable sketch. Then consider an unlined shaft that will be 4 m $\times$ 8 m in section and (a) choose the best opening shape (ellipse, ovaloidal, or rectangle) and best orientation from the rock mechanics perspective, assuming the given stress state is critical to the design, (b) suppose an identical second shaft is planned near the first. Show the orientation and location of the second shaft relative to the first such that interactions are likely to be acceptable and not unduly high. Sketch. Justify your answer.

Additional problems

- 3.78 Consider a circular shaft excavated in a stress field caused by gravity only. Inclination of the shaft axis with respect to the vertical varies. Suppose the shaft axis inclination

dips δ' degrees from the vertical. Evaluate algebraically the maximum and minimum shaft wall stress concentration factors at the extreme inclinations $\delta' = 0$ and $\delta' = 90$ degrees.

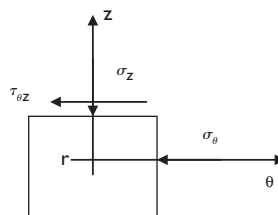
- 3.79 Consider a circular shaft excavated in a stress field caused by gravity only. Inclination of the shaft axis with respect to the vertical varies. Use the development in Example 3.36 to obtain plots of the maximum and minimum stress concentration factors with respect to the circumferential stress as functions of shaft inclination. Note: dip δ of the shaft cross-section is 90° -shaft inclination. Assume $K_o = 1/3$ and $S_v = 7.5$ MPa for plotting.
- 3.80 Consider a circular shaft excavated in a stress field caused by gravity only. Inclination of the shaft axis with respect to the vertical varies. Note: dip δ of the shaft cross-section is 90° -shaft inclination. Assume $K_o = 1/3$ and $S_v = 7.5$ MPa and then evaluate the safety of the shaft wall with respect to the axial stress when the dip of the shaft axis is 60° . Assume $C_o = 10$ MPa and $T_o = 1$ MPa. Are the factors of safety greater or less than in the vertical case?

Note: In the vertical case there are three factors of safety, one with respect to circumferential compression (FS_c), one with respect to tensile failure under circumferential stress, (FS_t), when tension is present, and the other with respect to vertical compression (FS_v).

- 3.81 Consider a circular shaft excavated in a stress field caused by gravity only. Inclination of the shaft axis with respect to the vertical varies. Note: dip δ of the shaft cross-section is 90° -shaft inclination. Assume $K_o = 1/3$ and $S_v = 7.5$ MPa and then evaluate the safety of the shaft wall with respect to the circumferential stress when the dip of the shaft axis is 60° . Assume $C_o = 10$ MPa and $T_o = 1$ MPa. Are the factors of safety greater or less than in the vertical case?

Note: In the vertical case there are three factors of safety, one with respect to circumferential compression (FS_c), one with respect to tensile failure under circumferential stress, (FS_t), when tension is present, and the other with respect to vertical compression (FS_v).

- 3.82 Consider an element of shaft wall in the inclined case where a circumferential normal stress and shear stress act as shown in the sketch. Inclination of the shaft axis with respect to the vertical varies. Evaluate the safety of the shaft wall at the low point of the shaft section with respect to a Mohr-Coulomb failure criterion when the dip of the shaft axis varies from the vertical to the horizontal and plot. Assume: $K_o = 1/3$, $S_v = 7.5$ MPa, and $\nu = 0.25$ and further suppose $C_o = 10$ MPa and $T_o = 1$ MPa.



Sketch for Problem 3.82 looking down the radial direction at a shaft wall point.